THE COMPLETE GUIDE

Zero to Hero AI

The Complete Machine Learning Knowledge Architecture

18 Chapters · 6,972 Concepts

From Python fundamentals to cutting-edge LLMs —
A structured curriculum covering Mathematics, Machine Learning,
Deep Learning, NLP, Large Language Models, and Production Systems.

Built from 23,641 pages across 31 textbooks · 12 breakthrough research deep-dives

Validated against 5,000+ real interview questions

Version 9.0

The Architecture Behind This Book

This is not a book written from memory. It was **engineered** — every concept extracted, cross-validated, and organized by a systematic multi-stage pipeline. Here is exactly what went into building this curriculum.

Stage 1: Source Material Analysis

Master textbook compilations analyzed	**5 PDFs**
Pages read via font-analysis extraction	**10,659**
Open-access textbooks downloaded and processed	**26 books**
Additional textbook pages analyzed	**12,982**
Total pages processed	**23,641**

Books include: Goodfellow *Deep Learning*, Murphy *Probabilistic ML*, Sutton & Barto *RL*, Jurafsky *Speech & Language Processing*, Prince *Understanding DL*, Boyd *Convex Optimization*, MacKay *Information Theory*, and 19 more.

Stage 2: Deep Extraction

Per-book TOC trees generated	
Total tree nodes extracted	**2**
AI agents deployed for extraction & validation	
Heading detection method	Font ar
Noise patterns filtered	Pseudocode, publishers, page nu

Stage 3: Concept Expansion & Deduplication

Duplicate concepts identified & merged	**5,921**
Unique concepts after dedup	**6,407**
Learning domains organized	**18**
Parallel expansion agents	**4**
Domain coverage validated	All 52 key concepts

Stage 4: Cutting-Edge Research (2024–2025)

Deep breakthrough research documents	**12**
Total lines of research written	**14,567**
Blog posts & papers surveyed per topic	**15+**

New concepts injected into tree	**748**
Topics:	SSMs, MoE, RLHF/DPO, RAG, Agents, Diffusion, Flash Attention, LoRA, Multimodal, AI for Science

Stage 5: Interview Question Validation

Interview questions gathered	**5,000+**
Web sources searched	**80+**
Question categories	**19**
Difficulty levels	Basic → Expert
System design questions	**200+**
Coding & implementation questions	**250+**
Goal	Every question answerable from this book's content

Pipeline Summary

Total AI agents orchestrated	**60+**
Total web searches performed	**100+**
Python scripts written for this pipeline	**12**
Validation passes	**5**
Final knowledge tree	**6,407 concepts**
Across	**18 domains**
Depth	**4 levels**

Philosophy: No concept was added by guesswork. Every topic was either (a) extracted from a published textbook, (b) validated via web research against expert sources, or (c) grounded in real interview questions from FAANG and top AI companies. If a 5th-grader can't understand the title, it was rewritten.

How to Use This Book

This book maps the **complete knowledge architecture** for mastering AI and Machine Learning — organized as a **structured curriculum** where every chapter builds on what came before.

Reading Guide

- **Follow the order:** Chapters 1–18 are sequenced so you never encounter a concept before its prerequisites
- **Each chapter** lists prerequisites and learning outcomes at the top
- **Depth:** Topics → Sections → Subsections provide 3–4 levels of detail
- **Cutting-edge sections** (marked □) cover 2024–2025 breakthroughs
- **Interview prep:** Use the companion question bank (5,000+ questions) to test yourself

The Learning Path

The 18 chapters are organized into a natural progression — the way a senior engineer would learn ML from scratch:

Part: Foundations — Chapters 1–4

Programming, mathematics, data literacy, and core ML concepts.

Part: Classical Machine Learning — Chapters 5–6

The supervised and unsupervised algorithms that power most of production ML.

Part: Deep Learning — Chapters 7–10

From neurons to transformers — the architectures reshaping AI.

Part: Language & LLMs — Chapters 11–12

Natural language processing and the large language model revolution.

Part: Production Systems — Chapter 13

Taking ML from notebooks to reliable, scalable production systems.

Part: Probabilistic & Generative AI — Chapters 14–15

Modeling uncertainty and generating new data — the mathematical heart of modern AI.

Part: Advanced Topics — Chapters 16–17

Research frontiers, learning theory, and pushing the boundaries of what's possible.

Part: Responsible AI Chapter 18

Ethics, fairness, privacy, and the societal impact of AI systems.

Chapter Overview

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	Total	**6,972**

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CHAPTERS 1–4

Foundations

Programming, mathematics, data literacy, and core ML concepts.

CHAPTER 1

Python & Programming Foundations

Outcomes: Write Python programs · Use NumPy, Pandas, Matplotlib · Work in Jupyter notebooks

530 concepts · Max depth: 3

Getting Started with Python

What is Programming

- How Computers Execute Instructions
- High-Level vs Low-Level Languages
- Interpreted vs Compiled Languages
- Why Python for Data Science and AI

Installing Python

- Python Versions (Python 3.x)
- Installing on Windows, macOS, Linux
- The Python Interpreter (REPL)
- Running Python Scripts from Terminal
- IDEs and Text Editors (VS Code, PyCharm)

Basic Syntax

- Indentation and Code Blocks
- Comments (Single-line, Multi-line)
- Print Function and Output
- Input Function and User Input
- Variables and Assignment
- Naming Conventions (snake_case)
- Reserved Keywords

Data Types and Variables

Numeric Types

- Integers (int)
- Floating Point Numbers (float)
- Complex Numbers (complex)
- Arithmetic Operators (+, -, *, /, //, %,)
- Operator Precedence
- Type Conversion (int(), float())

Strings

- String Creation (Single, Double, Triple Quotes)
- String Indexing and Slicing
- String Methods (upper, lower, strip, split, join, replace, find)
- String Formatting (f-strings, .format(), %)
- Escape Characters
- String Immutability

- Raw Strings

Booleans and None

- Boolean Values (True, False)
- Comparison Operators (==, !=, <, >, <=, >=)
- Logical Operators (and, or, not)
- Truthy and Falsy Values
- None Type and Its Uses
- Identity Operators (is, is not)

Type System

- Dynamic Typing
- type() Function
- isinstance() Function
- Type Hints and Annotations

Data Structures

Lists

- Creating Lists
- Indexing and Slicing
- List Methods (append, extend, insert, remove, pop, sort, reverse)
- List Mutability
- Nested Lists
- List Copying (Shallow vs Deep)
- List Membership (in operator)

Tuples

- Creating Tuples
- Tuple Immutability
- Tuple Unpacking
- Named Tuples
- When to Use Tuples vs Lists

Dictionaries

- Creating Dictionaries
- Accessing and Modifying Values
- Dictionary Methods (keys, values, items, get, update, pop)
- Iterating Over Dictionaries
- Nested Dictionaries
- Dictionary Comprehensions
- defaultdict and OrderedDict

Sets

- Creating Sets
- Set Operations (union, intersection, difference, symmetric_difference)
- Set Methods (add, remove, discard)
- Frozen Sets
- Use Cases for Sets (Deduplication, Membership Testing)

Control Flow

Conditional Statements

- if Statement
- if-else Statement
- if-elif-else Chain
- Nested Conditionals
- Ternary Operator (Conditional Expression)
- Match-Case (Python 3.10+)

Loops

- for Loop
- Iterating Over Sequences (Lists, Strings, Ranges)
- range() Function
- while Loop
- break Statement
- continue Statement
- else Clause on Loops
- Nested Loops
- enumerate() Function
- zip() Function

Iterators and Iterables

- The Iterator Protocol (__iter__, __next__)
- iter() and next() Built-in Functions
- Iterable vs Iterator
- StopIteration Exception

Functions

Function Basics

- Defining Functions (def keyword)
- Function Parameters and Arguments
- Return Values
- Docstrings
- Multiple Return Values

Arguments and Parameters

- Positional Arguments
- Keyword Arguments
- Default Parameter Values
- *args (Variable Positional Arguments)
- kwargs (Variable Keyword Arguments)
- Argument Unpacking

Scope and Closures

- Local Scope
- Global Scope
- Enclosing (Nonlocal) Scope
- LEGB Rule
- global Keyword

- nonlocal Keyword
- Closures

Advanced Function Concepts

- First-Class Functions
- Higher-Order Functions
- Lambda Functions
- map(), filter(), reduce()
- Recursion
- Memoization

List Comprehensions and Generators

List Comprehensions

- Basic Syntax [expr for item in iterable]
- Conditional Filtering [expr for item in iterable if condition]
- Nested Comprehensions
- Set Comprehensions
- Dictionary Comprehensions

Generators

- Generator Functions (yield keyword)
- Generator Expressions
- Lazy Evaluation
- Memory Efficiency
- Chaining Generators
- send() and close() Methods
- itertools Module (chain, product, combinations, permutations)

Object-Oriented Programming

Classes and Objects

- Defining Classes (class keyword)
- Creating Instances (Objects)
- __init__ Method (Constructor)
- self Parameter
- Instance Attributes
- Class Attributes
- Instance Methods

Encapsulation

- Public Attributes
- Private Attributes (Name Mangling __)
- Protected Attributes (_)
- Property Decorator (@property)
- Getters and Setters

Inheritance

- Single Inheritance
- super() Function
- Method Overriding

- Multiple Inheritance
- Method Resolution Order (MRO)
- Abstract Base Classes (ABC)

Polymorphism

- Duck Typing
- Method Overriding for Polymorphism
- Operator Overloading (Dunder Methods)

Special Methods

- __str__ and __repr__
- __len__, __getitem__, __setitem__
- __eq__, __lt__, __gt__
- __add__, __mul__
- __enter__ and __exit__ (Context Managers)
- __call__

Advanced OOP

- Class Methods (@classmethod)
- Static Methods (@staticmethod)
- Dataclasses
- Composition vs Inheritance
- Mixins
- Metaclasses (type, Custom Metaclasses, __init_subclass__)
- Descriptor Protocol (__get__, __set__, __delete__)
- __slots__ and Inheritance Interactions
- Design Patterns (Singleton, Factory, Observer)

Decorators

Decorator Fundamentals

- Functions as First-Class Objects (Recap)
- Wrapper Functions
- The @ Syntax
- Simple Function Decorators
- functools.wraps

Advanced Decorators

- Decorators with Arguments
- Stacking Multiple Decorators
- Class Decorators
- Common Built-in Decorators (@property, @staticmethod, @classmethod)
- Practical Use Cases (Timing, Logging, Caching)

Error Handling

Exceptions

- What are Exceptions
- Common Exception Types (ValueError, TypeError, IndexError, KeyError, FileNotFoundError)
- try-except Blocks

- Catching Specific Exceptions
- Multiple except Clauses
- else Clause
- finally Clause

Raising and Custom Exceptions

- raise Statement
- Creating Custom Exception Classes
- Exception Chaining (from)
- Exception Hierarchy

Best Practices

- EAFP vs LBYL
- Avoiding Bare except
- Logging Exceptions
- Context Managers for Resource Cleanup

File I/O

Reading and Writing Files

- open() Function
- File Modes (r, w, a, rb, wb)
- Reading Files (read, readline, readlines)
- Writing Files (write, writelines)
- Context Managers (with statement)
- File Paths (Absolute and Relative)

Working with Different Formats

- CSV Files (csv module)
- JSON Files (json module)
- Text Files
- Binary Files
- pickle Module

File System Operations

- os Module (path, listdir, makedirs)
- pathlib Module (Path objects)
- shutil Module (copy, move, delete)
- glob Module (Pattern Matching)

Modules and Packages

Modules

- Importing Modules (import, from…import)
- Module Search Path
- Creating Your Own Modules
- __name__ == '__main__'
- Standard Library Overview (os, sys, math, random, datetime, collections, re)

Packages

- Package Structure (__init__.py)

- Relative vs Absolute Imports
- Namespace Packages

Virtual Environments and pip

- Why Virtual Environments
- Creating venvs (python -m venv)
- Activating and Deactivating
- pip install, pip freeze, pip list
- requirements.txt
- conda Environments (Overview)
- Poetry and pipenv (Overview)

Regular Expressions (re Module)

- Pattern Syntax and Metacharacters
- match vs search vs findall
- Groups, Capture Groups, Named Groups
- Substitution (re.sub)
- Lookahead and Lookbehind Assertions
- Compiling Patterns and Flags
- Common Patterns for Data Cleaning

NumPy

NumPy Arrays (ndarray)

- Creating Arrays (array, zeros, ones, arange, linspace)
- Array Attributes (shape, dtype, ndim, size)
- Array Data Types
- Reshaping Arrays
- Array Indexing and Slicing
- Boolean Indexing (Masking)
- Fancy Indexing

Array Operations

- Element-wise Operations
- Broadcasting Rules
- Universal Functions (ufuncs)
- Aggregation Functions (sum, mean, std, min, max)
- Axis Parameter

Linear Algebra with NumPy

- Dot Product (np.dot)
- Matrix Multiplication (@ operator)
- Transpose
- Inverse and Determinant
- Eigenvalues and Eigenvectors
- np.linalg Module

NumPy Utilities

- Random Number Generation (np.random)
- Sorting and Searching (sort, argsort, where)

- Stacking and Splitting Arrays (vstack, hstack, concatenate, split)
- Saving and Loading Arrays (save, load, savetxt)
- Vectorization and Performance

Advanced Topics

- np.einsum() for Tensor Operations
- Memory Layout (C-order vs Fortran-order)
- Structured Arrays and Record Arrays
- np.memmap for Out-of-Core Computation
- np.random.Generator vs Legacy API
- Views vs Copies (.view() and .copy())

Pandas

Core Data Structures

- Series
- DataFrame
- Index Objects
- Creating DataFrames (from dict, list, CSV, Excel)

Data Selection and Indexing

- Column Selection
- Row Selection with loc and iloc
- Boolean Indexing
- Setting and Resetting Index
- MultiIndex

Data Manipulation

- Adding and Removing Columns
- Renaming Columns
- Sorting (sort_values, sort_index)
- Applying Functions (apply, map, applymap)
- String Methods (.str accessor)
- DateTime Methods (.dt accessor)

Handling Missing Data

- Detecting Missing Values (isnull, notnull)
- Dropping Missing Values (dropna)
- Filling Missing Values (fillna)
- Interpolation

Grouping and Aggregation

- groupby Operations
- Aggregation Functions (agg)
- Transform
- Pivot Tables
- Crosstab
- Melt and Stack/Unstack

Merging and Joining

- merge (SQL-style Joins)

- concat (Stacking DataFrames)
- join Method
- Inner, Outer, Left, Right Joins

Data I/O

- Reading CSV (read_csv)
- Reading Excel (read_excel)
- Reading JSON (read_json)
- Reading SQL (read_sql)
- Writing to Various Formats (to_csv, to_excel, to_json)

Advanced Topics

- query() Method
- pipe() for Method Chaining
- Time Series (resample, rolling, shift)
- Categorical Dtype for Memory Optimization
- Copy-on-Write (CoW)
- Polars as Alternative to Pandas

SciPy

scipy.stats (Statistical Tests & Distributions)

scipy.optimize (Optimization Routines)

scipy.sparse (Sparse Matrices)

scipy.linalg (Linear Algebra)

scipy.spatial (Distance & Spatial Data Structures)

Data Visualization

Matplotlib

- Figure and Axes Objects
- Line Plots
- Scatter Plots
- Bar Charts
- Histograms
- Subplots
- Customizing Plots (Labels, Titles, Legends, Colors)
- Saving Figures

Seaborn

- Statistical Plots Overview
- Distribution Plots (histplot, kdeplot, boxplot, violinplot)
- Categorical Plots (countplot, barplot, stripplot)
- Relational Plots (scatterplot, lineplot)
- Heatmaps and Correlation Matrices
- Pairplot
- Styling and Themes
- FacetGrid for Multi-panel Figures

Interactive Visualization (Plotly, Bokeh)

ML-Specific Plots (Confusion Matrix, ROC, Learning Curves)

Jupyter Notebooks

Jupyter Environment

- Installing Jupyter (pip install jupyter)
- Launching Notebook Server
- JupyterLab vs Jupyter Notebook
- Cell Types (Code, Markdown, Raw)
- Running Cells and Execution Order
- Kernel Management (Restart, Interrupt)

Effective Notebook Usage

- Markdown for Documentation
- Magic Commands (%timeit, %matplotlib inline, %%writefile)
- Inline Plotting
- Exporting Notebooks (HTML, PDF, Python Script)
- Best Practices for Reproducibility
- Google Colab Overview

Version Control with Git

Git Basics

- What is Version Control
- Installing Git
- git init
- git add (Staging)
- git commit
- git status and git log
- git diff
- .gitignore File

Branching and Merging

- Creating Branches
- Switching Branches (checkout/switch)
- Merging Branches
- Resolving Merge Conflicts

Remote Repositories

- GitHub/GitLab Overview
- git clone
- git push and git pull
- git remote
- Pull Requests (Concept)
- Forking a Repository

Scikit-learn

Estimator API (fit, predict, transform)

Pipeline and ColumnTransformer

- Pipeline Construction and Usage

- ColumnTransformer for Mixed Types
- FeatureUnion vs ColumnTransformer

Preprocessing with Scikit-learn

- StandardScaler, MinMaxScaler, RobustScaler
- LabelEncoder, OneHotEncoder, OrdinalEncoder

Model Selection and Evaluation

- train_test_split()
- cross_val_score() and cross_validate()
- GridSearchCV and RandomizedSearchCV

Custom Transformers and Estimators

- Writing Custom Transformers (TransformerMixin)
- Writing Custom Estimators (BaseEstimator)
- Integrating with Pipeline

Model Persistence

- pickle and joblib
- skops for Safe Serialization

Parallel Processing (n_jobs, joblib backend)

Memory Management and CPython Internals

Python Memory Model (Private Heap)

Reference Counting

Generational Garbage Collection

Circular References and gc Module

pymalloc (Small Object Allocator)

Memory Leak Detection and Debugging

__del__ Method and Its Pitfalls

weakref Module

tracemalloc for Memory Profiling

String Interning and Small Integer Caching

Concurrency and Parallelism

Global Interpreter Lock (GIL)

- What the GIL Is and Why It Exists
- Impact on Multi-threaded Programs
- PEP 703 and Free-Threaded Python (3.13+)

Threading

- threading Module
- I/O-Bound vs CPU-Bound Tasks
- Thread Safety and Locks

Multiprocessing

- multiprocessing Module
- Process Pools
- Shared State (Value, Array, Manager, Queue)
- Serialization Challenges (pickle)

concurrent.futures

- ThreadPoolExecutor
- ProcessPoolExecutor
- Future Objects and Callbacks

joblib for Scikit-learn Parallelization

Async Programming

Coroutines (async def, await)

Event Loop (asyncio)

asyncio.gather(), wait(), TaskGroup()

Mixing Async with Threads (run_in_executor)

Async Patterns for ML Services

Concurrency vs Parallelism

Modern Python Features

Walrus Operator (:=, Python 3.8+)

Type Parameter Syntax (Python 3.12)

TypeAlias, TypeGuard, Self Type Hints

Improved Error Messages (Python 3.11+)

ExceptionGroup and except* (Python 3.11)

tomllib Module (Python 3.11)

Faster CPython Project (Performance Improvements)

Algorithms and Data Structures for ML

Searching (Binary Search, Linear Search)

Sorting (Quicksort, Merge Sort, Bubble Sort)

Hash Maps (Implementation and Complexity)

Trees and Tries

Heaps and Priority Queues

ML Algorithm Implementations from Scratch

- K-Means Clustering (NumPy)
- Gradient Descent for Linear Regression
- Decision Tree (ID3/CART)

Testing, Debugging, and Profiling

Debugging

- Print Debugging vs pdb
- breakpoint() (Python 3.7+)
- Common NumPy/Pandas Debugging Pitfalls

Testing

- pytest Framework
- unittest.mock for Mocking
- Parameterized Tests
- Test-Driven Development (TDD)
- Property-Based Testing (hypothesis)

Profiling

- cProfile
- line_profiler
- memory_profiler

ML Implementation Patterns

Numerical Stability

- Log-Sum-Exp Trick
- Numerically Stable Softmax
- Numerically Stable Sigmoid
- Gradient Checking (Numerical vs Analytical)
- NaN/Inf Detection and Handling

From-Scratch Implementation Skills

- Forward and Backward Pass Implementation
- Automatic Differentiation (Micrograd-Style)
- Custom Loss Function Implementation
- Custom Optimizer Implementation
- Efficient Matrix Operations (im2col, einsum)

Data Pipeline Coding

- MapReduce Implementation Patterns
- Streaming Aggregation (Sliding Window)
- Online Learning Update Loops
- Distributed Gradient Computation
- A/B Test Statistical Engine (Sample Size, Sequential Testing)

ML Debugging Patterns

- Dead Neuron Detection
- Gradient Explosion/Vanishing Diagnosis
- Data Leakage Detection in Code
- Memory Profiling for Training
- Distribution Shift Detection in Code

CHAPTER 2

Mathematical Foundations

Prerequisites: Python & Programming Foundations

Outcomes: Linear algebra for ML · Calculus for optimization · Probability & statistics

466 concepts · Max depth: 4

Mathematical Notation and Preliminaries

Sets and Set Operations

- Set Definition and Membership
- Union, Intersection, Complement
- Subsets and Supersets
- Cartesian Product

Functions and Mappings

- Domain, Range, Codomain
- Injective, Surjective, Bijective
- Composition of Functions
- Inverse Functions

Summation and Product Notation

- Sigma Notation
- Product Notation (Pi)
- Common Summation Identities

Logarithms and Exponentials

- Exponential Functions
- Natural Logarithm (ln)
- Log Rules (Product, Quotient, Power)
- Change of Base Formula
- Logarithms in ML (Log-Likelihood, Cross-Entropy)

Linear Algebra

Vectors

- Scalars, Vectors, Matrices, Tensors
- Vector Notation and Representation
- Vector Addition and Scalar Multiplication
- Dot Product (Inner Product)
- Vector Norm (L1, L2, Lp, L-infinity)
- Unit Vectors and Normalization
- Orthogonality and Angle Between Vectors
- Cross Product (3D)
- Linear Combinations
- Span of Vectors

- Linear Independence

Matrices

- Matrix Notation and Dimensions
- Matrix Addition and Scalar Multiplication
- Matrix Multiplication
- Transpose
- Identity Matrix
- Diagonal and Symmetric Matrices
- Matrix Inverse
- Invertibility Conditions
- Orthogonal Matrices
- Positive Definite and Semi-Definite Matrices

Systems of Linear Equations

- Matrix Form of Linear Systems ($Ax = b$)
- Gaussian Elimination
- Row Echelon Form
- Existence and Uniqueness of Solutions
- Homogeneous Systems

Vector Spaces

- Definition of Vector Space
- Subspaces
- Basis and Dimension
- Column Space and Row Space
- Null Space (Kernel)
- Rank of a Matrix

Determinants

- Definition and Computation (2x2, 3x3)
- Properties of Determinants
- Cofactor Expansion
- Geometric Interpretation (Area/Volume Scaling)
- Determinant and Invertibility

Eigenvalues and Eigenvectors

- Definition ($Av = \lambda v$)
- Characteristic Equation
- Computing Eigenvalues and Eigenvectors
- Eigendecomposition (Spectral Decomposition)
- Diagonalization
- Geometric Interpretation
- Eigenvalues of Symmetric Matrices
- Applications in PCA

Singular Value Decomposition (SVD)

- Definition ($A = U\Sigma V^T$)
- Left and Right Singular Vectors
- Singular Values

- Relationship to Eigendecomposition
- Low-Rank Approximation
- Applications (Dimensionality Reduction, Recommender Systems)

Projections and Least Squares

- Vector Projections
- Projection onto Subspaces
- Projection Matrix
- Least Squares Solution
- Normal Equation ($A^TAx = A^Tb$)
- Geometric Interpretation of Least Squares

Matrix Calculus Essentials

- Gradient of Scalar w.r.t. Vector
- Jacobian Matrix
- Hessian Matrix
- Common Matrix Derivatives

Condition Number and Numerical Stability

Graph Laplacian and Spectral Clustering

Power Iteration Method

Eckart-Young Theorem

Randomized SVD

Non-Negative Matrix Factorization (NMF)

PCA Comprehensive

- PCA Step-by-Step Procedure
- Centering and Standardization Requirements
- Scree Plot and Explained Variance Ratio
- PCA vs Factor Analysis
- Variance Maximization Derivation
- Reconstruction Error Minimization Derivation
- Kernel PCA
- Incremental PCA
- Probabilistic PCA
- Sparse PCA

Covariance Matrix

- Definition and Properties
- Correlation Matrix
- Positive Semi-Definiteness
- Whitening Transform
- Mahalanobis Distance

Cholesky Decomposition

- Definition (LL^T Factorization)
- Efficient Sampling from Multivariate Gaussians
- Applications in Gaussian Processes

QR Decomposition

- Gram-Schmidt Process
- Householder Reflections
- Least Squares via QR
- Numerical Stability

Calculus

Limits and Continuity

- Intuition of Limits
- Limit Definition
- One-sided Limits
- Continuity
- Limits at Infinity

Derivatives (Single Variable)

- Definition as Rate of Change
- Derivative as Slope of Tangent
- Power Rule
- Product Rule
- Quotient Rule
- Chain Rule
- Derivatives of Common Functions (Polynomial, Exponential, Logarithmic, Trigonometric)
- Higher-Order Derivatives
- Implicit Differentiation

Applications of Derivatives

- Critical Points
- Local Minima and Maxima
- First and Second Derivative Tests
- Concavity and Inflection Points
- Optimization Problems
- Taylor Series and Approximation

Multivariable Calculus

- Functions of Multiple Variables
- Partial Derivatives
- Gradient Vector
- Directional Derivatives
- Gradient as Direction of Steepest Ascent
- Hessian Matrix (Second-Order Partial Derivatives)
- Multivariable Chain Rule
- Total Derivative

Integrals

- Definite Integrals (Area Under Curve)
- Indefinite Integrals (Antiderivatives)
- Fundamental Theorem of Calculus
- Integration Techniques (Substitution, Parts)

- Multiple Integrals (Double, Triple)
- Applications in Probability (PDF Integration)

Key Functions in ML

- Sigmoid Function and Its Derivative
- Softmax Function
- ReLU and Its Derivative
- Tanh Function
- Log-Sum-Exp Trick

Backpropagation

- Backpropagation Algorithm (Step by Step)
- Chain Rule Through Multiple Layers
- Deriving Backprop for a Two-Layer Network
- Vanishing Gradient Problem
- Exploding Gradient Problem and Gradient Clipping
- Backpropagation Through Time (BPTT)
- Computational Graphs (Static vs Dynamic)

Automatic Differentiation

- Forward-Mode vs Reverse-Mode AD
- Symbolic vs Numerical vs Automatic Differentiation
- Autograd in PyTorch and TensorFlow

Softmax Gradient Derivation

Cross-Entropy + Softmax Combined Gradient

L2 Regularization as Weight Decay (Derivation)

OLS Closed-Form Solution (Derivation)

Probability

Probability Foundations

- Conditional Probability — Definition P(A|B), Multiplication Rule, Independence, Law of Total Probability, Bayes' Theorem, Prior, Likelihood, Posterior, *(+1 more)*
- Information Theory Basics — Entropy, Cross-Entropy, KL Divergence, Mutual Information, Connection to ML Loss Functions

Distributions

- Common Distributions — Discrete Distributions, Continuous Distributions
- Conjugate Priors (Bernoulli-Beta, Gaussian-Gaussian, Poisson-Gamma)
- Student's t-Distribution
- Chi-Squared Distribution and Relationship to Gamma
- Gaussian Mixture Models (GMM)
- Probability Chain Rule for Joint Distributions

Statistical Inference

- MLE and MAP Estimation — MLE Formula and Intuition, MLE Derivation for Gaussian, MLE Derivation for Bernoulli, MAP Formula and Comparison with MLE, MAP as Regularization (Gaussian Prior → L2, Laplacian → L1), Limitations of Point Estimates vs Full Bayesian
- Bayesian Model Comparison and Marginal Likelihood

- Likelihood Ratio Test
- Frequentist vs Bayesian Hypothesis Testing
- Multiple Comparisons (Bonferroni, Benjamini-Hochberg)
- Bootstrap Confidence Intervals
- Confidence Intervals for ML Metrics (Accuracy, AUC)

Probability Fundamentals

- Sample Space and Events
- Probability Axioms
- Complementary Events
- Frequentist vs Bayesian Interpretation
- Addition Rule (Union of Events)
- Mutually Exclusive Events

Random Variables

- Discrete Random Variables
- Continuous Random Variables
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Joint Distributions
- Marginal Distributions
- Conditional Distributions

Expectation and Moments

- Expected Value (Mean)
- Variance and Standard Deviation
- Covariance
- Correlation Coefficient
- Properties of Expectation (Linearity)
- Law of Large Numbers
- Central Limit Theorem

Moment-Generating Function (MGF)

Characteristic Function

Skewness and Kurtosis

Bayesian Credible Intervals

Concentration Inequalities

- Markov’s Inequality
- Chebyshev’s Inequality
- Hoeffding’s Inequality
- Chernoff Bound
- McDiarmid’s Inequality
- Sub-Gaussian Random Variables

Statistics

Descriptive Statistics

- Measures of Central Tendency (Mean, Median, Mode)
- Measures of Spread (Range, IQR, Variance, Standard Deviation)
- Percentiles and Quartiles
- Skewness and Kurtosis
- Data Visualization for Statistics (Box Plots, Histograms)

Inferential Statistics

- Population vs Sample
- Sampling Methods (Random, Stratified, Systematic)
- Sampling Distribution
- Standard Error
- Point Estimation
- Bias and Consistency of Estimators

Confidence Intervals

- Concept and Interpretation
- Confidence Level (90%, 95%, 99%)
- CI for Mean (Known and Unknown Variance)
- CI for Proportions
- t-Distribution
- Sample Size and Margin of Error

Hypothesis Testing

- Null and Alternative Hypotheses
- Test Statistic
- p-value
- Significance Level (alpha)
- Type I and Type II Errors
- Power of a Test
- One-tailed vs Two-tailed Tests
- z-Test
- t-Test (One-sample, Two-sample, Paired)
- Chi-Squared Test
- ANOVA (Analysis of Variance)
- Multiple Testing Problem (Bonferroni Correction)

A/B Testing

- Experiment Design
- Control and Treatment Groups
- Randomization
- Sample Size Calculation
- Statistical Significance in A/B Tests
- Practical Significance vs Statistical Significance
- Common Pitfalls (Peeking, Multiple Comparisons)

Correlation and Regression (Statistical View)

- Pearson Correlation
- Spearman Rank Correlation

- Correlation vs Causation
- Simple Linear Regression
- R-squared
- Residual Analysis

Optimization

Optimization Fundamentals

- What is Optimization
- Objective Function (Cost/Loss Function)
- Minimization vs Maximization
- Local vs Global Optima
- Saddle Points
- Convex vs Non-Convex Functions

Gradient Descent

- Intuition (Ball Rolling Down Hill)
- The Update Rule ($\theta = \theta - \alpha\nabla f(\theta)$)
- Learning Rate
- Batch Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-Batch Gradient Descent
- Convergence Criteria
- Learning Rate Scheduling
- Effect of Learning Rate (Too Large, Too Small)

Advanced Gradient Methods

- Momentum
- Nesterov Accelerated Gradient
- AdaGrad
- RMSProp
- Adam Optimizer
- AdamW (Weight Decay)
- Gradient Clipping

Convex Optimization

- Convex Sets
- Convex Functions
- Properties (Global Minimum is Unique)
- Jensen’s Inequality
- Examples of Convex Problems in ML

Constrained Optimization

- Equality Constraints
- Inequality Constraints
- Lagrange Multipliers
- KKT Conditions
- Lagrangian Function
- Dual Problem
- Application in SVM

Regularization as Optimization

- L1 Regularization (Lasso)
- L2 Regularization (Ridge)
- Elastic Net
- Regularization as Constrained Optimization
- Effect on Loss Landscape

Convergence Rate Theory (Convex, Strongly Convex)

Natural Gradient Descent and Fisher Information Matrix

Second-Order Optimization Methods

- Newton's Method
- L-BFGS (Limited-Memory BFGS)
- Quasi-Newton Methods
- Hessian Approximation
- Conjugate Gradient Method

Information Theory

Joint Entropy

Conditional Entropy H(Y|X)

Cross-Entropy as Loss Function (Detailed Treatment)

Relationship: H(P,Q) = H(P) + D_KL(P||Q)

KL Divergence Properties (Asymmetry, Non-negativity)

KL Divergence in ML (VAEs, Knowledge Distillation, Policy Optimization)

Jensen-Shannon Divergence

Mutual Information for Feature Selection

Information Bottleneck Method

Entropy Estimation from Finite Samples

Data Processing Inequality

Probability Puzzles and Brain Teasers

Birthday Problem and Collision Analysis

- Birthday Paradox Formula and Derivation
- Poisson Approximation for Collisions
- Birthday Attack in Cryptography
- Hash Collision Probability Estimation
- Generalized Birthday Problem (N Categories, k Samples)
- Non-Uniform Birthday Distributions

Monty Hall and Bayesian Reasoning Puzzles

- Classic Monty Hall (3 Doors)
- Generalized n-Door Monty Hall
- Random Host Variant
- Connection to Bayesian Updating

- Likelihood Ratios and Sequential Hypothesis Testing
- Base Rate Neglect and Medical Testing

Expected Value Problems

- Dice Rolling Expected Values
- Coin Flipping Sequences (Consecutive Heads, Pattern Waiting)
- Card Drawing Problems
- St. Petersburg Paradox
- Optimal Re-Roll Strategies
- Stick Breaking Problems

Conditional Probability Puzzles

- Boy or Girl Paradox
- Three Prisoners Problem
- 100 Lockers Problem (Number Theory Connection)
- Airplane Seating Problem
- Three Cards Problem
- Matching Socks Problem

Geometric Probability

- Buffon's Needle Problem
- Meeting Problem (Uniform Arrivals)
- Stick Breaking Triangle Inequality
- Bertrand's Paradox (Chord Length)
- Monte Carlo Estimation of Pi
- Random Points in Geometric Regions
- Expected Distances in Unit Shapes

Markov Chain Puzzles

- Gambler's Ruin Problem
- Absorbing States and Absorption Probabilities
- PageRank as Markov Chain
- Mixing Time and MCMC Connection
- Card Shuffling as Markov Chain
- Weather Model Stationary Distribution

Random Walk Problems

- Polya's Recurrence Theorem (1D, 2D, 3D)
- First Passage Times
- Cover Time on Graphs
- Random Walk and Graph Laplacian Eigenvalues
- DeepWalk Connection to Word2Vec
- Label Propagation via Random Walks

Coupon Collector Problem

- Expected Time via Harmonic Series
- Variance of Collection Time
- Non-Uniform Coupon Probabilities
- Connection to Mini-Batch Sampling Coverage
- Concentration Inequalities for Coupon Collector

Secretary Problem and Optimal Stopping

- 1/e Optimal Strategy and Proof
- Connection to Explore-Exploit Tradeoff
- Connection to Multi-Armed Bandits
- Connection to Early Stopping in Training
- k-Secretary Variant

Fermi Estimation for ML

- GPU Memory and Parameter Counting
- FLOPs Estimation for Training
- Dataset Size and Token Estimation
- Inference Latency and Energy Estimation
- Vocabulary Size and Tokenizer Design
- Hardware Throughput Estimation

Matrix Puzzles and Linear Algebra Riddles

- Eigenvalue Transformations (A^2, A^{-1}, A+I)
- Idempotent and Projection Matrices
- Condition Number and Optimization Convergence
- Sherman-Morrison Formula for Online Learning
- Nuclear Norm and Low-Rank Completion
- Numerical Stability: SVD vs Eigendecomposition for PCA
- Sparse Matrix Computation (Randomized SVD, Lanczos)

CHAPTER 3

Data Fundamentals

Prerequisites: Python & Programming Foundations, Mathematical Foundations

Outcomes: Load, clean, explore data · Feature engineering · Avoid data leakage

286 concepts · Max depth: 4

Data Types and Sources

Types of Data

- Structured vs Unstructured vs Semi-Structured
- Numerical Data (Continuous, Discrete)
- Categorical Data (Nominal, Ordinal)
- Text Data
- Time Series Data
- Image Data
- Tabular Data

Data Sources

- CSV and TSV Files
- JSON and XML
- Databases (SQL, NoSQL)
- APIs and Web Scraping
- Public Datasets (Kaggle, UCI, Government)
- Data Warehouses and Data Lakes

Loading Data

- Loading CSV with Pandas
- Loading Excel Files
- Loading JSON Data
- Connecting to SQL Databases
- Handling Large Files (Chunking)
- Data Type Inference and Explicit Type Casting

Columnar and Binary Data Formats

- Apache Parquet
- Apache Arrow and Feather
- HDF5
- Protocol Buffers
- Format Selection Guidelines

Data Labeling and Annotation

Annotation Strategies and Tools

Inter-Annotator Agreement (Cohen’s Kappa)

Labeling Quality Control

Weak Supervision and Programmatic Labeling (Snorkel)

Active Learning for Efficient Labeling

Exploratory Data Analysis (EDA)

Initial Data Inspection

- Shape and Size of Data
- Column Names and Data Types (dtypes)
- head(), tail(), sample()
- info() and describe()
- Checking for Duplicates
- Unique Values and Value Counts

Univariate Analysis

- Distribution of Numerical Features (Histograms, KDE)
- Central Tendency and Spread
- Skewness and Kurtosis
- Distribution of Categorical Features (Bar Plots, Pie Charts)
- Identifying Rare Categories

Bivariate and Multivariate Analysis

- Scatter Plots (Numerical vs Numerical)
- Box Plots (Numerical vs Categorical)
- Correlation Matrix and Heatmap
- Pair Plots
- Grouped Statistics
- Cross-tabulation
- Feature vs Target Relationships

Data Quality Assessment

- Missing Value Patterns (MCAR, MAR, MNAR)
- Data Integrity Checks
- Inconsistencies and Errors
- Class Imbalance Detection
- Feature Cardinality

Data Drift Detection During EDA

EDA for High-Dimensional Data (1000+ Features)

Automated EDA Tools (pandas-profiling, sweetviz, D-Tale)

Data Cleaning

Handling Missing Values

- Types of Missingness (MCAR, MAR, MNAR)
- Dropping Rows or Columns
- Imputation with Mean, Median, Mode
- Forward Fill and Backward Fill
- Interpolation
- KNN Imputation
- Iterative Imputation (MICE)

- Adding Missing Indicator Features
- When to Drop vs Impute

Handling Outliers

- Detecting Outliers (IQR Method)
- Z-Score Method
- Visual Detection (Box Plots, Scatter Plots)
- Domain Knowledge for Outlier Detection
- Removing Outliers
- Capping/Winsorizing
- Log Transformation
- When to Keep Outliers

Data Type Corrections

- Converting String to Numeric
- Parsing Dates
- Fixing Inconsistent Categories
- Handling Mixed Types in Columns

Handling Duplicates

- Identifying Exact Duplicates
- Near-Duplicate Detection
- Deciding What to Drop

Text Data Cleaning

- Trimming Whitespace
- Case Normalization
- Removing Special Characters
- Standardizing Formats (Phones, Addresses)

Model-Based Outlier Detection

- Isolation Forest
- DBSCAN-Based Outlier Detection
- Local Outlier Factor (LOF)

Feature Engineering

Encoding Categorical Variables

- Label Encoding
- Ordinal Encoding
- One-Hot Encoding
- Binary Encoding
- Target Encoding (Mean Encoding)
- Frequency Encoding
- Handling High Cardinality Features
- When to Use Which Encoding

Feature Scaling and Normalization

- Why Scaling Matters
- Min-Max Scaling (Normalization to [0,1])
- Standardization (Z-score Scaling)

- Robust Scaling (Using Median and IQR)
- MaxAbs Scaling
- Log Transformation
- Power Transforms (Box-Cox, Yeo-Johnson)
- When to Scale (Distance-Based Models, Gradient-Based Models)
- When Not to Scale (Tree-Based Models)

Creating New Features

- Mathematical Combinations (Ratios, Products, Differences)
- Polynomial Features
- Binning/Discretization
- Date/Time Feature Extraction (Day, Month, Hour, Day of Week)
- Aggregation Features (Group Statistics)
- Interaction Features
- Domain-Specific Feature Engineering
- Text Feature Extraction (Length, Word Count)

Feature Selection

- Filter Methods — Variance Threshold, Correlation-Based Selection, Chi-Squared Test, Mutual Information, ANOVA F-test
- Wrapper Methods — Forward Selection, Backward Elimination, Recursive Feature Elimination (RFE)
- Embedded Methods — L1 Regularization (Lasso) for Selection, Tree-Based Feature Importance, Permutation Importance
- Dimensionality Reduction — PCA (Principal Component Analysis), t-SNE (for Visualization), UMAP (for Visualization), Truncated SVD

Variance Inflation Factor (VIF)

SHAP-Based Feature Selection

Curse of Dimensionality

Domain-Specific

- Geospatial Feature Engineering (Distance, Clustering, Region Encoding)
- Recommendation System Features (User-Item Interactions)
- Automated Feature Engineering (Featuretools)
- Feature Stores (Concept and Tools)
- TF-IDF, N-grams, and Embedding Features from Text

Cyclical Feature Encoding (sin/cos)

Entity Embeddings for Categorical Variables

Hash Encoding

Data Splitting and Leakage

Train/Test/Validation Splits

- Purpose of Each Split
- Typical Split Ratios (70/15/15, 80/10/10)
- Random Splitting
- Stratified Splitting (Preserving Class Distribution)

- Time-Based Splitting (for Time Series)
- Group-Based Splitting (Preventing Group Leakage)
- sklearn train_test_split

Cross-Validation Strategies

- K-Fold Cross-Validation
- Stratified K-Fold
- Leave-One-Out Cross-Validation (LOOCV)
- Repeated K-Fold
- Time Series Cross-Validation (Rolling Window)
- Group K-Fold

Data Leakage

- What is Data Leakage
- Target Leakage
- Train-Test Contamination
- Leakage from Preprocessing (Fitting on Full Data)
- Temporal Leakage
- How to Detect Leakage
- Pipelines to Prevent Leakage (sklearn Pipeline)
- Common Sources of Leakage in Practice

Data Visualization for Communication

Choosing the Right Chart

- Bar Charts for Comparisons
- Line Charts for Trends
- Scatter Plots for Relationships
- Histograms for Distributions
- Heatmaps for Matrices
- Pie Charts (and Why to Avoid Them)

Visualization Best Practices

- Clear Titles and Labels
- Appropriate Scale and Axis Limits
- Color Usage and Accessibility
- Avoiding Misleading Visualizations
- Annotation and Highlighting
- Plot Size and Aspect Ratio

Handling Imbalanced Datasets

What is Class Imbalance (Definition, Examples)

Why Accuracy Fails for Imbalanced Data

Metrics for Imbalanced Data

- Precision, Recall, F1-Score
- Precision-Recall Tradeoff
- ROC Curve vs Precision-Recall Curve
- AUC-ROC and AUC-PR

Resampling Techniques

- Random Oversampling and Undersampling
- SMOTE (Synthetic Minority Over-sampling)
- SMOTE Risks and Limitations
- ADASYN

Algorithmic Approaches

- Class Weights in Loss Function
- Cost-Sensitive Learning
- Focal Loss
- Ensemble Methods (BalancedRandomForest, EasyEnsemble)

Extreme Imbalance and Anomaly Detection

Data Augmentation

Why Data Augmentation Works (Regularization Effect)

Image Augmentation (Flip, Rotate, Crop, Color Jitter, Noise)

Text Augmentation (Synonym Replacement, Back-Translation, Random Insertion)

Tabular Augmentation (SMOTE, Noise Injection, Mixup)

Advanced Techniques

- Mixup and CutMix
- Test-Time Augmentation (TTA)
- Generative Augmentation (GANs, Diffusion Models)

Risks and Limitations of Augmentation

SQL for Machine Learning

SQL Fundamentals

- SELECT, WHERE, GROUP BY, HAVING, ORDER BY
- JOIN Types (INNER, LEFT, RIGHT, FULL OUTER)
- NULL Handling (IS NULL, COALESCE, IFNULL)
- Subqueries and Nested Queries

Window Functions

- ROW_NUMBER, RANK, DENSE_RANK
- LAG and LEAD
- Running Totals and Moving Averages
- PARTITION BY and ORDER BY in Windows

Common Table Expressions (CTEs)

- Basic CTEs vs Subqueries
- Recursive CTEs

Advanced SQL Patterns

- Consecutive Event Detection
- Sessionization
- Pivoting and Unpivoting
- Feature Engineering in SQL (Rolling Aggregates)

Query Optimization (Indexing, Query Plans, Partitioning)

Data Pipelines and Infrastructure

ETL vs ELT

Idempotency in Pipelines

Pipeline Orchestration Tools (Airflow, Prefect, Dagster)

Feature Stores (Feast, Tecton)

Data Versioning (DVC, LakeFS, Delta Lake)

Data Quality Frameworks (Great Expectations, Pandera)

Schema Evolution

Real-Time vs Batch Inference Pipelines

Data Drift Monitoring in Production

Model Versioning vs Data Versioning

Reproducible ML Experiments

Data Ethics and Quality

Bias in Data

- Sampling Bias
- Survivorship Bias
- Label Noise
- Algorithmic Fairness

Data Quality

- Data Quality Dimensions (Completeness, Consistency, Accuracy)
- Data Quality Measurement and Monitoring

Privacy and Ethics

- Differential Privacy
- PII Handling in ML Datasets
- Ethics of Web-Scraped Data
- Fairness Detection and Mitigation

Time Series Preprocessing

Lag Features

Rolling Statistics (Mean, Std, Min, Max)

Differencing for Stationarity

Handling Irregular Time Series

Seasonal Decomposition

CHAPTER 4

ML Fundamentals & Core Concepts

Prerequisites: Mathematical Foundations, Data Fundamentals

Outcomes: Understand ML paradigms · Model evaluation · Bias-variance tradeoff

278 concepts · Max depth: 4

Introduction to Machine Learning

What is Machine Learning

- Definition (Learning from Data)
- ML vs Traditional Programming
- When to Use ML (and When Not To)
- Brief History of ML
- AI vs ML vs Deep Learning vs Data Science

Types of Machine Learning

- Supervised Learning — Definition (Labeled Data), Classification (Discrete Output), Regression (Continuous Output), Examples (Spam Detection, House Prices, Medical Diagnosis)
- Unsupervised Learning — Definition (No Labels), Clustering, Dimensionality Reduction, Anomaly Detection, Association Rules, Examples (Customer Segmentation, Topic Modeling)
- Reinforcement Learning — Agent, Environment, Actions, Rewards, Exploration vs Exploitation, Examples (Game Playing, Robotics)
- Semi-Supervised Learning — Small Labeled + Large Unlabeled Data, Label Propagation, Pseudo-Labeling, Use Cases (Medical Imaging, NLP)
- Self-Supervised Learning — Creating Labels from Data Itself, Pretext Tasks, Contrastive Learning, Examples (BERT Masked Language Model, Image Inpainting)

Key ML Terminology

- Features (Input Variables)
- Target (Output Variable)
- Training Data and Test Data
- Model
- Parameters vs Hyperparameters
- Training (Fitting)
- Prediction (Inference)
- Loss Function
- Generalization

The ML Workflow (End-to-End)

Problem Formulation

- Defining the Business Problem
- Translating to ML Problem (Classification vs Regression)
- Choosing Success Metrics
- Feasibility Assessment

- Baseline Definition

Data Collection and Preparation

- Data Gathering
- Data Cleaning (Recap)
- EDA (Recap)
- Feature Engineering (Recap)
- Data Splitting (Recap)

Model Training

- Choosing Candidate Models
- Training the Model (fit)
- Monitoring Training (Loss Curves)
- Iterating on Features and Models

Model Evaluation

- Evaluating on Validation Set
- Comparing Multiple Models
- Error Analysis
- Final Evaluation on Test Set

Model Deployment and Monitoring

- Saving and Loading Models (pickle, joblib)
- Serving Predictions (APIs, Batch)
- Monitoring Model Performance
- Model Drift and Retraining
- A/B Testing in Production

Data Leakage

- Train-Test Contamination
- Target Leakage
- Temporal Leakage
- Feature Leakage from Preprocessing
- Prevention Strategies

Bias-Variance Tradeoff

Understanding Bias

- Definition (Error from Assumptions)
- High Bias = Underfitting
- Models with High Bias (Linear Regression on Non-linear Data)
- Symptoms of High Bias (Low Training Accuracy)

Understanding Variance

- Definition (Sensitivity to Training Data)
- High Variance = Overfitting
- Models with High Variance (Deep Trees, High-Degree Polynomials)
- Symptoms of High Variance (Gap Between Train and Test Accuracy)

The Tradeoff

- Total Error = Bias^2 + Variance + Irreducible Error

- Bias-Variance Tradeoff Curve
- Finding the Sweet Spot (Model Complexity)
- Strategies to Reduce Bias (More Complex Models, More Features)
- Strategies to Reduce Variance (More Data, Regularization, Simpler Models)

Overfitting and Underfitting

Overfitting

- Definition (Memorizing Training Data)
- Causes (Too Complex Model, Too Little Data, Noise)
- Detection (Train vs Validation Performance Gap)
- Learning Curves for Detecting Overfitting

Underfitting

- Definition (Model Too Simple)
- Causes (Insufficient Model Capacity, Too Few Features)
- Detection (Poor Performance on Both Train and Validation)
- Learning Curves for Detecting Underfitting

Remedies for Overfitting

- Regularization (L1, L2)
- Reducing Model Complexity
- Getting More Training Data
- Data Augmentation
- Early Stopping
- Dropout (for Neural Networks)
- Ensemble Methods
- Feature Selection (Removing Noisy Features)

Remedies for Underfitting

- Increasing Model Complexity
- Adding More Features
- Feature Engineering
- Reducing Regularization
- Training Longer

Advanced Regularization Techniques

- Batch Normalization as Implicit Regularization
- Weight Decay vs L2 Regularization (Differ with Adam)
- Multi-Task Learning as Regularization
- Parameter Norm Penalties vs Constrained Optimization
- Lottery Ticket Hypothesis

Learning Theory Foundations

- PAC Learning Framework
- VC Dimension (Vapnik-Chervonenkis)
- Structural Risk Minimization (SRM)
- Rademacher Complexity
- Double Descent Phenomenon
- Regularization as Bayesian Prior (L2 = Gaussian, L1 = Laplace)

Cross-Validation

Why Cross-Validation

- Limitations of Single Train/Test Split
- More Reliable Performance Estimates
- Better Use of Limited Data

Cross-Validation Techniques

- Hold-Out Validation
- K-Fold Cross-Validation
- Stratified K-Fold
- Leave-One-Out (LOOCV)
- Repeated K-Fold
- Nested Cross-Validation
- Time Series Split
- Group K-Fold

Interpreting CV Results

- Mean and Standard Deviation of Scores
- Fold-by-Fold Analysis
- Comparing Models with CV
- Statistical Tests for Model Comparison

Evaluation Metrics

Classification Metrics

- Confusion Matrix (TP, TN, FP, FN)
- Accuracy
- Precision
- Recall (Sensitivity, True Positive Rate)
- Specificity (True Negative Rate)
- F1 Score (Harmonic Mean of Precision and Recall)
- F-beta Score
- ROC Curve
- AUC-ROC (Area Under ROC Curve)
- Precision-Recall Curve
- Average Precision (AP)
- Log Loss (Cross-Entropy Loss)
- Cohen’s Kappa
- Matthews Correlation Coefficient (MCC)

Multi-Class Metrics

- Macro Averaging
- Micro Averaging
- Weighted Averaging
- Multi-Class Confusion Matrix
- One-vs-All and One-vs-One
- Top-K Accuracy

Regression Metrics

- Mean Absolute Error (MAE)

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (Coefficient of Determination)
- Adjusted R-squared
- Mean Absolute Percentage Error (MAPE)
- Median Absolute Error

Choosing the Right Metric

- Metric Selection Based on Business Goals
- Imbalanced Classes (Precision, Recall, F1 over Accuracy)
- Cost-Sensitive Evaluation
- Threshold Selection for Classification
- Custom Metrics

Ranking Metrics (NDCG, MAP, MRR)

Offline vs Online Metrics (and When They Disagree)

Balanced Accuracy

Calibration (Reliability Diagrams, Brier Score, Platt Scaling, Isotonic Regression)

Model Selection

Choosing Between Models

- No Free Lunch Theorem
- Starting Simple (Baselines)
- Model Complexity vs Data Size
- Interpretability vs Performance
- Training Time vs Prediction Time
- Data Characteristics (Linearity, Feature Types, Size)

Model Comparison Methods

- Cross-Validation Comparison
- Statistical Significance of Differences
- Paired t-Test for Model Comparison
- Wilcoxon Signed-Rank Test
- Multiple Model Comparison (Friedman Test)

Ensemble Methods (Overview)

- Why Ensembles Work (Diversity)
- Bagging (Bootstrap Aggregating)
- Boosting
- Stacking
- Voting (Hard and Soft)
- When to Use Ensembles

Practical Model Selection Workflow

- Start with Baseline (Mean, Most Frequent)
- Try Simple Models First (Logistic Regression, Decision Tree)
- Move to Complex Models (Random Forest, Gradient Boosting)
- Compare Performance Across Metrics

- Consider Deployment Constraints
- Document Model Selection Rationale

Hyperparameter Search Methods

- Grid Search
- Random Search
- Bayesian Optimization (Optuna, Hyperopt)
- Halving and Successive Elimination
- Early Stopping in HPO

Model Interpretability Overview

- SHAP (SHapley Additive exPlanations)
- LIME (Local Interpretable Explanations)
- Partial Dependence Plots
- Feature Importance Methods
- Interpretability vs Performance Tradeoffs

ML Optimization Algorithms

Gradient Descent Fundamentals

- Batch Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-Batch Gradient Descent
- Convergence Properties
- Learning Rate Sensitivity

Momentum-Based Methods

- Classical Momentum
- Nesterov Accelerated Gradient (NAG)

Adaptive Learning Rate Methods

- AdaGrad (Adaptive Gradient)
- RMSprop
- Adam (Adaptive Moment Estimation)
- Adam Bias Correction Terms
- AdamW (Decoupled Weight Decay)
- LAMB (Layer-wise Adaptive Moments for Batch Training)
- SAM (Sharpness-Aware Minimization)

Learning Rate Schedules

- Step Decay
- Exponential Decay
- Cosine Annealing
- Warmup Schedules (Linear, Gradual)
- Cyclical Learning Rates
- One-Cycle Policy and Super-Convergence
- Linear Scaling Rule for Large Batch

Optimization Challenges

- Saddle Points vs Local Minima
- Loss Landscape Smoothness

- Hessian Conditioning and Convergence
- Vanishing and Exploding Gradients
- Gradient Clipping (Norm and Value)
- Reasons for Non-Convergence

Second-Order Methods

- Newton's Method
- L-BFGS
- Natural Gradient Descent
- Fisher Information Matrix
- Why Rarely Used in Deep Learning

Advanced Training Techniques

- Mixed-Precision Training (FP16/BF16)
- Loss Scaling for Mixed Precision
- Batch Size and Learning Rate Interaction

CHAPTERS 5–6

Classical Machine Learning

The supervised and unsupervised algorithms that power most of production ML.

CHAPTER 5

Supervised Learning

Prerequisites: ML Fundamentals & Core Concepts

Outcomes: Regression & classification · Ensemble methods · SVMs

397 concepts · Max depth: 4

Linear Regression

Simple Linear Regression

- Least Squares Estimation
- Coefficient Interpretation
- Residual Analysis
- R-squared and Adjusted R-squared
- Assumptions (Linearity, Independence, Homoscedasticity, Normality)
- Confidence and Prediction Intervals

Multiple Linear Regression

- Matrix Formulation (Normal Equation)
- Multicollinearity and VIF
- Feature Selection (Forward, Backward, Stepwise)
- Interaction Terms
- Partial F-tests
- Categorical Variable Encoding (Dummy Variables)

Polynomial Regression

- Degree Selection
- Overfitting vs Underfitting Tradeoff
- Basis Function Expansion
- Cross-Validation for Model Selection

Regularization

- Ridge Regression (L2) — L2 Penalty Term, Closed-Form Solution, Bias-Variance Tradeoff, Hyperparameter Tuning (Lambda), Relationship to Bayesian Prior
- Lasso Regression (L1) — L1 Penalty Term, Sparsity and Feature Selection, Coordinate Descent Algorithm, LARS (Least Angle Regression), Solution Path
- ElasticNet — Combined L1 and L2 Penalties, Mixing Parameter (Alpha), When to Use Over Ridge or Lasso, Group Effect for Correlated Features

Math Prerequisites

- Linear Algebra (Matrix Operations, Inverses)
- Calculus (Partial Derivatives, Gradient)
- Statistics (Hypothesis Testing, Distributions)
- Optimization (Convexity, Gradient Descent)

Geometric Interpretation (Projection onto Column Space)

Weighted Least Squares (WLS)

Generalized Least Squares (GLS)

Influence Diagnostics (Cook's Distance, Leverage, Hat Matrix)

Gauss-Markov Theorem (BLUE Property)

Logistic Regression

Binary Logistic Regression

- Sigmoid Function
- Log-Odds (Logit) Interpretation
- Maximum Likelihood Estimation
- Cross-Entropy Loss
- Decision Boundary
- Odds Ratio Interpretation

Multinomial Logistic Regression

- Softmax Function
- One-vs-Rest Strategy
- Categorical Cross-Entropy
- Reference Category Selection

Ordinal Logistic Regression

- Proportional Odds Model
- Cumulative Logit
- Threshold Parameters

Optimization

- Gradient Descent
- Newton-Raphson Method
- Iteratively Reweighted Least Squares (IRLS)
- Stochastic Gradient Descent

Regularized Logistic Regression

- L1 Regularization (Sparse Models)
- L2 Regularization
- ElasticNet Penalty

Evaluation Metrics

- Confusion Matrix
- Precision, Recall, F1-Score
- ROC Curve and AUC
- Precision-Recall Curve
- Calibration (Brier Score)
- Threshold Selection

Complete Separation and Quasi-Separation

Probit Regression (Comparison with Logistic)

k-Nearest Neighbors (k-NN)

Algorithm Fundamentals

- Instance-Based (Lazy) Learning
- Distance Metrics (Euclidean, Manhattan, Minkowski)
- Choosing k (Odd Values, Cross-Validation)
- Weighted k-NN (Distance-Weighted Voting)

Computational Considerations

- Curse of Dimensionality
- KD-Trees for Efficient Search
- Ball Trees
- Locality-Sensitive Hashing (LSH)
- Feature Scaling Importance

Variants

- k-NN for Classification
- k-NN for Regression
- Radius-Based Neighbors
- Condensed Nearest Neighbors
- Edited Nearest Neighbors

When to Use

- Small to Medium Datasets
- Non-Linear Decision Boundaries
- Multi-Class Problems
- Low-Dimensional Data

Handling Missing Data in KNN

KNN and Kernel Density Estimation Relationship

Naive Bayes

Theoretical Foundation

- Bayes Theorem
- Conditional Independence Assumption
- Prior and Posterior Probabilities
- Likelihood Computation
- MAP Decision Rule

Gaussian Naive Bayes

- Continuous Feature Assumption
- Mean and Variance Estimation per Class
- Gaussian PDF for Likelihood
- Use Case: Real-Valued Features

Multinomial Naive Bayes

- Count-Based Features
- Laplace Smoothing (Additive Smoothing)
- Use Case: Text Classification (Bag of Words)
- Document Term Frequencies

Bernoulli Naive Bayes

- Binary Feature Vectors
- Presence/Absence Modeling
- Use Case: Binary Text Features
- Explicit Penalty for Non-Occurrence

Complement Naive Bayes

- Imbalanced Class Handling
- Complement Class Estimation

When to Use

- High-Dimensional Data
- Text Classification and Spam Filtering
- Fast Training and Prediction Required
- Limited Training Data

Handling Correlated Features

Naive Bayes vs Logistic Regression Comparison

Incremental / Out-of-Core Learning

Decision Trees

CART (Classification and Regression Trees)

- Binary Splits
- Recursive Partitioning
- Classification Trees (Discrete Output)
- Regression Trees (Continuous Output, Mean Prediction)

Splitting Criteria

- Entropy and Information Gain — Shannon Entropy Formula, Information Gain Calculation, Gain Ratio (Normalized), ID3 and C4.5 Algorithms
- Gini Impurity — Gini Index Formula, Weighted Gini for Splits, Comparison with Entropy, CART Default Criterion
- Variance Reduction (Regression) — MSE Reduction, MAE Criterion

Pruning

- Pre-Pruning (Early Stopping) — Max Depth, Min Samples Split, Min Samples Leaf, Max Features, Min Impurity Decrease
- Post-Pruning — Cost-Complexity Pruning (CCP), Reduced Error Pruning, Minimal Cost-Complexity Subtree, Alpha Parameter Selection via CV

Tree Properties

- Interpretability
- Feature Importance
- Handling Missing Values
- No Feature Scaling Required
- Instability (High Variance)

Conditional Inference Trees (Unbiased Splitting)

Minimum Description Length (MDL) Principle

Oblique (Multivariate) Decision Trees

Ensemble Methods

Bagging (Bootstrap Aggregating)

- Bootstrap Sampling
- Aggregation (Voting/Averaging)
- Variance Reduction
- Out-of-Bag (OOB) Error Estimation
- Parallel Training

Random Forests

- Feature Subspace Sampling (sqrt(p), log2(p))
- Tree Decorrelation
- Feature Importance (MDI, Permutation)
- Proximity Matrix
- Hyperparameters (n_estimators, max_features, max_depth)
- Extremely Randomized Trees (Extra-Trees)
- When to Use (High-Dimensional, Tabular Data)

Boosting

- AdaBoost — Adaptive Reweighting of Samples, Weak Learner (Decision Stumps), Exponential Loss, Learner Weight Computation, Sequential Training
- Gradient Boosting — Functional Gradient Descent, Fitting Residuals (Pseudo-Residuals), Arbitrary Differentiable Loss Functions, Learning Rate (Shrinkage), Subsampling (Stochastic Gradient Boosting), Tree Depth Control
- XGBoost — Regularized Objective (L1 + L2 on Leaves), Second-Order Taylor Approximation, Weighted Quantile Sketch (Approximate Split Finding), Sparsity-Aware Split Finding, Column Block and Cache-Aware Access, Parallel and Distributed Computing, *(+6 more)*
- LightGBM — Leaf-Wise (Best-First) Tree Growth, Gradient-Based One-Side Sampling (GOSS), Exclusive Feature Bundling (EFB), Histogram-Based Splitting, Categorical Feature Handling, Speed and Memory Efficiency
- CatBoost — Ordered Boosting (Prevents Target Leakage), Native Categorical Feature Encoding, Symmetric (Oblivious) Decision Trees, Random Permutations, GPU Training Support

Stacking (Stacked Generalization)

- Base Learners (Level-0)
- Meta-Learner (Level-1)
- Cross-Validated Predictions for Training
- Blending (Holdout-Based Variant)

Voting Ensembles

- Hard Voting (Majority)
- Soft Voting (Probability Averaging)
- Weighted Voting

Support Vector Machines (SVMs)

Linear SVM

- Maximum Margin Hyperplane
- Support Vectors

- Margin Calculation
- Primal Formulation
- Dual Formulation (Lagrange Multipliers)
- KKT Conditions

Soft Margin SVM

- Slack Variables
- C Parameter (Regularization)
- Hinge Loss
- Tradeoff: Margin Width vs Misclassification

Kernel Trick

- Implicit Feature Mapping
- Kernel Function (Mercer's Condition)
- Common Kernels — Linear Kernel, Polynomial Kernel, Radial Basis Function (RBF/Gaussian), Sigmoid Kernel
- Kernel Matrix (Gram Matrix)
- Kernel Hyperparameter Selection (Gamma, Degree)

Multi-Class SVM

- One-vs-One (OvO)
- One-vs-Rest (OvR)
- Directed Acyclic Graph SVM (DAGSVM)
- Crammer-Singer Formulation

SVM for Regression (SVR)

- Epsilon-Insensitive Loss
- Epsilon Tube
- Nu-SVR

Optimization Algorithms

- Sequential Minimal Optimization (SMO)
- Quadratic Programming
- Chunking
- LibSVM / LibLinear Implementations

Math Prerequisites

- Convex Optimization
- Lagrangian Duality
- Functional Analysis (Kernel Spaces)
- Linear Algebra (Inner Products, Projections)

Generalized Linear Models (GLMs)

Framework

- Exponential Family Distributions
- Link Function
- Linear Predictor
- Variance Function
- Canonical Link

Common GLMs

- Gaussian (Identity Link → Linear Regression)
- Binomial (Logit Link → Logistic Regression)
- Poisson (Log Link → Count Data)
- Gamma (Inverse Link → Positive Continuous)
- Negative Binomial (Overdispersed Counts)
- Tweedie (Insurance/Zero-Inflated)

Estimation

- Maximum Likelihood via IRLS
- Deviance and Model Comparison
- Wald, Score, and Likelihood Ratio Tests
- Quasi-Likelihood Methods

Diagnostics

- Residual Types (Pearson, Deviance, Anscombe)
- Leverage and Influence
- Overdispersion Detection
- Goodness-of-Fit (AIC, BIC)

Advanced Supervised Learning Topics

Multi-Label Classification (Approaches, Metrics)

Online Learning vs Batch Learning

Active Learning (Uncertainty Sampling, Query-by-Committee)

Label Noise Handling (Noise-Robust Losses, Confident Learning)

Discriminative vs Generative Models

Adversarial Robustness (Adversarial Training, PGD)

Covariate Shift Detection and Correction

Transfer Learning for Tabular Data

Classifier Calibration (Platt Scaling, Isotonic Regression, Temperature Scaling)

Bayesian vs Frequentist Approach Comparison

Multi-Modal Data Fusion (Images + Text + Tabular)

Few-Shot and Semi-Supervised Learning Strategies

Class Imbalance Handling

- SMOTE and Variants (ADASYN, Borderline-SMOTE)
- Random Oversampling / Undersampling
- Tomek Links and Edited Nearest Neighbors
- Cost-Sensitive Learning (class_weight)
- Threshold Tuning
- Combined Strategies

Time Series Forecasting

Time Series Fundamentals

- Components: Trend, Seasonality, Cyclicity, Noise
- Additive vs Multiplicative Decomposition
- Stationarity Definition and Importance
- Stationarity Tests (ADF Test, KPSS Test)
- Making Series Stationary (Differencing, Detrending, Log Transform)
- Autocorrelation Function (ACF)
- Partial Autocorrelation Function (PACF)
- Seasonal Decomposition (STL)

Classical Statistical Models

- ARIMA — AR (Autoregressive) Component — p, I (Integrated / Differencing) Component — d, MA (Moving Average) Component — q, Parameter Selection via ACF/PACF and AIC/BIC, Residual Diagnostics (Ljung-Box Test), Forecasting and Confidence Intervals
- SARIMA — Seasonal Parameters (P, D, Q, m), When Seasonal Component is Needed
- ARIMAX / SARIMAX — Exogenous Variables, Covariate Integration
- Exponential Smoothing — Simple Exponential Smoothing (Level Only), Holt's Method (Level + Trend), Holt-Winters (Level + Trend + Seasonality), ETS Framework, Relationship to ARIMA Models

Modern Forecasting Methods

- Prophet — Additive Regression Model, Trend Modeling and Changepoints, Fourier Series for Seasonality, Holiday and Special Event Effects, Advantages and Limitations
- ML for Time Series — Feature Engineering: Lag Features, Rolling Statistics, Structuring Time Series for Supervised Learning, XGBoost / LightGBM for Time Series, Data Leakage Risks in Time Series
- Neural Approaches — LSTM and GRU for Sequences, Temporal Convolutional Networks (TCN), Transformers for Time Series, Foundation Models for Time Series, Statistical vs Neural: When to Use Each

Evaluation and Validation

- Walk-Forward Validation
- Expanding Window vs Sliding Window CV
- Forecast Accuracy Metrics (MAE, RMSE, MAPE, SMAPE)
- Residual Diagnostics

Advanced Topics

- Multivariate Time Series Forecasting
- Ensembling Time Series Models
- Outlier Detection in Time Series
- Missing Value Handling in Time Series
- Transfer Learning for Time Series

Survival Analysis

Censoring and Survival Functions

Kaplan-Meier Estimator

Cox Proportional Hazards Model

Accelerated Failure Time Models

Deep Survival Models (DeepSurv)

CHAPTER 6

Unsupervised Learning

Prerequisites: ML Fundamentals & Core Concepts

Outcomes: Clustering · Dimensionality reduction · Anomaly detection

268 concepts · Max depth: 4

Clustering

k-Means Clustering

- Lloyd's Algorithm
- Objective: Minimize Within-Cluster Sum of Squares
- Random Initialization Problem
- Convergence Properties
- k-Means++ Initialization — Probabilistic Seeding, D^2 Weighting, Guaranteed O(log k) Approximation
- Mini-Batch k-Means
- Choosing k (Elbow Method, Silhouette Score, Gap Statistic)
- Limitations (Spherical Clusters, Equal Size Assumption)
- K-Modes and K-Prototypes (Categorical Data)
- Parallelizing K-Means for Large Datasets

DBSCAN

- Density-Based Approach
- Core Points, Border Points, Noise Points
- Epsilon Neighborhood
- MinPts Parameter
- Reachability and Connectivity
- Arbitrary Cluster Shapes
- No Need to Specify Number of Clusters
- OPTICS (Ordering Points, Density-Based)
- HDBSCAN (Hierarchical DBSCAN)
- k-Distance Graph for eps Parameter Selection

Hierarchical Clustering

- Agglomerative (Bottom-Up) — Single Linkage (Minimum Distance), Complete Linkage (Maximum Distance), Average Linkage (UPGMA), Ward's Method (Minimum Variance)
- Divisive (Top-Down) — DIANA Algorithm, Bisecting k-Means
- Dendrogram Visualization
- Cutting the Dendrogram (Choosing Clusters)
- Cophenetic Correlation
- Inversions in Dendrograms
- Scalable Variants (BIRCH, CURE)

Spectral Clustering

- Graph-Based Approach
- Similarity/Affinity Matrix Construction
- Graph Laplacian (Unnormalized, Normalized)
- Eigendecomposition of Laplacian
- Spectral Embedding
- k-Means on Embedded Space
- Ncut (Normalized Cut)
- When to Use (Non-Convex Clusters)

Gaussian Mixture Models (GMMs)

- Mixture of Gaussians
- Soft (Probabilistic) Cluster Assignments
- Parameters (Means, Covariances, Mixing Weights)
- EM Algorithm for GMMs
- E-Step: Responsibilities
- M-Step: Parameter Updates
- Covariance Types (Full, Tied, Diagonal, Spherical)
- Model Selection (BIC, AIC for Number of Components)
- Bayesian GMM (Dirichlet Process)

Cluster Evaluation Metrics

- Silhouette Score
- Davies-Bouldin Index
- Calinski-Harabasz Index
- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)
- Homogeneity, Completeness, V-Measure
- Internal vs External Metrics Taxonomy
- Dimensionality Reduction Evaluation (Trustworthiness, Continuity)
- Stability-Based Cluster Evaluation
- Business-Oriented Evaluation Frameworks

Dimensionality Reduction

Principal Component Analysis (PCA)

- Covariance Matrix Computation
- Eigendecomposition
- Singular Value Decomposition (SVD)
- Explained Variance Ratio
- Scree Plot
- Data Centering and Scaling
- Reconstruction Error
- Incremental PCA (Large Datasets)
- Sparse PCA
- Randomized PCA
- Probabilistic PCA
- PCA with Sparse Data
- Kaiser Criterion for Component Selection

- PCA and Linear Autoencoder Equivalence

Kernel PCA

- Non-Linear Extension of PCA
- Kernel Trick in Feature Space
- Centering in Feature Space
- Kernel Selection (RBF, Polynomial)
- Pre-Image Problem

t-SNE (t-distributed Stochastic Neighbor Embedding)

- Pairwise Similarity in High-D (Gaussian Kernel)
- Pairwise Similarity in Low-D (Student-t Distribution)
- KL Divergence Minimization
- Perplexity Parameter
- Crowding Problem Solution
- Barnes-Hut Approximation
- Non-Convex Optimization (Multiple Runs)
- Visualization Only (Not for Downstream Tasks)
- t-SNE vs MDS Cost Function Comparison
- Out-of-Sample Extension (Parametric t-SNE)

UMAP (Uniform Manifold Approximation and Projection)

- Topological Data Analysis Foundation
- Fuzzy Simplicial Sets
- Cross-Entropy Optimization
- n_neighbors Parameter (Local vs Global)
- min_dist Parameter
- Preserves More Global Structure than t-SNE
- Faster and Scalable
- Supports Supervised and Semi-Supervised Modes

Linear Discriminant Analysis (LDA)

- Supervised Dimensionality Reduction
- Maximizing Between-Class / Within-Class Scatter
- Fisher’s Criterion
- Generalized Eigenvalue Problem
- Maximum (C-1) Components
- Assumptions (Gaussian, Equal Covariance)

Independent Component Analysis (ICA)

- Statistical Independence (Not Just Uncorrelated)
- Non-Gaussianity Maximization
- FastICA Algorithm
- Negentropy Approximation
- Blind Source Separation
- Cocktail Party Problem

Autoencoders (as Dimensionality Reduction)

- Encoder-Decoder Architecture
- Bottleneck (Latent) Layer

- Reconstruction Loss
- Non-Linear PCA Analogy
- Undercomplete Autoencoders
- Denoising Autoencoders
- Variational Autoencoders (VAE)
- Sparse Autoencoders
- Contractive Autoencoders

Other Methods

- Multidimensional Scaling (MDS)
- Isomap (Geodesic Distances)
- Locally Linear Embedding (LLE)
- Factor Analysis
- Non-Negative Matrix Factorization (NMF)
- Random Projections (Johnson-Lindenstrauss)
- Laplacian Eigenmaps

Anomaly Detection

Statistical Methods

- Z-Score / Standard Deviation
- Grubbs' Test
- Mahalanobis Distance
- Gaussian Distribution Fitting

Density-Based Methods

- Local Outlier Factor (LOF)
- DBSCAN-Based (Noise Points as Anomalies)
- Kernel Density Estimation (KDE)

Isolation Forest

- Random Partitioning
- Anomaly Score (Path Length)
- No Density Estimation Needed
- Linear Time Complexity

One-Class SVM

- Learning Boundary of Normal Data
- Nu Parameter (Outlier Fraction)
- Kernel-Based Separation

Autoencoder-Based

- High Reconstruction Error as Anomaly Signal
- Training on Normal Data Only
- Threshold Selection

Evaluation

- Precision at k
- ROC-AUC for Anomaly Scoring
- Class Imbalance Challenges

Anomaly Detection Taxonomy (Supervised, Semi-Supervised, Unsupervised)

Anomaly Types (Point, Contextual, Collective)

Time Series Anomaly Detection

Concept Drift in Anomaly Detection Systems

Association Rules

Key Concepts

- Itemsets and Transactions
- Support (Frequency of Itemset)
- Confidence (Conditional Probability)
- Lift (Independence Ratio)
- Conviction

Algorithms

- Apriori Algorithm — Anti-Monotone Property, Candidate Generation and Pruning, Level-Wise Search, Minimum Support Threshold
- FP-Growth — FP-Tree Construction, Conditional Pattern Bases, No Candidate Generation, Memory-Efficient Representation
- Eclat (Vertical Data Format)

Applications

- Market Basket Analysis
- Recommendation Systems
- Web Usage Mining

Expectation-Maximization (EM) Algorithm

General Framework

- Latent Variable Models
- Incomplete Data Likelihood
- Evidence Lower Bound (ELBO)
- Jensen's Inequality

E-Step

- Compute Expected Sufficient Statistics
- Posterior Over Latent Variables
- Responsibility Computation

M-Step

- Maximize Expected Complete-Data Log-Likelihood
- Parameter Update Rules
- Closed-Form Solutions (When Available)

Properties

- Monotonic Likelihood Increase
- Convergence to Local Maximum
- Sensitive to Initialization
- Generalizes k-Means (Hard EM)

Applications

- Gaussian Mixture Models
- Hidden Markov Models (Baum-Welch)
- Missing Data Imputation
- Factor Analysis
- Latent Dirichlet Allocation

Self-Supervised and Representation Learning

Contrastive Learning

Word Embeddings (Word2Vec, GloVe)

Pre-Text Tasks

Feature Learning

SimCLR (Simple Contrastive Learning of Representations)

MoCo (Momentum Contrast)

BYOL (Bootstrap Your Own Latent)

Barlow Twins

UMAP for Visualizing Transformer Embeddings

Deep Clustering Methods (DEC, IDEC, DeepCluster)

Topic Modeling

Latent Dirichlet Allocation (LDA)

- Generative Process
- Document-Topic Distribution (θ)
- Topic-Word Distribution (β)
- Dirichlet Priors and Hyperparameters (α, β)
- Plate Notation
- Inference: Variational Inference (VI)
- Inference: Collapsed Gibbs Sampling
- Online LDA for Large Corpora
- Choosing the Number of Topics

Evaluation

- Perplexity
- Topic Coherence (C_v, NPMI, UCI)
- Human Evaluation of Topics
- Intrinsic vs Extrinsic Evaluation

NMF for Topic Modeling

- Non-Negative Matrix Factorization Approach
- Comparison with LDA
- When to Prefer NMF over LDA

Text Preprocessing for Topic Modeling

- Tokenization and Stopword Removal
- Lemmatization / Stemming

- TF-IDF vs Bag-of-Words Input

Extensions and Modern Approaches

- Dynamic Topic Models
- Correlated Topic Models
- Hierarchical LDA
- Neural Topic Models (ProdLDA, ETM)
- BERTopic (Embedding-Based)

Applications

- Document Classification and Clustering
- Content Recommendation
- Trend Analysis

CHAPTERS 7–10

Deep Learning

From neurons to transformers — the architectures reshaping AI.

CHAPTER 7

Deep Learning Foundations

Prerequisites: Supervised Learning, Mathematical Foundations

Outcomes: Neural network fundamentals · Backpropagation · Activation functions

236 concepts · Max depth: 4

Biological Inspiration

Biological Neurons

- Dendrites & Axons
- Synaptic Connections
- Action Potential
- Hebbian Learning

Artificial Neurons

- Weighted Inputs
- Summation Function
- Activation Threshold
- Output Signal

Biological vs Artificial Comparison

- Parallelism
- Learning Mechanisms
- Structural Differences

Perceptron

Single-Layer Perceptron

- Linear Decision Boundary
- Perceptron Learning Rule
- Convergence Theorem

Limitations

- XOR Problem
- Linear Separability Requirement

Multi-Layer Perceptron

- Input Layer
- Hidden Layers
- Output Layer
- Fully Connected Architecture
- Non-Linear Decision Boundaries

Bias Term

- Role in Shifting Decision Boundary
- Bias vs No-Bias Tradeoff

Dense vs Sparse Layers

- Fully Connected Architecture
- Sparse Activation Patterns

Architecture Design Heuristics

- Layer Count Selection
- Neuron Count Heuristics
- Grid Search vs Rules of Thumb

Parameter Sharing

- Weight Tying
- Shared Representations Across Layers
- Use in CNNs and RNNs

Activation Functions

Sigmoid

- Formula & Range (0,1)
- Saturation Problem
- Non-Zero Centered Output
- Use in Binary Classification

Tanh

- Formula & Range (-1,1)
- Zero-Centered
- Saturation at Extremes

ReLU

- Formula: max(0, x)
- Sparsity Activation
- Dying ReLU Problem
- Computational Efficiency

ReLU Variants

- Leaky ReLU
- Parametric ReLU (PReLU)
- ELU
- SELU

GELU

- Gaussian Error Linear Unit
- Smooth Approximation
- Use in Transformers

Swish

- Self-Gated Activation
- x * sigmoid(βx)
- Smooth Non-Monotonic

Softmax

- Probability Distribution Output

- Multi-Class Classification
- Temperature Scaling

Activation Function Selection Criteria

- Computational Cost Comparison (FLOPs, Latency)
- Smoothness & Differentiability Properties
- Architecture-Specific Choices (CNNs vs Transformers)
- Empirical Ablation Study Design

Custom Activation Functions

- Design Principles
- Monotonicity vs Non-Monotonicity
- Task-Specific Activations

Discrete Relaxation Techniques

- Gumbel-Softmax (Concrete Distribution)
- Temperature Annealing for Discrete Variables
- Applications in VAEs and Discrete Latent Models

Loss Functions

Regression Losses

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Huber Loss
- Log-Cosh Loss

Classification Losses

- Binary Cross-Entropy
- Categorical Cross-Entropy
- Sparse Categorical Cross-Entropy
- Focal Loss

Margin-Based Losses

- Hinge Loss
- Squared Hinge Loss
- Triplet Loss
- Contrastive Loss

Loss Function Properties

- Convexity
- Differentiability
- Sensitivity to Outliers

Loss vs Cost Function Distinction

- Per-Sample Loss
- Batch-Aggregated Cost
- Regularization Terms in Cost

Backpropagation

Forward Pass

- Input Propagation
- Layer-wise Computation
- Loss Computation

Chain Rule

- Composite Function Derivatives
- Local Gradients
- Upstream Gradients

Backward Pass

- Output Layer Gradients
- Hidden Layer Gradients
- Parameter Gradients
- Gradient Accumulation
- Gradient Verification — Numerical Gradient Checking, Finite Difference Method, Relative Error Threshold

Computational Graph

- Nodes and Edges
- Forward Mode Differentiation
- Reverse Mode Differentiation
- Automatic Differentiation
- Static vs Dynamic Computation Graphs — Define-and-Run (TF 1.x), Define-by-Run (PyTorch), Tracing vs Eager Execution, Trade-offs for Debugging and Performance

Backpropagation Through Time (BPTT)

- Unrolled Computation Graph
- Long-Range Dependency Challenge
- Truncated BPTT
- Sequence Length vs Memory Tradeoff

Jacobian & Hessian in Backprop

- Jacobian Matrix Definition & Role
- Hessian Matrix
- Second-Order Optimization (Newton's Method)
- Hessian-Vector Products
- Fisher Information Matrix
- Efficient Computation via Automatic Differentiation

Differentiation Methods Comparison

- Symbolic Differentiation
- Numerical Differentiation (Finite Differences)
- Automatic Differentiation (Forward & Reverse Mode)
- Trade-offs: Accuracy, Speed, Complexity

Gradient Problems

Vanishing Gradients

- Cause: Saturating Activations
- Effect on Deep Networks

- Sigmoid/Tanh Gradient Shrinkage
- Solutions: ReLU, Skip Connections

Exploding Gradients

- Cause: Large Weight Magnitudes
- Numerical Overflow
- Solutions: Gradient Clipping
- Proper Initialization

Weight Initialization

Zero Initialization Problem

- Symmetry Breaking
- Identical Gradients

Random Initialization

- Small Random Values
- Uniform vs Normal Distribution

Xavier/Glorot Initialization

- Variance Preservation
- Fan-in and Fan-out
- For Sigmoid/Tanh

He/Kaiming Initialization

- ReLU-Aware Scaling
- Variance = 2/fan_in
- For ReLU Networks

Other Strategies

- Orthogonal Initialization
- LSUV
- Fixup Initialization

Initialization & Normalization Interaction

- BN Reducing Initialization Sensitivity
- When Initialization Still Matters with BN

Universal Approximation Theorem

Statement

- Single Hidden Layer Sufficiency
- Arbitrary Width Requirement
- Continuous Function Approximation

Implications

- Existence vs Learnability
- Width vs Depth Tradeoff
- Practical Limitations
- Over-Parameterization — Implicit Regularization, Generalization Despite Excess Capacity, Interpolation Threshold

- Double Descent Phenomenon — Classical vs Modern Interpolation Regime, Epoch-Wise Double Descent, Model-Wise Double Descent
- Lottery Ticket Hypothesis — Sparse Subnetwork Discovery, Iterative Magnitude Pruning, Trainable Matching Tickets
- Weight Matrix Rank & Expressiveness — Low-Rank Approximations, Rank Deficiency Effects, Intrinsic Dimensionality

Depth Efficiency

- Exponential Advantage of Depth
- Hierarchical Feature Learning
- Compositional Functions

Deep Learning Theory

- Neural Tangent Kernel (NTK) — Infinite Width Limit, Kernel Regime vs Feature Learning Regime, Lazy Training
- Information Bottleneck — Compression Phase, Fitting Phase, Mutual Information in Layers
- Grokking — Delayed Generalization, Memorization-to-Generalization Transition, Weight Decay Role
- Generalization Theory — PAC-Bayes Bounds, Compression-Based Bounds, Flatness-Based Bounds

CHAPTER 8

Training Deep Networks

Prerequisites: Deep Learning Foundations

Outcomes: Optimizer selection · Regularization · PyTorch/TF proficiency

458 concepts · Max depth: 5

Optimization Algorithms

Gradient Descent Variants

- Batch Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-Batch Gradient Descent
- Batch Size Effects

Momentum-Based

- Classical Momentum
- Exponential Moving Average of Gradients
- Nesterov Accelerated Gradient
- Lookahead Gradient

Adaptive Learning Rate

- AdaGrad — Per-Parameter Learning Rates, Accumulated Squared Gradients, Diminishing Learning Rate Problem
- RMSprop — Exponential Decay of Squared Gradients, Leaky AdaGrad
- Adam — First Moment Estimate (Mean), Second Moment Estimate (Variance), Bias Correction, Hyperparameters: β1, β2, ε
- AdamW — Decoupled Weight Decay, L2 vs Weight Decay Distinction
- Other Optimizers — AdaFactor, LAMB, RAdam, Lookahead, LION, LARS (Layer-wise Adaptive Rate Scaling), *(+1 more)*

Optimizer Theory & Analysis

- Sharp vs Flat Minima
- Implicit Regularization of SGD
- Adam vs SGD Generalization Gap
- Natural Gradient Descent
- Fisher Information Matrix
- Large Batch Training Challenges

Learning Rate Schedules

Step Decay

- Fixed Step Reduction
- Milestone-Based

Cosine Annealing

- Cosine Decay

- Cosine with Warm Restarts
- Half-Cycle Cosine

Warmup

- Linear Warmup
- Gradual Warmup
- Warmup + Decay Combination

Cyclical Learning Rates

- Triangular Policy
- One-Cycle Policy
- Super-Convergence

Other Schedules

- Exponential Decay
- Polynomial Decay
- Reduce on Plateau
- Inverse Square Root

Learning Rate Finder

- Leslie Smith's Method
- Log-Scale LR Sweep
- Loss vs LR Curve Interpretation
- Selecting Max LR

LR & Batch Size Interaction

- Linear Scaling Rule
- Square Root Scaling Rule
- When Linear Scaling Breaks Down
- Gradient Accumulation Effective LR

LR Schedule Diagnostics

- Signs of Suboptimal Schedule
- Switching Optimizers Mid-Training (Adam→SGD)
- Elastic Scaling Adjustments

Normalization Techniques

Batch Normalization

- Mini-Batch Statistics
- Learnable Parameters (γ, β)
- Training vs Inference Mode
- Running Mean/Variance
- Internal Covariate Shift
- BN Placement Debate — Before Activation (Original), After Activation (Common Practice), Empirical Comparisons
- BN & Convergence — Learning Rate Tolerance, Faster Convergence Mechanism, Loss Landscape Smoothing (Santurkar 2018)
- BN Failure Modes — Small Batch Size Issues, Batch Size = 1 Problem, Distributed Training Statistics
- Internal Covariate Shift Debate — Original Hypothesis (Ioffe & Szegedy 2015), Critique

(Santurkar et al. 2018), Alternative Explanations
- Normalization Interactions — BN + Dropout Interaction, BN as Implicit Regularizer, BN + Weight Decay Effects

Layer Normalization

- Per-Sample Normalization
- Batch-Size Independent
- Use in Transformers

Other Normalizations

- Instance Normalization
- Group Normalization
- RMS Normalization
- Weight Normalization

Regularization

Dropout

- Random Neuron Deactivation
- Inverted Dropout
- Dropout Rate Selection
- Monte Carlo Dropout
- Spatial Dropout
- DropConnect
- Ensemble Interpretation — Exponential Sub-Network Averaging, Approximate Model Averaging, Weight Scaling at Inference
- Dropout Effects on Training Dynamics — Impact on Training vs Validation Curves, Interaction with Batch Normalization

Weight Decay

- L2 Regularization
- L1 Regularization (Sparsity)
- Elastic Net
- Decoupled Weight Decay

Early Stopping

- Validation Monitoring
- Patience Parameter
- Best Model Checkpointing

Data Augmentation as Regularization

Label Smoothing

Stochastic Depth

Gradient Management

Gradient Clipping

- Clip by Value
- Clip by Norm
- Global Norm Clipping

Gradient Accumulation

- Simulating Large Batches
- Memory Efficiency
- Effective Batch Size Calculation — Accumulation Steps × Micro-Batch × GPUs, Learning Rate Adjustment
- Gradient Accumulation Pitfalls — Batch Normalization Statistics Mismatch, LR Scaling Requirements

Mixed Precision Training

FP16 vs FP32

- Memory Savings
- Compute Speedup

Loss Scaling

- Static Loss Scaling
- Dynamic Loss Scaling

BFloat16

FP8 Training

Automatic Mixed Precision (AMP)

FP32 Master Weights & Safe Operations

- Operations Requiring FP32 (Softmax, Loss, Norm)
- Master Weight Copy
- Convergence Impact Analysis

PyTorch

Tensors

- Creation & Operations
- GPU Transfer (.to(device))
- Tensor Shapes & Broadcasting
- Tensor Datatypes

Autograd

- Computation Graph
- requires_grad
- .backward()
- .grad Attribute
- torch.no_grad()
- Gradient Accumulation

nn.Module

- Custom Module Definition
- forward() Method
- Parameter Registration
- Module Composition
- nn.Sequential

Data Pipeline

- Dataset Class
- DataLoader
- Transforms
- Samplers
- num_workers
- pin_memory
- Large-Scale Data Loading — WebDataset / TFRecord Format, Streaming Datasets, Sharded Data Storage, Prefetching & Pipeline Parallelism

Training Loop

- Forward Pass
- Loss Computation
- optimizer.zero_grad()
- loss.backward()
- optimizer.step()
- Evaluation Mode (model.eval())
- Model Saving/Loading

Model Compilation & Export

- torch.compile (Dynamo + Inductor)
- torch.jit.script vs torch.jit.trace
- Graph-Mode Optimizations
- ONNX Export & Runtime
- TorchScript Limitations

Training Reproducibility

- Random Seed Management
- Deterministic Algorithms
- CUDA Non-Determinism
- Reproducible Data Loading

TensorFlow & Keras

Keras Sequential API

Keras Functional API

tf.data Pipeline

Custom Training with GradientTape

Callbacks

SavedModel Format

TensorBoard

JAX

Functional Transformations (jit, grad, vmap)

XLA Compilation and Acceleration

Flax and Haiku (Neural Network Libraries)

Optax (Optimizers)

JAX vs PyTorch vs TensorFlow

Distributed Training

Data Parallelism

- Replicated Model
- Gradient All-Reduce
- Synchronous vs Asynchronous
- Communication Primitives — All-Reduce Operation, Ring-AllReduce Algorithm, All-Gather, Reduce-Scatter, Parameter Server Architecture
- Communication Optimization — Gradient Compression, Communication-Computation Overlap, Bucket Fusion, Network Bandwidth & Latency Effects
- Scaling Challenges — Linear Scaling Rule, Square Root Scaling, Straggler Problem & Mitigation, Batch Size Scaling Limits
- Fault Tolerance & Elasticity — Checkpoint-Based Recovery, PyTorch Elastic (TorchElastic), Dynamic Worker Count, Graceful Degradation

Model Parallelism

- Pipeline Parallelism
- Tensor Parallelism
- Expert Parallelism

Frameworks

- PyTorch DDP
- PyTorch FSDP
- DeepSpeed
- Megatron-LM
- Horovod

Memory Optimization

- Gradient Checkpointing
- ZeRO Optimization
- Offloading (CPU/NVMe)
- Activation Recomputation

CUTTING-EDGE Modern Training Infrastructure

Data Pipeline & Preprocessing

- Fully Sharded Data Parallel (FSDP)
- Knowledge distillation via synthetic data
- Textbook-quality data (Phi series)
- Data quality vs data quantity trade-off

PEFT & Adapter Methods

- Parameter-Efficient Fine-Tuning (PEFT)
- Low-Rank Adaptation (LoRA)
- LoRA rank selection and ablation
- LoRA alpha scaling factor
- LoRA weight merging (adapter fusion)

- Multi-LoRA serving (S-LoRA, LoRAX)
- QLoRA (Quantized LoRA)
- DoRA (Weight-Decomposed LoRA)
- Prefix tuning (learnable key/value prefixes)
- Prompt tuning (soft token embeddings)
- AdaLoRA (adaptive rank allocation)

Distributed Training Systems

- DeepSpeed ZeRO (Stage 1, 2, 3, Infinity) — ZeRO-Infinity, ZeRO Communication Analysis, FSDP vs DeepSpeed Comparison

Training Optimization

- Full Fine-Tuning (gradient computation, optimizer states)
- Paged optimisers (CPU offloading)
- Optimizer state partitioning
- Gradient partitioning
- Gradient checkpointing (activation recomputation)
- Gradient accumulation (effective batch size)

Low-rank matrix factorisation (A and B matrices)

NormalFloat4 (NF4) quantisation

Double quantisation

Weight decomposition (magnitude vs direction)

$(IA)^3$ (learned activation scaling)

Parameter partitioning / sharding

CPU and NVMe offloading

Mixed precision training (FP32, FP16, BF16, FP8)

Automatic Mixed Precision (AMP) / loss scaling

Open weights vs open source (OSI definition)

Superficial Alignment Hypothesis (LIMA)

Mixture-of-Experts (MoE) sparse activation

Training Stability & Loss Landscape

Loss Curve Analysis

- Healthy Training Curves
- Overfitting Detection
- Underfitting Detection
- Divergence Patterns

Loss Landscape Geometry

- Flat vs Sharp Minima
- Loss Landscape Visualization (Li et al. 2018)
- Filter-Wise Normalization
- Mode Connectivity
- Linear Mode Connectivity

Training Dynamics

- Edge of Stability Phenomenon
- Hessian Spectrum & Condition Number
- SGD Noise & Implicit Regularization
- Batch Size Effect on Exploration
- Gradient Norm Monitoring

Training Failure Diagnosis

- NaN/Inf Loss Debugging
- Loss Spikes Root Causes
- Oscillating Loss Patterns
- Gradient Explosion Detection

Training Debugging Methodology

Systematic Debugging Checklist

- Data Pipeline Verification
- Overfit-to-One-Batch Test
- Gradient Flow Checks
- Learning Rate Sanity
- Loss Function Correctness

Common Training Failures

- Training Loss Not Decreasing
- Stuck at Random Baseline
- Validation Diverges from Training
- Loss Metric Disagreement
- Numerical Instability Sources

Advanced Debugging

- GAN Mode Collapse Diagnosis
- Transformer Loss Spikes
- Large Model Pre-Training Instabilities
- Distributed Training Accuracy Drop

Monitoring & Visualization

- Gradient Norm Plots
- Activation Distribution Monitoring
- Weight Distribution Tracking
- Training Dashboard Design (Loss, LR, Grad Norm, GPU Util)

Experiment Tracking and Logging

- Weights & Biases (W&B)
- MLflow Tracking
- TensorBoard
- Experiment Comparison and Reproducibility
- Hyperparameter Logging Best Practices

Hardware for Deep Learning

GPU Fundamentals

- GPU Architecture (SMs, CUDA Cores)
- CUDA Programming Model
- GPU Memory (VRAM) vs System RAM
- Tensor Cores (Mixed-Precision Matrix Multiply)
- GPU Memory Hierarchy (Registers, Shared Memory, L1/L2, HBM)

TPU Architecture

- Systolic Array Design
- TPU vs GPU Comparison
- TPU Pod Training
- XLA Compilation

Training Memory Analysis

- Memory Breakdown (Parameters, Gradients, Optimizer States, Activations)
- Memory Calculation for Large Models
- Batch Size vs Memory Relationship
- FP32 vs FP16 vs BF16 Memory Footprint

GPU Optimization

- CUDA Streams (Overlapping Transfer & Compute)
- Pinned Memory
- GPU Memory Fragmentation
- Operator Fusion / Kernel Fusion
- Roofline Model (Compute-Bound vs Memory-Bound)

Multi-GPU Communication

- NCCL
- NVLink vs PCIe
- InfiniBand / RoCE
- Communication Bandwidth Bottlenecks

Profiling & Performance Analysis

- NVIDIA Nsight Systems/Compute
- PyTorch Profiler
- GPU Utilization Monitoring
- Bottleneck Identification (Data Loading, Compute, Communication)

Hardware Comparison

- NVIDIA A100 vs H100
- AMD MI300X
- Google TPU v4/v5
- Apple Silicon for ML

Advanced Training Techniques

Multi-Task Learning

- Hard Parameter Sharing
- Soft Parameter Sharing
- Loss Balancing Strategies
- Task Weighting (Uncertainty-Based, GradNorm)

Self-Supervised Pre-Training

- Contrastive Methods (SimCLR, MoCo, BYOL)
- Masked Image Modeling (MAE)
- Masked Language Modeling
- Pre-Text Tasks

Exponential Moving Average (EMA)

- Polyak Averaging
- EMA Decay Rate
- EMA for Stable Evaluation
- Teacher-Student EMA (DINO)

Adversarial Training

- FGSM Perturbations
- PGD Training
- Adversarial Robustness
- Clean vs Robust Accuracy Tradeoff

Noisy Label Handling

- Label Noise Types (Symmetric, Asymmetric)
- Co-Teaching
- MixUp for Noise Robustness
- Confident Learning

Progressive Training

- Progressive Resizing
- Gradual Complexity Increase
- Efficiency Gains

Hyperparameter Optimization

Grid Search & Random Search

Bayesian Optimization (GP-Based)

Population-Based Training (PBT)

Optuna / Ray Tune Frameworks

Budget-Constrained HPO Strategies

CHAPTER 9

CNNs & Computer Vision

Prerequisites: Training Deep Networks

Outcomes: CNN architectures · Image classification · Object detection

353 concepts · Max depth: 4

Convolution Operation

1D, 2D, 3D Convolutions

Filters/Kernels

- Kernel Size
- Number of Filters
- Learnable Weights
- Edge Detection Filters
- Gabor Filters

Padding

- Valid Padding (No Padding)
- Same Padding
- Causal Padding

Stride

- Output Size Calculation
- Downsampling Effect

Dilation (Atrous Convolution)

Depthwise Separable Convolution

Transposed Convolution (Deconvolution)

Group Convolution

Convolution vs Cross-Correlation

- Mathematical Definition
- Kernel Flipping
- Practical Equivalence in CNNs

Upsampling Methods

- Bilinear Interpolation
- Nearest-Neighbor Upsampling
- Pixel Shuffle / Sub-Pixel Convolution
- Comparison with Transposed Convolution

Deformable Convolution

- Learnable Offsets
- Deformable Convolution v2

- Use Cases: Geometric Transformations

Computational Analysis

- FLOPs Calculation Formula
- Parameter Count Formula
- Memory Footprint per Layer
- 5×5 vs Stacked 3×3 Equivalence

Pooling Layers

Max Pooling

Average Pooling

Global Average Pooling

Adaptive Pooling

Spatial Pyramid Pooling

Feature Maps & Receptive Field

Feature Hierarchy

- Low-Level Features (Edges, Textures)
- Mid-Level Features (Parts, Patterns)
- High-Level Features (Objects, Scenes)

Receptive Field

- Local Receptive Field
- Effective Receptive Field
- Growth with Depth

Channel Dimension

Feature Visualization

Channel Attention Mechanisms

- Squeeze-and-Excitation (SE) Block
- Channel Recalibration
- SE Integration in Architectures
- CBAM (Convolutional Block Attention Module)

Classic Architectures

LeNet-5

- Digit Recognition Pioneer
- Conv-Pool-Conv-Pool-FC

AlexNet

- ImageNet Breakthrough (2012)
- ReLU Activation
- Dropout Usage
- GPU Training

VGGNet

- 3x3 Filter Philosophy

- Depth Increase (16/19 layers)
- Uniform Architecture

GoogLeNet/Inception

- Inception Module
- 1x1 Convolutions (Bottleneck)
- Auxiliary Classifiers
- Inception v2/v3/v4
- Convolution Factorization — $n \times n \rightarrow 1 \times n + n \times 1$ Decomposition, Parameter Reduction, Inception v3 Factorized Modules

ResNet

- Residual Connections (Skip)
- Identity Mapping
- Degradation Problem Solution
- Bottleneck Blocks
- ResNet-50/101/152
- Pre-Activation ResNet

DenseNet

- Dense Connections
- Feature Reuse
- Growth Rate
- Transition Layers

EfficientNet

- Compound Scaling
- Width/Depth/Resolution Scaling
- Neural Architecture Search
- MBConv Blocks

Vision Transformers (ViT)

- Patch Embedding
- Position Embeddings
- CLS Token
- Self-Attention on Patches
- DeiT (Data-Efficient)
- Swin Transformer

MobileNet

- MobileNetV1 (Depthwise Separable)
- MobileNetV2 (Inverted Residuals, Linear Bottleneck)
- MobileNetV3 (NAS + SE Blocks)
- Width Multiplier & Resolution Multiplier

ShuffleNet

- Channel Shuffle Operation
- Pointwise Group Convolution
- ShuffleNet V2 Design Guidelines

ConvNeXt

- Modernized ResNet Design
- Transformer-Inspired Macro Design
- Larger Kernels (7×7)
- Layer Normalization in CNNs
- GELU in CNNs
- Patchify Stem

MLP-Mixer

- Token-Mixing MLP
- Channel-Mixing MLP
- Patch-Based Input
- Comparison with ViT and CNNs

Transfer Learning

Pre-trained Models (ImageNet)

Feature Extraction

- Frozen Backbone
- Custom Classification Head

Fine-Tuning

- Unfreezing Layers
- Discriminative Learning Rates
- Gradual Unfreezing

Domain Adaptation

Transfer Learning Failure Modes

- Negative Transfer
- Catastrophic Forgetting
- Domain Gap Assessment
- When NOT to Transfer

Fine-Tuning Best Practices

- Freezing BatchNorm Layers
- Layer-Wise LR Decay
- Few-Shot Fine-Tuning Strategies
- Class Imbalance Handling

Knowledge Distillation

- Teacher-Student Framework
- Soft Labels (Temperature Scaling)
- Feature-Based Distillation
- Relation-Based Distillation

Self-Supervised Pre-Training for Vision

- Contrastive Learning (SimCLR, MoCo)
- Masked Image Modeling (MAE, BEiT)
- BYOL / DINO
- Comparison with Supervised Pre-Training

Image Data Augmentation

Geometric Transforms (Flip, Rotate, Crop, Scale)

Color and Intensity Transforms

Cutout, CutMix, and Mixup

RandAugment and AutoAugment

Test-Time Augmentation (TTA)

Augmentation Libraries (Albumentations, torchvision)

Object Detection

Two-Stage Detectors

- R-CNN
- Fast R-CNN
- Faster R-CNN
- Region Proposal Network (RPN)
- RoI Pooling
- Feature Pyramid Network (FPN)

One-Stage Detectors

- YOLO (v1-v8)
- SSD
- RetinaNet
- Focal Loss
- DETR (Detection Transformer)
- Anchor-Free Detectors — CenterNet (Keypoint-Based), FCOS (Fully Convolutional One-Stage), CornerNet, Benefits over Anchor-Based
- YOLO Version Evolution — YOLOv5 (Ultralytics), YOLOv7 (E-ELAN), YOLOv8 (Anchor-Free, Decoupled Head), Key Improvements per Version

Core Concepts

- Anchor Boxes
- IoU (Intersection over Union)
- Non-Maximum Suppression (NMS)
- mAP (Mean Average Precision)
- Bounding Box Regression
- Bounding Box Formats — (x, y, w, h) Center Format, (x1, y1, x2, y2) Corner Format, Format Conversion
- ROI Align — Bilinear Interpolation, Misalignment Fix from ROI Pooling, Mask R-CNN Dependency
- Anchor Box Generation — K-Means Clustering on GT Boxes, Aspect Ratio Priors, Scale Selection
- Soft-NMS — Gaussian Penalty Function, Linear Decay, Handling Crowded Scenes
- Hungarian Matching (DETR) — Bipartite Matching Algorithm, Cost Matrix Construction, Set Prediction Formulation

Semantic Segmentation

FCN (Fully Convolutional Network)

U-Net

- Encoder-Decoder Structure
- Skip Connections
- Medical Image Applications

DeepLab

- Atrous/Dilated Convolution
- ASPP Module
- CRF Post-Processing

Instance Segmentation

- Mask R-CNN
- Per-Instance Masks

Panoptic Segmentation

Segment Anything Model (SAM)

Segmentation Metrics

- Pixel Accuracy
- Mean IoU (mIoU)
- Dice Coefficient / F1 Score
- Boundary F1 Score

Segmentation Loss Functions

- Pixel-wise Cross-Entropy
- Dice Loss
- Focal Loss for Segmentation
- Combined Losses
- Class Imbalance Handling (Weighted Loss, Oversampling)

DeepLab Evolution

- DeepLabV1 (CRF)
- DeepLabV2 (ASPP)
- DeepLabV3 (Improved ASPP, No CRF)
- DeepLabV3+ (Encoder-Decoder + ASPP)

MaskFormer & Mask2Former

- Per-Segment Mask Classification
- Transformer Decoder for Masks
- Unified Semantic/Instance/Panoptic

Lightweight Segmentation Models

- BiSeNet (Bilateral)
- ENet (Efficient)
- ICNet
- Real-Time Segmentation Tradeoffs

Segmentation Training Practices

- Patch-Based vs Full-Image Training
- Synchronized Augmentation (Image + Mask)

- Sliding Window Inference
- Multi-Scale Inference

CUTTING-EDGE Vision-Language Models

VLM Architectures

- CLIP (Contrastive Language-Image Pre-training)
- ColPali

Multimodal Fusion

- Early Fusion
- Late Fusion
- Cross-Attention Fusion

VLM Tasks

- OCR-Free Document Understanding

VLM Training

- Visual Instruction Tuning

DINOv2

- Self-Supervised Vision Foundation Model
- Teacher-Student with EMA
- Multi-Crop Strategy
- Self-Distillation Loss
- Downstream Task Performance without Fine-Tuning
- Comparison with CLIP Features

Hybrid CNN-Transformer Architectures

- CoAtNet
- CvT (Convolutional Vision Transformer)
- LeViT
- When Hybrids Outperform Pure Architectures

Additional Topics

- InfoNCE Loss
- Vision Transformer (ViT)
- Patch Embeddings
- Visual Tokens
- Perceiver Resampler
- Q-Former (Query Transformer)
- Native Multimodal
- Residual Vector Quantization (RVQ)
- Lost in the Middle
- Needle-in-a-Haystack
- Late Interaction Retrieval
- Multimodal Tokenization

Generative Vision Models

Generative Adversarial Networks (GANs)

- Generator & Discriminator
- Adversarial Loss
- DCGAN
- StyleGAN
- Training Instability & Mode Collapse
- Wasserstein GAN

Diffusion Models

- Forward Diffusion Process
- Reverse Denoising
- DDPM
- Score Matching
- Latent Diffusion (Stable Diffusion)
- Classifier-Free Guidance

Neural Radiance Fields (NeRF)

- Volumetric Rendering
- Positional Encoding
- Novel View Synthesis
- Instant-NGP

Image Super-Resolution

- SRCNN
- ESRGAN
- Perceptual Loss
- Upscaling Architectures

CNN Interpretability & Visualization

Grad-CAM / Grad-CAM++

Saliency Maps

Feature Map Visualization

Style Transfer (Gram Matrix)

Neural Style Transfer (Gatys et al.)

Activation Maximization

Video & Temporal Vision

Video Understanding

- 3D CNNs (C3D, I3D)
- Video Transformers (TimeSFormer, ViViT)
- Two-Stream Networks
- Temporal Modeling

Optical Flow

- FlowNet
- RAFT
- Dense Motion Estimation

Multi-Object Tracking

- Detection-Based Tracking
- SORT / DeepSORT
- Re-Identification
- TrackFormer

Pose Estimation

Top-Down vs Bottom-Up

Heatmap-Based Methods

OpenPose

HRNet

Keypoint Detection

Visual Search & Retrieval

Embedding-Based Retrieval

Metric Learning (Triplet Loss, ArcFace)

Approximate Nearest Neighbor Search

Re-Ranking

CHAPTER 10

Sequence Models & Attention

Prerequisites: Training Deep Networks

Outcomes: RNN/LSTM/GRU · Attention mechanism · Transformer architecture

290 concepts · Max depth: 4

Recurrent Neural Networks

Vanilla RNN

- Hidden State
- Weight Sharing Across Time
- Unrolling Through Time
- Backpropagation Through Time (BPTT)

Bidirectional RNN

- Forward & Backward Pass
- Context from Both Directions
- Concatenated Hidden States

Deep RNNs

- Stacked RNN Layers
- Residual Connections in RNNs

RNN Limitations

- Short-Term Memory
- Vanishing Gradients in Sequences
- Sequential Processing (No Parallelism)

RNN Training Techniques

- Truncated BPTT
- Gradient Clipping (by Value vs by Norm)
- Variational Dropout (Gal & Ghahramani)
- Zoneout Regularization
- Padding and Masking for Variable-Length Sequences

RNN Variants

- Echo State Networks / Reservoir Computing
- Jordan Network
- Clockwork RNN
- Highway LSTM
- IndRNN
- Minimal Gated Unit

Sequence-to-Sequence Extensions

- Copy Mechanism / Pointer Network

- Pointer-Generator Network
- Non-Autoregressive Decoding
- Minimum Risk Training

LSTM

Cell State (Long-Term Memory)

Gates

- Forget Gate
- Input Gate
- Output Gate

Gate Mechanics

- Sigmoid Activation for Gates
- Tanh for Candidate Values
- Element-Wise Operations

LSTM Variants

- Peephole Connections
- Coupled Forget-Input Gate

GRU

Update Gate

Reset Gate

Simplified vs LSTM

- Fewer Parameters
- No Separate Cell State
- Comparable Performance

Sequence-to-Sequence

Encoder-Decoder Architecture

- Encoder: Input Compression
- Context Vector
- Decoder: Output Generation

Teacher Forcing

- Training with Ground Truth
- Exposure Bias
- Scheduled Sampling

Beam Search Decoding

- Beam Width
- Length Normalization
- Top-K Sampling
- Nucleus Sampling (Top-P)

Applications

- Machine Translation
- Text Summarization

- Dialogue Systems

Attention Mechanism

Motivation

- Information Bottleneck Problem
- Long-Range Dependencies
- Alignment Learning

Bahdanau Attention (Additive)

- Alignment Score (MLP)
- Context Vector Computation
- Soft Attention Weights

Luong Attention (Multiplicative)

- Dot-Product Score
- General Score
- Concat Score
- Global vs Local Attention

Attention Visualization

Hard vs Soft Attention

- Differentiable (Soft) Attention
- Stochastic (Hard) Attention
- REINFORCE for Hard Attention
- Straight-Through Estimator

Monotonic Attention

- Online / Streaming Applications
- Monotonic Alignment

Attention Theory

- Attention and Kernel Methods
- Attention and Hopfield Networks
- Attention as Explanation (Limitations)
- Attention Rollout

Self-Attention

Query, Key, Value

- Linear Projections
- QKV from Same Input
- Attention Score Computation

Scaled Dot-Product Attention

- QK^T / √d_k
- Softmax Normalization
- Value Weighting
- Scaling Factor Importance

Multi-Head Attention

- Parallel Attention Heads
- Different Representation Subspaces
- Head Concatenation
- Output Projection
- Number of Heads Selection

Transformer Architecture

Encoder Block

- Multi-Head Self-Attention
- Add & Norm (Residual + LayerNorm)
- Position-Wise Feed-Forward Network
- Stack of N Encoders

Decoder Block

- Masked Multi-Head Self-Attention
- Cross-Attention (Encoder-Decoder)
- Feed-Forward Network
- Causal Masking
- Auto-Regressive Generation

Positional Encoding

- Sinusoidal Encoding
- Learned Positional Embeddings
- Relative Position Encoding (RoPE)
- ALiBi

Key Properties

- Parallelizable Training
- $O(n^2)$ Attention Complexity
- No Recurrence
- Long-Range Dependencies

Efficient Attention Variants

- Linear Attention
- Sparse Attention
- Flash Attention
- Multi-Query Attention
- Grouped-Query Attention (GQA)
- Sliding Window Attention
- Longformer — Local Windowed Attention, Global Attention Tokens, Linear Complexity
- BigBird — Random Attention, Window Attention, Global Attention, Theoretical Properties
- Linformer — Low-Rank Key-Value Projection, Linear Complexity Approximation
- Performer — FAVOR+ Algorithm, Random Feature Maps, Kernel Approximation

BERT Internals

Encoder-Only Architecture

Masked Language Modeling (MLM)

Next Sentence Prediction (NSP)

WordPiece Tokenization

[CLS] and [SEP] Tokens

Segment Embeddings

Bidirectional Context

Fine-Tuning for Downstream Tasks

GPT Internals

Decoder-Only Architecture

Causal Language Modeling (CLM)

Autoregressive Generation

BPE Tokenization

Left-to-Right Context Only

In-Context Learning

Emergent Abilities with Scale

CUTTING-EDGE State Space Models & Mamba

SSM Foundations

- State Space Models (SSM)
- Discretization Methods
- Linear Recurrence
- HiPPO Framework
- HiPPO-LegS Matrix
- S4D (Diagonal State Spaces)
- Selective State Spaces
- Input-Dependent Discretization
- State Space Duality (SSD)
- Trapezoidal Discretization
- Gated Linear Recurrence

Mamba Architecture

- Parallel Associative Scan
- Blelloch Scan Algorithm
- Cross-Scan (SS2D)
- Selective Scan Algorithm

SSM Variants & Extensions

- S5 / DSS

Hybrid SSM-Attention

- SSM-Attention Equivalence
- Hybrid SSM-Transformer

Softplus Activation — smooth positive function for Δ

- softplus(x) = log(1 + exp(x))

Additional Topics

- Linear Time-Invariant (LTI) Systems
- Matrix Exponential
- Convolution Kernel View
- Semiseparable Matrices
- Legendre Polynomial Basis
- DPLR Parameterization
- Cauchy Kernel
- Woodbury Matrix Identity
- Kernel Fusion
- SRAM vs HBM Optimization
- SiLU / Swish Gating
- Depthwise Convolution in SSMs
- Block Decomposition
- Multi-Head SSM
- Complex-Valued States
- MIMO SSM
- Mixture of Experts (MoE) + SSM

Transformer Interpretability & Analysis

Probing

- Probing Classifiers
- Layer-wise Analysis
- Linguistic Knowledge in Layers
- BERTology

Mechanistic Interpretability

- Transformer Circuits (Elhage et al.)
- QK Circuits
- OV Circuits
- Induction Heads
- Circuit Discovery

Representation Analysis

- Superposition Hypothesis
- Scaling Monosemanticity (Anthropic)
- Logit Lens
- Tuned Lens
- Residual Stream Hypothesis

Knowledge Storage & Editing

- Knowledge in FFN Layers
- ROME (Rank-One Model Editing)
- MEMIT
- Factual Associations

Attention Analysis

- Attention as Explanation (Limitations)
- Attention Rollout
- Attention Head Pruning
- Attention Visualization

Theoretical Properties

- Turing Completeness of Transformers
- Circuit Complexity (TC0)
- Depth vs Computational Power
- Lottery Ticket Hypothesis
- Grokking Phenomenon
- Reversal Curse
- Softmax Bottleneck

Sub-Quadratic Architectures

Transformer-XL

- Segment-Level Recurrence
- Relative Positional Encoding
- Context Fragmentation Solution

RetNet (Retentive Network)

- Retention Mechanism
- Recurrent / Parallel / Chunkwise Modes

RWKV

- Linear Attention Variant
- Training as Transformer, Inference as RNN
- WKV Operator

Griffin

- Gated Linear Recurrences
- Local Attention Hybrid

xLSTM

- Exponential Gating
- Matrix Memory (mLSTM)
- sLSTM Variant

Test-Time Training (TTT) Layers

- Self-Supervised at Inference
- TTT-Linear
- TTT-MLP

Hyena

- Long Convolutions
- Implicit Parametrization
- Gating Mechanism

Temporal Convolutional Networks (TCNs)

- Dilated Causal Convolutions
- Large Receptive Fields
- Parallelizable Training

Online Softmax (Memory-Efficient Attention)

CHAPTERS 11–12

Language & LLMs

Natural language processing and the large language model revolution.

CHAPTER 11

Natural Language Processing

Prerequisites: Sequence Models & Attention

Outcomes: Text preprocessing · Embeddings · NLP tasks

311 concepts · Max depth: 5

Text Preprocessing

Tokenization

- Word Tokenization
- Sentence Tokenization
- Whitespace Tokenization
- Regex-Based Tokenization
- Language-Specific Challenges

Text Normalization

- Lowercasing
- Punctuation Removal
- Number Handling
- Unicode Normalization
- Contraction Expansion

Stemming

- Porter Stemmer
- Snowball Stemmer
- Over-Stemming/Under-Stemming

Lemmatization

- Dictionary-Based
- Morphological Analysis
- POS-Aware Lemmatization
- Stemming vs Lemmatization

Stop Words

- Standard Stop Word Lists
- Domain-Specific Stop Words
- When to Remove vs Keep

Classical Text Representations

Bag-of-Words (BoW)

- Vocabulary Construction
- Count Vectorization
- Binary BoW
- N-gram Extension

- Sparsity Issue

TF-IDF

- Term Frequency (TF)
- Inverse Document Frequency (IDF)
- TF-IDF Weighting
- Document Similarity
- Limitations

Word Embeddings

Word2Vec

- CBOW (Continuous Bag-of-Words) — Context Predicts Target, Window Size
- Skip-gram — Target Predicts Context, Better for Rare Words
- Training Tricks — Negative Sampling, Hierarchical Softmax, Subsampling Frequent Words
- Word Analogies
- Vector Arithmetic

GloVe

- Global Co-occurrence Matrix
- Matrix Factorization
- Log-Bilinear Model
- Combining Local and Global

FastText

- Subword Information
- Character N-grams
- Out-of-Vocabulary Handling
- Morphologically Rich Languages

Embedding Properties

- Dimensionality Selection
- Cosine Similarity
- Bias in Embeddings
- Contextual vs Static

Subword Tokenization

Byte-Pair Encoding (BPE)

- Iterative Merge Operations
- Vocabulary Size Control
- Merge Rules Learning
- Used in GPT Models

WordPiece

- Likelihood-Based Merging
- ## Prefix for Continuations
- Used in BERT

SentencePiece

- Language-Agnostic

- Treats Input as Raw Stream
- BPE/Unigram Modes
- No Pre-Tokenization Needed

Unigram Language Model

- Probabilistic Tokenization
- Iterative Pruning
- Multiple Segmentations

Byte-Level BPE

- UTF-8 Byte Sequences
- No Unknown Tokens
- Used in GPT-2/3

NLP Tasks

Text Classification

- Sentiment Analysis — Binary Sentiment, Multi-Class Sentiment, Aspect-Based Sentiment
- Spam Detection
- Intent Classification
- Topic Classification
- Classical Neural NLP Models — TextCNN (Kim 2014), BiLSTM-CRF, ELMo
- Extractive Summarization Methods — TextRank, LexRank, BertSumExt

Named Entity Recognition (NER)

- Entity Types (PER, ORG, LOC)
- BIO/IOB Tagging Scheme
- Sequence Labeling Approach
- Nested NER

POS Tagging

- Part-of-Speech Tags
- Universal POS Tagset
- Sequence Labeling
- HMM-Based
- CRF-Based

Topic Modeling

- LDA (Latent Dirichlet Allocation) — Document-Topic Distribution, Topic-Word Distribution, Gibbs Sampling, Number of Topics Selection
- NMF (Non-Negative Matrix Factorization)
- BERTopic

Information Extraction

- Relation Extraction — Supervised RE, Distant Supervision, Neural RE Models
- Event Extraction — Event Triggers, Event Arguments, Document-Level EE
- Named Entity Linking (NEL) — Entity Disambiguation, Knowledge Base Linking, NEL vs NER
- Temporal Information Extraction — Date/Duration Extraction, Temporal Relations
- Coreference Resolution — Mention Detection, Clustering-Based Methods, End-to-End Neural Coreference

Natural Language Inference (NLI)

- Entailment, Contradiction, Neutral
- NLI Datasets (SNLI, MultiNLI)
- NLI for Zero-Shot Classification
- Hypothesis-Based Approach

Semantic Parsing

- Text-to-SQL — Schema Understanding, Complex Query Handling, Spider Benchmark, Domain Adaptation
- Logical Form Generation
- AMR Parsing

Dialogue Systems

- Task-Oriented Dialogue — NLU Component, Dialogue State Tracking (DST), Dialogue Policy, NLG Component
- Open-Domain Dialogue — Retrieval-Based, Generative-Based, Hybrid Approaches
- Conversational QA

Syntactic Analysis

- Dependency Parsing
- Constituency Parsing
- Universal Dependencies
- Semantic Role Labeling
- Coreference Resolution

NLP Benchmarks

- GLUE and SuperGLUE
- SQuAD
- CoNLL Tasks
- WMT (Machine Translation)
- BIG-bench

Text Generation

Language Modeling

- Next Token Prediction
- Perplexity Metric
- Autoregressive Generation

Decoding Strategies

- Greedy Decoding
- Beam Search
- Temperature Sampling
- Top-K Sampling
- Top-P (Nucleus) Sampling
- Repetition Penalty

Summarization

- Extractive Summarization
- Abstractive Summarization
- ROUGE Metrics

Question Answering

- Extractive QA
- Generative QA
- Open-Domain QA
- Reading Comprehension
- SQuAD Benchmark

CUTTING-EDGE RAG for NLP Applications

Retrieval Methods

- Retrieval-Augmented Generation (RAG)
- Dense Passage Retrieval (DPR)
- Sparse Retrieval (BM25/TF-IDF)
- Hybrid Search
- Embedding Models
- HyDE (Hypothetical Document Embeddings)
- Bi-Encoder Retrieval
- Multi-Query Retrieval
- Parent Document Retrieval
- Retrieval Evaluation Tokens
- Contextual Retrieval

Chunking & Indexing

- Vector Databases
- Chunking Strategies
- Semantic Chunking

Re-ranking & Filtering

- Cross-Encoder Re-rankers
- Answer Relevance

Advanced RAG Patterns

- GraphRAG
- Knowledge Graph Construction
- Multi-hop Reasoning
- Corrective RAG (CRAG)
- Self-RAG

RAG Evaluation

- RAGAS Evaluation Framework
- Faithfulness Metric
- Hallucination Detection

Additional Topics

- Query Transformation
- Query Decomposition
- ColBERT Late Interaction
- Reciprocal Rank Fusion (RRF)
- Contextual Compression
- Community Detection (Leiden Algorithm)

- Agentic RAG
- Query Routing
- Multi-Agent RAG
- Multimodal RAG
- Table Understanding
- Context Precision/Recall
- RAG vs Fine-Tuning
- Fusion-in-Decoder (FiD)

NLP Practical Methods

Explainability for NLP

- LIME for Text
- SHAP for Text
- Attention-Based Explanations

Data Augmentation for NLP

- Back-Translation
- Synonym Replacement
- Easy Data Augmentation (EDA)
- Contextual Word Augmentation
- Paraphrase Generation

Active Learning for NLP

- Uncertainty Sampling
- Query-by-Committee
- Annotation Efficiency

Multilingual NLP

- mBERT
- XLM-RoBERTa
- mT5 / mBART
- NLLB
- Cross-Lingual Transfer Learning
- Zero-Shot Cross-Lingual

NLP Security

- Prompt Injection Attacks
- Adversarial Attacks on NLP
- Privacy Concerns
- Federated Learning for NLP

Machine Translation

Historical Approaches

- Rule-Based MT
- Statistical MT — Noisy Channel Model, Phrase-Based SMT, Alignment Models

Neural Machine Translation

- Encoder-Decoder for MT
- Attention-Based NMT

- Multilingual NMT — mBART, NLLB, Zero-Shot Translation, Off-Target Translation
- Document-Level MT
- Low-Resource MT

MT Evaluation

- BLEU Score
- METEOR
- chrF
- TER
- COMET
- Quality Estimation (QE)

CHAPTER 12

Large Language Models

Prerequisites: Natural Language Processing, Sequence Models & Attention

Outcomes: Pre-training & fine-tuning · Prompt engineering · RAG & agents

789 concepts · Max depth: 5

Pre-training Objectives

Masked Language Modeling (MLM)

- Random Token Masking (15%)
- 80-10-10 Strategy
- Bidirectional Context
- Whole Word Masking

Causal Language Modeling (CLM)

- Next Token Prediction
- Left-to-Right Generation
- Autoregressive Factorization

Denoising Objectives

- Span Corruption (T5)
- Text Infilling (BART)
- Sentence Permutation
- Document Rotation

Prefix Language Modeling

Replaced Token Detection (ELECTRA)

Pretraining Data Engineering

- Data Deduplication — Exact Deduplication, Fuzzy Deduplication (MinHash, SimHash)
- Data Quality Filtering — Perplexity-Based Filtering, Language Detection, Toxicity Filtering, Heuristic Filters (URL, Repetition, Length)
- Multilingual Data Handling — Language Balancing Strategies, Cross-Lingual Transfer, Tokenizer Fertility Across Languages
- Continual Pretraining — Domain-Adaptive Pretraining, Data Replay for Forgetting Prevention
- Optimizers for LLM Training — Adam / AdamW, Adafactor, Training Instability Mitigation

BERT Family

BERT-Base/Large

- 110M / 340M Parameters
- 12/24 Transformer Layers
- MLM + NSP Training

RoBERTa

- Removed NSP
- Dynamic Masking
- Larger Batches, More Data
- Longer Training

ALBERT

- Factorized Embedding
- Cross-Layer Parameter Sharing
- Sentence Order Prediction

DistilBERT

- Knowledge Distillation
- 40% Smaller, 60% Faster
- Teacher-Student Training

DeBERTa

- Disentangled Attention
- Enhanced Mask Decoder
- Relative Position Encoding

XLNet

- Permutation Language Modeling
- Two-Stream Self-Attention
- Transformer-XL Integration

GPT Family

GPT-2

- 1.5B Parameters
- WebText Training Data
- Zero-Shot Task Transfer
- BPE Tokenizer

GPT-3

- 175B Parameters
- In-Context Learning
- Few-Shot Prompting
- Emergent Abilities
- API-Based Access

GPT-4

- Multimodal (Text + Vision)
- Improved Reasoning
- Longer Context Window
- RLHF Alignment
- System Prompts

InstructGPT

- Instruction Following
- RLHF Pipeline

- Human Preference Data

Other Foundation Models

T5 (Text-to-Text)

- Unified Text-to-Text Framework
- Encoder-Decoder Architecture
- Span Corruption Pre-training
- Task Prefixes

LLaMA Family

- LLaMA-1 (7B-65B)
- LLaMA-2 (7B-70B)
- LLaMA-3
- Open Weights
- RoPE Positional Encoding
- SwiGLU Activation
- RMSNorm

Mistral

- Mistral-7B
- Sliding Window Attention
- Grouped-Query Attention
- Mixtral (Mixture of Experts)

Other Models

- PaLM / Gemini
- Claude
- Falcon
- Phi (Small Language Models)
- Qwen
- Additional Architectures (add to Other Models) — RWKV (RNN-Transformer Hybrid), Switch Transformer (Simplified MoE)

Scaling Laws

Kaplan Scaling Laws

- Loss vs Parameters
- Loss vs Data
- Loss vs Compute
- Power Law Relationships

Chinchilla Scaling

- Compute-Optimal Training
- Tokens $\approx 20\times$ Parameters
- Data-Constrained Regimes

Emergent Abilities

- Phase Transitions
- Unpredictable Capabilities
- Scale Thresholds

Chinchilla Scaling Laws

Prompt Engineering

Zero-Shot Prompting

- Task Description Only
- Role Prompting
- Instruction Clarity

Few-Shot Prompting

- Example Selection
- Example Ordering
- Format Consistency
- Number of Examples

Chain-of-Thought (CoT)

- Step-by-Step Reasoning
- Zero-Shot CoT ('Let's think...')
- Few-Shot CoT
- Self-Consistency

Advanced Techniques

- ReAct (Reasoning + Acting)
- Tree of Thoughts
- Reflexion
- Automatic Prompt Optimization
- Meta-Prompting
- Additional Prompting Techniques (add to Advanced Techniques) — Graph-of-Thought Prompting, Least-to-Most Prompting, Prompt Chaining / Task Decomposition, DSPy (Programmatic LLM Framework), LATS (Language Agent Tree Search)

Structured Output

- JSON Mode
- Function Calling
- Output Parsing

Retrieval-Augmented Generation (RAG)

RAG Architecture

- Retriever Component
- Generator Component
- Retrieval-Then-Generate Pipeline
- Context Window Stuffing

Document Processing

- Chunking Strategies — Fixed-Size Chunks, Semantic Chunking, Recursive Character Splitting, Sentence-Based Splitting, Overlap Between Chunks
- Document Parsing — PDF Extraction, HTML Parsing, Table Extraction, Metadata Extraction

Vector Stores

- Embedding Models
- Vector Databases — Pinecone, Weaviate, Chroma, FAISS, Milvus, Qdrant

- Indexing Strategies — HNSW, IVF, Product Quantization
- Similarity Search (Cosine, L2)

Retrieval Enhancement

- Reranking — Cross-Encoder Rerankers, Cohere Rerank, ColBERT
- Hybrid Search (Dense + Sparse)
- Query Expansion
- HyDE (Hypothetical Document)
- Multi-Query Retrieval
- Parent Document Retrieval

Advanced RAG

- Agentic RAG
- Self-RAG
- CRAG (Corrective RAG)
- Graph RAG
- Multi-Modal RAG

RAG Evaluation

- Retrieval Metrics — Recall@k, NDCG (Normalized Discounted Cumulative Gain), MRR (Mean Reciprocal Rank), MAP (Mean Average Precision)
- Generation Metrics — Answer Faithfulness, Answer Relevance, Context Precision, Context Recall
- RAGAS Framework
- Diagnosing RAG Failures — Retrieval vs Generation Failure Analysis, Top-k Impact on Quality
- Production RAG Monitoring — A/B Testing RAG Systems, Continuous Evaluation Pipeline, Latency and Cost Monitoring

Vector Store Production (add to Vector Stores)

- Metadata Filtering Combined with Similarity Search
- Vector Database Updates and Incremental Indexing
- Scalability and High Availability
- Privacy and Security (Embedding Inversion Attacks)
- Vector Database Benchmarking

Embeddings & Semantic Search

Sentence Embeddings

- Sentence-BERT
- Contrastive Learning
- Instructor Embeddings
- E5 / BGE Models
- OpenAI Embeddings

Semantic Search Pipeline

- Query Encoding
- Document Encoding
- Approximate Nearest Neighbor
- Relevance Scoring

Training Embeddings

- Contrastive Loss
- InfoNCE Loss
- Hard Negative Mining
- In-Batch Negatives

Embedding Production Topics

- Matryoshka Representation Learning
- Bi-Encoder vs Cross-Encoder
- Embedding Dimensionality Tradeoffs
- Domain-Specific Embedding Fine-Tuning
- Retrieval Evaluation Metrics — Recall@k, NDCG, MRR, MAP

Tokenizer Training

Vocabulary Construction

BPE Training Algorithm

Vocabulary Size Selection

Special Tokens

Multilingual Tokenization

Tokenizer Evaluation (Fertility)

Evaluation

Benchmarks

- MMLU
- HellaSwag
- HumanEval
- GSM8K
- TruthfulQA
- MT-Bench
- LMSYS Chatbot Arena

Metrics

- Perplexity
- BLEU
- ROUGE
- BERTScore
- Human Evaluation
- LLM-as-Judge

Evaluation Challenges

- Contamination/Data Leakage
- Task Saturation
- Prompt Sensitivity
- Reproducibility

CUTTING-EDGE Reasoning & Inference-Time Scaling

Reasoning Foundations

- Chain-of-Thought (CoT) Prompting
- Self-Consistency
- Scratchpad Reasoning
- Reasoning Tokens / Thinking Tokens
- Beam Search over Reasoning Steps
- Reasoning Distillation

Inference-Time Scaling

- Test-Time Compute
- Inference-Time Scaling Laws
- Adaptive Compute Allocation
- Training-Inference Compute Tradeoff

Reasoning Evaluation

- Monte Carlo Tree Search (MCTS)

Additional Topics

- Tree of Thoughts (ToT)
- Process Reward Model (PRM)
- Outcome Reward Model (ORM)
- GRPO (Group Relative Policy Optimization)
- PPO (Proximal Policy Optimization)
- Rule-Based Rewards
- Self-Verification
- Backtracking
- Best-of-N Sampling
- OpenAI o1/o3
- DeepSeek-R1
- DeepSeek-R1-Zero
- R1-Distill
- Mixture-of-Experts (MoE)
- System 1 vs System 2 Thinking
- Generation-Verification Asymmetry
- The Aha Moment

CUTTING-EDGE MoE in Large Language Models

MoE Foundations

- Mixture of Experts (MoE)
- Gating Network / Router
- Top-k Routing
- Expert Collapse
- Expert Capacity
- Capacity Factor
- Expert Parallelism
- Fine-Grained Expert Segmentation
- Shared Experts
- Expert Choice Routing

- Noisy Top-k Gating
- Expert Offloading
- Topology-Aware Routing
- Expert Specialization
- Expert Merging / Aggregation
- Simplified Top-1 Routing
- Expert Capacity and Token Dropping
- Switch Transformer (Simplified Top-1 Routing)

MoE Training

- Auxiliary Load-Balancing Loss
- Router Z-Loss
- Auxiliary-Loss-Free Balancing

MoE Efficiency

- Sparse Activation
- Block-Sparse Operations

MoE Research Frontiers

- Soft MoE
- Vision MoE (V-MoE)

Conditional Computation

Token Dropping

All-to-All Communication

Multi-head Latent Attention (MLA)

Multi-Token Prediction

Dense-MoE Hybrid

MoE Distillation

Grouped Query Attention (GQA)

Trillion Parameter Scale

CUTTING-EDGE Post-Training Alignment (RLHF/DPO/GRPO)

RLHF Foundations

- RLHF (Reinforcement Learning from Human Feedback)
- Reward Model
- PPO (Proximal Policy Optimization)
- Process Reward Models
- Iterative RLHF
- RLHF — Step 1: SFT on Demonstrations, Step 2: Reward Model Training, Step 3: PPO Optimization, Human Preference Collection, KL Divergence Constraint

Direct Preference Optimization

- Bradley-Terry Model
- Implicit Reward
- Closed-Form Solution

- IPO (Identity Preference Optimization)
- ORPO (Odds Ratio Preference Optimization)
- SimPO (Simple Preference Optimization)
- Preference Data
- DPO (Direct Preference Optimization) — No Reward Model Needed, Bradley-Terry Model, Implicit Reward, Simpler Training Pipeline

GRPO & Group Methods

- GRPO (Group Relative Policy Optimization)
- RLAIF (Reinforcement Learning from AI Feedback)

Alignment Theory

- Post-Training Alignment
- Reward Hacking
- Goodhart's Law
- Alignment Tax
- Alignment Bonus
- Self-Play for Alignment
- Other Alignment Methods — RLAIF (AI Feedback), Constitutional AI, KTO (Kahneman-Tversky), ORPO, IPO

Safety & Constitutional AI

- Safety & Guardrails — Red Teaming, Content Filtering, Harmlessness Training, Refusal Behavior

KL Divergence Penalty

Reference Policy

KTO (Kahneman-Tversky Optimization)

Group-Relative Advantage

Scalable Oversight

Weak-to-Strong Generalization

AI Debate

Emergent Reasoning

SFT (Supervised Fine-Tuning)

Exposure Bias

Sycophancy

CUTTING-EDGE Agentic AI Systems

Core Agent Architecture

- Agentic AI
- Agent Loop
- Reasoning
- Planning
- Reflection
- ReAct Pattern

- Plan-and-Execute Architecture
- Reactive Agent
- Multi-Agent Systems
- Agent Debate
- Code Agent
- Computer Use / GUI Agents
- Agent Alignment
- Generative Agents
- Agent Frameworks — LangChain, LlamaIndex, AutoGPT, CrewAI, OpenAI Assistants, Agent Frameworks
- Agent Orchestration & Production — Orchestration Patterns, State Management, Observability & Monitoring, Production Safeguards

Tool Use & Function Calling

- Tool Use
- Function Calling
- Toolformer

Memory & Knowledge

- Memory
- Model Context Protocol (MCP)
- Episodic Memory
- Semantic Memory
- Working Memory (Scratchpad)
- Vector Store Memory

Agent Safety & Evaluation

- Sandboxing
- Guardrails — Runtime safety mechanisms — action gating, rate limiting, input/output filtering, kill switches
- Trajectory Evaluation

Perception

JSON Schema

MCP Primitives

Hierarchical Task Networks (HTN)

Task Decomposition

Reflexion

Supervisor-Worker Pattern

SWE-bench

Instrumental Convergence

Skill Library

Defense in Depth

MemGPT

CUTTING-EDGE Quantization & Model Compression

Quantization Methods

- Quantization
- Post-Training Quantization (PTQ)
- Quantization-Aware Training (QAT)
- Symmetric Quantization
- Asymmetric Quantization
- Group Quantization
- GPTQ
- AWQ (Activation-Aware Weight Quantization)
- GGUF Format
- Arithmetic Intensity
- Mixed-Precision Inference
- Pruning-Quantization Interaction

Knowledge Distillation

- Knowledge Distillation
- Black-Box Distillation
- White-Box Distillation
- Temperature Scaling (Distillation)
- KL Divergence Distillation Loss
- Feature-Based Distillation (FitNets)
- Relational Knowledge Distillation
- Self-Distillation
- Data-Free Knowledge Distillation
- Online / Mutual Distillation
- Ensemble Compression via Distillation
- Distillation Failure Modes

Pruning & Sparsity

- Movement Pruning for Fine-Tuned Models
- Structured Pruning of Attention Heads
- Channel Pruning in CNNs
- SparseGPT (One-Shot LLM Pruning)
- Activation Sparsity and MoE Connection
- Lottery Ticket Hypothesis (Detailed Treatment)
- Iterative Magnitude Pruning

Speculative Decoding

- Speculative Decoding
- Rejection Sampling (in speculative decoding)

Attention Optimization

- Multi-Query Attention (MQA)
- Paged Attention
- FlashAttention

Additional Topics

- SmoothQuant
- K-Quants

- Soft Labels / Dark Knowledge
- Small Language Models (SLMs)
- Data Quality Scaling
- Medusa Decoding
- Memory-Bandwidth Bound
- Continuous Batching
- Weight Outliers
- Straight-Through Estimator (STE)
- Synthetic Data Training
- CPU Inference (SIMD)
- Unified Memory Architecture
- Model Sharding
- Binary Neural Networks
- ONNX Runtime Optimizations
- Accuracy-Efficiency Tradeoff Analysis
- Soft Targets and Temperature Scaling Theory
- DistilBERT and TinyBERT Techniques

CUTTING-EDGE Parameter-Efficient Fine-Tuning

DeepSpeed ZeRO (Stage 1, 2, 3, Infinity)

Full Fine-Tuning (SFT)

- All Parameters Updated
- Instruction Tuning
- Task-Specific Adaptation
- Catastrophic Forgetting Risk

Parameter-Efficient (PEFT)

- LoRA — Low-Rank Decomposition, A and B Matrices, Rank Selection, Target Modules (Q, K, V, O), Merging Adapters
- QLoRA — 4-bit Quantized Base Model, NormalFloat (NF4) Datatype, Double Quantization, Paged Optimizers
- Adapters — Bottleneck Adapters, Parallel Adapters
- Prefix Tuning
- Prompt Tuning (Soft Prompts)
- IA3

Fine-Tuning Data

- Instruction Datasets
- Data Quality vs Quantity
- Synthetic Data Generation
- Data Formatting (Alpaca, ShareGPT)

CUTTING-EDGE Modern LLM Architecture Decisions

Positional Encoding

- Rotary Position Embedding (RoPE)
- ALiBi

Attention Variants

- Multi-Head Attention (MHA)
- Multi-Head Latent Attention (MLA)
- Gated Linear Units (GLU)
- Parallel attention + FFN
- PagedAttention

Architecture Design

- Vocabulary size trade-off
- RMSNorm
- Pre-norm vs Post-norm
- DeepNorm
- SwiGLU activation
- Tied vs untied embeddings
- Tensor parallelism
- Pipeline parallelism

Context & Memory

- Context length extension
- Position Interpolation
- YaRN (Yet Another RoPE extensioN)
- NTK-Aware Scaling
- Attention Sink Phenomenon
- Lost-in-the-Middle Problem
- Needle-in-a-Haystack Evaluation

Chinchilla scaling laws — Optimal compute allocation

- parameters and tokens scale equally

Additional Topics

- Byte-Pair Encoding (BPE)
- SentencePiece
- Byte-level BPE
- tiktoken
- Tokenizer-free models
- Depth vs width trade-off
- Cosine learning rate schedule
- WSD schedule
- Gradient clipping
- Beyond-Chinchilla training
- MuP (Maximal Update Parameterization)
- Data mixing ratios
- Curriculum learning
- Post-training quantization
- BF16 training
- Prefill vs decode phases

Transformer Training Fundamentals

AdamW Optimizer

- Decoupled Weight Decay
- Adam vs AdamW

Learning Rate Schedules

- Warmup Phase
- Inverse Square Root Decay
- Cosine Annealing
- WSD Schedule

Embedding Scaling (sqrt d_model)

Padding Masks

Training Instabilities

- Loss Spikes
- Gradient Clipping Strategies
- Numerical Stability

Distributed Training for LLMs

- Data Parallelism
- Fully Sharded Data Parallel (FSDP)
- Gradient Accumulation and Micro-Batching
- Megatron-LM and 3D Parallelism
- Communication Optimization

Mixed Precision Training

- FP16 and BF16 Training
- Loss Scaling
- FP8 Training (Emerging)
- Automatic Mixed Precision (AMP)

Activation and Gradient Checkpointing

- Memory-Compute Tradeoff
- Selective Checkpointing
- Offloading Strategies

Production LLM Systems

LLM Serving Architecture

- Batching Strategies
- KV Cache Management
- Load Balancing
- vLLM

Application Design

- Prompt Versioning
- Monitoring & Observability
- Fallback Strategies
- Cost Management
- Model Cascading

Code Generation Models

- Codex
- StarCoder
- DeepSeek-Coder
- Code-Specific Pre-training
- Fill-in-the-Middle (FIM)

Speech Models

- Whisper
- wav2vec 2.0
- Speech-to-Text Pipeline

Model Merging

- SLERP
- TIES Merging
- DARE

Inference Optimization

Decoding Strategies

- Greedy Decoding
- Beam Search — Beam Width Tradeoffs, Length Normalization
- Sampling Methods — Top-k Sampling, Nucleus (Top-p) Sampling, Temperature Parameter, Typical Sampling
- Repetition Control — Repetition Penalty, Frequency Penalty, Presence Penalty
- Advanced Decoding — Contrastive Decoding, Minimum Bayes Risk (MBR) Decoding, Constrained Decoding (JSON/Grammar)
- Stop Sequences and Stopping Criteria

Inference Serving Systems

- Serving Frameworks — vLLM (PagedAttention, Continuous Batching), TGI (Text Generation Inference), TensorRT-LLM, Triton Inference Server
- Serving Optimization — Prefix Caching, Disaggregated Serving (Prefill/Decode Separation), Model Pruning (Structured, Unstructured), Roofline Model Analysis
- Production Infrastructure — Load Balancing for LLM Serving, Autoscaling Strategies, Latency Budgets (P50, P99), Cost-per-Token Optimization

LLM Safety & Responsible AI

Hallucination

- Types of Hallucination (Intrinsic, Extrinsic)
- Causes (Training Data, Decoding, Knowledge Gaps)
- Mitigation Strategies (RAG, Grounding, Self-Consistency)
- Hallucination Evaluation Metrics
- Factuality Benchmarks (TruthfulQA, FActScore)

Toxicity & Bias

- Toxicity Detection (Perspective API, Classifiers)
- Bias Types (Gender, Racial, Cultural)
- Bias Detection Methods
- Debiasing Techniques

Prompt Injection & Jailbreaking

- Direct Prompt Injection
- Indirect Prompt Injection
- Jailbreaking Techniques (DAN, Many-Shot, Payload Splitting)
- Defenses (Input Sanitization, Canary Tokens, Output Filtering)
- System-Level Defense Architecture

Content Safety Systems

- Real-Time Content Filtering
- Safety Classifiers (Llama Guard, NeMo Guardrails)
- Multi-Modal Safety

Watermarking & Detection

- Statistical Watermarking (Green/Red List)
- AI-Generated Text Detection
- Robustness of Watermarks

Regulatory & Governance

- EU AI Act
- NIST AI Risk Management Framework
- Responsible AI Frameworks

Emerging AI Patterns (2024-2025)

Compound AI Systems

Model Routing (Query Complexity-Based)

Evaluation-Driven Development for LLM Apps

LLM OS Concept (Agents + Memory + Tools)

Agent Benchmarks

- SWE-bench
- WebArena
- GAIA

Reliability-Autonomy Tradeoff

Agent Protocol Standardization

Long-Context LLMs vs RAG Tradeoff

ML/LLM Model Lifecycle Management

Full-Duplex Voice AI

- Real-Time Speech-to-Speech Models
- Streaming Audio Generation

CHAPTER 13

Production Systems

Taking ML from notebooks to reliable, scalable production systems.

CHAPTER 13

ML Systems and Production

Prerequisites: Supervised Learning, Deep Learning Foundations

Outcomes: Deploy & serve models · MLOps · Monitoring & scaling

575 concepts · Max depth: 5

ML System Design

Requirements and Scoping

- Problem Formulation
- Business Metrics vs ML Metrics
- Feasibility Assessment
- Data Availability Analysis
- Baseline Selection
- Success Criteria Definition

System Architecture

- Online vs Offline Components
- Prediction Latency Requirements
- Throughput Planning
- Fault Tolerance Design
- Microservices vs Monolithic ML Systems

Metrics Design

- Offline Metrics (Accuracy, F1, AUC, NDCG)
- Online Metrics (CTR, Revenue, Engagement)
- Proxy Metrics
- Guardrail Metrics
- A/B Testing and Statistical Significance
- Multi-Armed Bandits
- Experimentation Methods (expanded) — Sample Size Calculation, Multiple Comparisons / Family-Wise Error Rate, Interleaving Experiments for Ranking, Holdout Groups, Network Effects and Interference in A/B Tests, Switchback Experiment Design, *(+1 more)*

System Design Case Studies

- Fraud and Trust & Safety — Real-Time Fraud Detection (Payments), Content Moderation at Scale, Fake Review Detection, Graph-Based Fraud (GNNs for Coordinated Fraud)
- Recommendation and Personalization — Movie / Video Recommendation (Netflix-style), E-Commerce Personalization Engine, News Feed Ranking (Relevance, Freshness, Diversity), Cold-Start Problem Solutions
- Search and Ranking — E-Commerce Search Ranking (Learning-to-Rank), Ad Click-Through Rate (CTR) Prediction, Real-Time Bidding (RTB)
- Predictive Systems — Churn Prediction (SaaS), Credit Scoring with Regulatory Compliance, Dynamic Pricing (Ride-Sharing, E-Commerce), Anomaly Detection for Infrastructure Monitoring

- Specialized Systems — Autonomous Vehicle Perception Pipeline, Voice Assistant Intent Classification, Spam Detection with Adversarial Robustness, GDPR-Compliant Recommendation System

Recommendation System Design

- Multi-Stage Architecture — Candidate Generation (Retrieval), Ranking Stage, Re-Ranking and Business Rules, Blending and Diversification
- Retrieval Methods — Two-Tower (Dual-Encoder) Model, Approximate Nearest Neighbor (ANN) Retrieval, Collaborative Filtering at Scale, Content-Based Retrieval, Graph-Based Retrieval (GraphSAGE, PinSage)
- Deep Ranking Models — Wide & Deep Learning, DeepFM (Deep Factorization Machines), DCN (Deep & Cross Network), DIN (Deep Interest Network), DIEN (Deep Interest Evolution Network), Multi-Task Learning for Ranking (MMOE, PLE)
- Cold-Start Strategies — New User Cold Start, New Item Cold Start, Popularity-Based Fallbacks, Content Feature Bootstrapping, Bandit-Based Exploration
- Real-Time Personalization — Session-Based Recommendations, Real-Time Feature Updates, Contextual Features (Time, Location, Device), Online Model Updates
- Multi-Objective Optimization — Engagement vs Revenue Trade-offs, Scalarization Methods, Pareto-Optimal Solutions, Diversity and Serendipity Constraints
- Evaluation for Recommender Systems — Offline Metrics (Precision@K, Recall@K, NDCG@K, MAP), Online A/B Testing for Recommendations, Interleaving Experiments, Long-Term User Satisfaction Metrics

Search & Ranking System Design

- Search Architecture — Query Understanding Pipeline, Query Intent Classification, Query Expansion and Reformulation, Spell Correction and Auto-Complete, Inverted Index + ANN Hybrid Retrieval
- Learning to Rank — Pointwise Methods, Pairwise Methods (RankNet, LambdaRank), Listwise Methods (LambdaMART, ListNet), Neural Learning to Rank, Position Bias and Debiasing
- Hybrid Retrieval — Sparse + Dense Fusion, Cross-Encoder Re-Ranking, Reciprocal Rank Fusion, ColBERT (Late Interaction)
- Search Quality — Relevance Feedback Loops, Click Models (Position-Based, Cascade), Search Result Diversification, Freshness vs Relevance Trade-offs

Ads & Monetization Systems

- CTR Prediction — Feature Engineering for Ads (User, Ad, Context Features), Deep CTR Models (Wide & Deep, DeepFM, DIN), Calibration for CTR (Platt Scaling, Isotonic Regression), Multi-Task Prediction (CTR + CVR)
- Ad Auction and Bidding — Second-Price Auction and GSP, VCG Mechanism, Real-Time Bidding (RTB) Architecture, Bid Optimization and Pacing, Budget Allocation Across Campaigns
- Ad Targeting — Behavioral Targeting, Contextual Targeting, Lookalike Audience Modeling, Privacy-Preserving Targeting (Post-Cookie Era)
- Revenue Optimization — eCPM and Revenue Metrics, Conversion Attribution Models, Ad Fraud Detection, Ad Load Optimization

Content Moderation & Safety Systems

- Moderation Pipeline Architecture — Text Moderation (Toxicity, Hate Speech, Spam), Image Moderation (NSFW, Violence, Graphic Content), Video Moderation (Keyframe Extraction + Classification), Multimodal Moderation, Cascading Classifiers for Efficiency
- Human-in-the-Loop Moderation — Confidence-Based Routing to Human Review, Appeal Workflow Design, Active Learning for Moderation Models, Annotator Quality and Agreement

- Emerging Content Safety — Deepfake Detection, AI-Generated Content Detection, Near-Duplicate and Copy Detection, Coordinated Inauthentic Behavior Detection

Fraud Detection & Anomaly Systems

- Real-Time Fraud Pipeline — Feature Engineering for Fraud (Velocity, Aggregation, Graph Features), Rule Engine + ML Hybrid Architecture, Real-Time Scoring Infrastructure, Feedback Delay and Label Latency
- Advanced Fraud Techniques — Graph-Based Fraud Detection (Fraud Rings), Unsupervised Anomaly Detection (Autoencoders, Isolation Forest), Concept Drift in Fraud Patterns, Adversarial Adaptation by Fraudsters
- Device and Identity Intelligence — Device Fingerprinting, IP Reputation and Geolocation, Behavioral Biometrics, Account Takeover Detection

Data Engineering

Data Storage

- Data Lakes vs Data Warehouses
- Object Storage (S3, GCS, Azure Blob)
- Relational Databases (PostgreSQL, MySQL)
- NoSQL Databases (MongoDB, Cassandra, DynamoDB)
- Vector Databases (Pinecone, Weaviate, Milvus)
- Time-Series Databases

Data Formats

- Columnar Formats (Parquet, ORC)
- Row-Based Formats (CSV, JSON, Avro)
- TFRecord and Protocol Buffers
- HDF5 and Zarr
- Arrow (In-Memory Format)

Data Pipelines

- ETL vs ELT
- Batch Processing (Spark, Hadoop)
- Stream Processing (Kafka, Flink, Kinesis)
- Orchestration (Airflow, Dagster, Prefect)
- Data Validation (Great Expectations, TFDV)
- Data Versioning (DVC, LakeFS)

Data Quality and Observability

- Data Profiling and Schema Validation
- Anomaly Detection in Data Pipelines
- Freshness and Completeness Checks
- Great Expectations and Data Quality Tools

Training Data Management

Data Collection and Labeling

- Labeling Strategies (Manual, Programmatic, Active Learning)
- Annotation Tools and Platforms
- Weak Supervision (Snorkel)
- Label Quality and Inter-Annotator Agreement

- Data Flywheel

Sampling Strategies

- Random Sampling
- Stratified Sampling
- Importance Sampling
- Curriculum Learning
- Active Learning Strategies

Data Augmentation

- Image Augmentation (Geometric, Color, Cutout, Mixup)
- Text Augmentation (Back-Translation, Paraphrasing)
- Audio Augmentation (SpecAugment, Time Stretch)
- Synthetic Data Generation
- AutoAugment and RandAugment

Class Imbalance

- Oversampling (SMOTE, ADASYN)
- Undersampling
- Class Weights
- Focal Loss
- Threshold Tuning

Model Development and Debugging

Experiment Management

- Hyperparameter Search (Grid, Random, Bayesian)
- Reproducibility (Seeds, Deterministic Training)
- Ablation Studies
- Experiment Tracking (MLflow, W&B, Neptune)

Debugging ML Models

- Overfitting Diagnosis
- Underfitting Diagnosis
- Data Leakage Detection
- Error Analysis
- Slice-Based Evaluation
- Training Curve Analysis

Testing ML Systems

- Unit Tests for Data Pipelines
- Integration Tests
- Model Validation Tests
- Behavioral Testing (CheckList)
- Shadow Mode Evaluation

Benchmarking

Model Benchmarks

- Standard Benchmarks (ImageNet, GLUE, SuperGLUE)
- LLM Benchmarks (MMLU, HumanEval, HellaSwag)

- Leaderboards and Reproducibility
- Benchmark Contamination

System Benchmarks

- Latency Profiling
- Throughput Measurement
- Memory Footprint Analysis
- Cost-Performance Trade-offs
- MLPerf

Deployment Paradigms

Batch Inference

- Offline Prediction Pipelines
- Scheduled Batch Jobs
- Precomputation Strategies
- Result Caching

Real-Time Inference

- Synchronous Serving
- Latency Optimization
- Request Batching
- Model Warm-Up
- Auto-Scaling

Edge Deployment

- On-Device Inference
- Model Compression for Edge
- Hardware Constraints
- Offline Capability
- Edge-Cloud Hybrid Architectures

Deployment Strategies

- Blue-Green Deployment
- Canary Deployment
- Rolling Updates
- Feature Flags for ML
- Rollback Procedures

Serverless Inference

- AWS Lambda for ML
- Google Cloud Functions for ML
- Azure Functions for ML
- Cold Start Challenges
- When Serverless is Appropriate for ML

Model Serving

Serving Infrastructure

- REST API Design for ML
- gRPC for ML Serving

- WebSocket Streaming
- Load Balancing
- Containerization (Docker) — Docker for ML
- Kubernetes for ML — Kubernetes for ML (expanded)

Model Servers

- TorchServe
- NVIDIA Triton Inference Server
- TensorFlow Serving
- BentoML
- Seldon Core
- vLLM (LLM Serving)

Serving Optimization

- Model Caching
- Dynamic Batching
- Model Ensembles in Production
- Multi-Model Serving
- GPU Sharing and Scheduling

ML Reliability Engineering

- Circuit Breaker Pattern for ML Serving
- Retry and Fallback Patterns
- Graceful Degradation
- High Availability Design for ML Inference
- Chaos Engineering for ML Infrastructure
- Partial Failure Handling in Multi-Model Pipelines
- Request Queuing and Backpressure
- Inference Timeout Handling
- SLAs and SLOs for ML Serving Systems

Model Serialization Formats

- ONNX Format
- TensorFlow SavedModel
- TorchScript
- Pickle and Joblib
- GGUF for LLMs
- Cross-Framework Compatibility

API Security and Access Control

- Authentication and Authorization
- Rate Limiting
- API Gateway Patterns
- RBAC for ML Services

MLOps

CI/CD for ML

- Continuous Training Pipelines
- Automated Model Validation

- Model Promotion Gates
- Infrastructure as Code for ML
- GitOps for ML

Experiment Tracking and Model Registry

- MLflow
- Weights & Biases
- Model Versioning
- Model Lineage
- Artifact Management
- Model Cards

Feature Stores

- Online vs Offline Feature Store
- Feature Computation and Serving
- Feature Reuse and Discovery
- Point-in-Time Correctness
- Feast, Tecton, Hopsworks

Pipeline Orchestration

- Kubeflow Pipelines
- Apache Airflow for ML
- Metaflow
- ZenML
- Vertex AI Pipelines

Monitoring and Observability

Data Drift Detection

- Statistical Tests (KS Test, Chi-Square)
- Population Stability Index (PSI)
- Feature Distribution Monitoring
- Schema Drift
- Upstream Data Changes

Concept Drift

- Gradual vs Sudden Drift
- Prediction Drift
- Label Drift
- Windowed Performance Analysis
- Drift Adaptation Strategies

Model Performance Monitoring

- Real-Time Metric Dashboards
- Alerting and Thresholds
- Segment-Level Monitoring
- Feedback Loop Integration
- Ground Truth Collection

Observability Tools

- Evidently AI

- Whylabs / whylogs
- Arize AI
- Prometheus + Grafana for ML
- Custom Logging Pipelines

Monitoring Advanced Topics

- Delayed Ground Truth Handling
- Embedding-Based Drift Detection for LLMs
- Multivariate Drift Detection
- Fairness Monitoring in Production
- Distributed Tracing (OpenTelemetry, Jaeger)
- SLAs and SLOs for ML Serving
- Observability vs Monitoring Distinction
- Monitoring Tools: Alibi Detect, Fiddler

Scaling and Hardware

GPU Computing

- GPU Architecture (CUDA Cores, Tensor Cores)
- GPU Memory Management
- NVIDIA GPU Ecosystem (A100, H100)
- Mixed Precision Training (FP16, BF16)
- GPU Profiling

TPU and Custom Hardware

- TPU Architecture
- TPU Programming Model (XLA)
- Custom AI Accelerators
- Apple Neural Engine
- Intel Habana Gaudi

Distributed Training

- Data Parallelism
- Model Parallelism (Tensor, Pipeline)
- Fully Sharded Data Parallel (FSDP)
- DeepSpeed ZeRO
- Collective Communication (AllReduce, AllGather)
- Gradient Accumulation
- Megatron-LM Parallelism

Infrastructure Scaling

- Multi-Node Training Clusters
- High-Speed Interconnects (NVLink, InfiniBand)
- Spot/Preemptible Instance Strategies
- Training Cost Optimization

Model Optimization

Quantization

- Post-Training Quantization (PTQ)
- Quantization-Aware Training (QAT)

- INT8, INT4, Binary Quantization
- GPTQ, AWQ, GGUF
- Dynamic vs Static Quantization

Pruning

- Unstructured Pruning
- Structured Pruning
- Magnitude-Based Pruning
- Lottery Ticket Hypothesis
- Iterative Pruning with Retraining

Knowledge Distillation

- Teacher-Student Framework
- Soft Label Distillation
- Feature-Based Distillation
- Self-Distillation
- DistilBERT, TinyBERT Examples

Runtime Optimization

- ONNX Runtime
- TensorRT
- OpenVINO
- Graph Optimization and Operator Fusion
- Compiler-Based Optimization (TVM, XLA)

Cloud ML Platforms

AWS SageMaker

- SageMaker Training Jobs
- SageMaker Endpoints
- SageMaker Pipelines
- SageMaker Feature Store
- SageMaker Model Monitor

GCP Vertex AI

- Vertex AI Training
- Vertex AI Prediction
- Vertex AI Pipelines
- Vertex AI Feature Store
- AutoML on Vertex

Azure Machine Learning

- Azure ML Workspaces
- Azure ML Pipelines
- Azure ML Endpoints
- Azure ML Responsible AI Dashboard
- Azure OpenAI Service

AI Engineering with Foundation Models

LLM Integration Patterns

- API-Based Model Access
- Prompt Engineering in Production
- Retrieval-Augmented Generation (RAG)
- Function Calling and Tool Use
- Agents and Orchestration (LangChain, LlamaIndex)

Fine-Tuning Foundation Models

- Full Fine-Tuning
- Parameter-Efficient Fine-Tuning (LoRA, QLoRA)
- Instruction Tuning
- RLHF and DPO
- Domain Adaptation
- RLHF Deep Dive — Three-Stage Pipeline, Reward Modeling, Alternatives to RLHF

LLM Application Architecture

- Vector Store Integration
- Embedding Pipelines
- Chunking Strategies
- Guardrails and Output Validation
- Caching and Cost Optimization
- Evaluation Frameworks (RAGAS, LLM-as-Judge)

LLM Production Operations

- Cost Optimization — Prompt Caching (KV Cache Sharing Across Requests), Model Routing (Small Model → Large Model Cascade), Semantic Caching (Similar Query Reuse), API vs Self-Hosted Cost Comparison, Token Budget Management
- Output Quality Control — Output Parsing and Structured Generation, Deterministic Generation Strategies, Automated Evaluation Pipelines (LLM-as-Judge), Regression Testing for Prompts, Hallucination Detection in Production
- Long Document Processing — Chunking Strategies (Fixed, Semantic, Recursive), Hierarchical Summarization, Map-Reduce over Documents, Long-Context Models vs RAG Trade-offs

Federated Learning

Core Algorithms

- Federated Averaging (FedAvg)
- FedProx (Proximal Term for Heterogeneity)
- SCAFFOLD (Variance Reduction for Client Drift)
- FedMA (Matched Averaging for Heterogeneous Architectures)

Data Heterogeneity

- Non-IID Data Challenges
- FedBN (Batch Normalization for Heterogeneity)
- Personalized Federated Learning
- Local Fine-Tuning and Per-FedAvg
- Multi-Task Federated Learning

Communication Efficiency

- Gradient Compression and Sparsification
- Federated Distillation

- Asynchronous Aggregation
- Straggler Mitigation

Privacy and Security

- Differential Privacy in FL
- Secure Aggregation Protocols
- Secure Multi-Party Computation (SMPC)
- Model Poisoning Attacks and Defenses
- Data Poisoning Attacks and Defenses
- Byzantine-Robust Aggregation

FL Variants

- Cross-Device vs Cross-Silo FL
- Horizontal Federated Learning
- Vertical Federated Learning
- Federated Transfer Learning

Deployment and Evaluation

- FL Frameworks (Flower, PySyft, TFF)
- Mobile/Edge Device Challenges
- Evaluation Metrics Beyond Accuracy

On-Device ML

Mobile Deployment

- TensorFlow Lite
- CoreML (Apple)
- NNAPI (Android)
- PyTorch Mobile
- On-Device Training

Edge Inference

- Edge Hardware (Jetson, Coral, Raspberry Pi)
- Model Optimization for Edge
- Real-Time Constraints
- Power-Efficient Inference
- Model Splitting (Edge-Cloud)

Embedded ML

- TinyML
- Microcontroller Deployment
- Memory-Constrained Model Design
- Sensor Fusion on Edge

CHAPTERS 14–15

Probabilistic & Generative AI

Modeling uncertainty and generating new data — the mathematical heart of modern AI.

CHAPTER 14

Probabilistic & Bayesian ML

Prerequisites: Mathematical Foundations, Supervised Learning

Outcomes: Bayesian inference · Graphical models · MCMC methods

347 concepts · Max depth: 4

Bayesian Inference

Foundations

- Bayes' Theorem
- Prior Distribution (Encoding Beliefs)
- Likelihood Function
- Posterior Distribution
- Evidence (Marginal Likelihood)
- Bayesian vs Frequentist Interpretation

MAP Estimation

- Mode of Posterior
- Relationship to MLE
- Regularization as MAP with Priors
- L2 Regularization ↔ Gaussian Prior
- L1 Regularization ↔ Laplace Prior
- Point Estimate vs Full Posterior

Conjugate Priors

- Definition and Convenience
- Beta-Binomial Conjugacy
- Dirichlet-Multinomial Conjugacy
- Gaussian-Gaussian (Known Variance)
- Gamma-Poisson Conjugacy
- Normal-Inverse-Gamma (Unknown Mean and Variance)
- Exponential Family and Conjugacy

Predictive Distributions

- Prior Predictive
- Posterior Predictive
- Marginalization Over Parameters
- Uncertainty Quantification

Model Comparison

- Bayes Factors
- Bayesian Information Criterion (BIC)
- Bayesian Model Averaging
- Bayesian Occam's Razor

Bayesian Linear Regression

Prior on Weights

- Gaussian Prior (Zero Mean, Precision Matrix)
- Prior Hyperparameters

Posterior on Weights

- Closed-Form Gaussian Posterior
- Posterior Mean and Covariance
- Sequential Bayesian Update

Predictive Distribution

- Gaussian Predictive (Mean and Variance)
- Epistemic vs Aleatoric Uncertainty
- Credible Intervals

Evidence Approximation (Empirical Bayes)

- Type-II Maximum Likelihood
- Automatic Relevance Determination (ARD)
- Hyperparameter Optimization

Graphical Models

Directed Graphical Models (Bayesian Networks)

- DAG Structure
- Conditional Probability Tables (CPTs)
- Factorization of Joint Distribution
- d-Separation
- Conditional Independence
- Plate Notation
- Structure Learning — Score-Based (BIC, MDL), Constraint-Based (PC Algorithm), Hybrid Methods
- Parameter Learning — MLE with Complete Data, EM with Missing Data, Bayesian Parameter Estimation

Undirected Graphical Models (Markov Random Fields)

- Potential Functions (Factors)
- Partition Function (Z)
- Gibbs Distribution
- Markov Properties (Pairwise, Local, Global)
- Clique Factorization (Hammersley-Clifford)
- Ising Model
- Conditional Random Fields (CRFs)
- Boltzmann Machines

Factor Graphs

- Unifying Directed and Undirected
- Factor Nodes and Variable Nodes
- Message Passing on Factor Graphs

Comparing Directed vs Undirected

- Moralization
- I-Maps and D-Maps
- Expressiveness Tradeoffs

Exact Inference

Variable Elimination

- Elimination Ordering
- Intermediate Factor Computation
- Complexity vs Treewidth

Belief Propagation (Sum-Product)

- Message Passing on Trees
- Forward-Backward Algorithm (Chains)
- Marginal Computation
- Max-Product (MAP Inference)

Junction Tree Algorithm

- Triangulation (Chordal Graph)
- Clique Tree Construction
- Calibration
- Running Intersection Property

Approximate Inference: Variational Methods

Variational Inference (VI)

- KL Divergence Minimization
- Evidence Lower Bound (ELBO)
- ELBO = Expected Log-Likelihood − KL(q || prior)
- Optimization of Variational Parameters

Mean Field Variational Inference

- Fully Factorized Approximation
- Coordinate Ascent Updates
- Optimal Factor Form
- Underestimates Uncertainty

Stochastic Variational Inference (SVI)

- Natural Gradients
- Mini-Batch Updates
- Scalability to Large Datasets

Black-Box Variational Inference

- Score Function Estimator (REINFORCE)
- Reparameterization Trick
- Automatic Differentiation Variational Inference (ADVI)

Variational Autoencoders (VAEs)

- Amortized Inference (Encoder Network)
- Generative Model (Decoder Network)

- Latent Space Regularization
- Reparameterization Trick for Backprop

Expectation Propagation

- Moment Matching
- Iterative Refinement
- Comparison with VI

Normalizing Flows

- Change of Variables Formula
- Planar and Radial Flows
- Real NVP and GLOW
- Autoregressive Flows (MAF, IAF)
- Continuous Normalizing Flows (Neural ODEs)
- Improving Variational Approximations with Flows

Laplace Approximation

- Second-Order Taylor Expansion at MAP
- Gaussian Approximation to Posterior
- Hessian Computation
- Comparison with VI and MCMC
- Laplace Approximation for Neural Networks

Approximate Inference: Sampling Methods (MCMC)

Monte Carlo Fundamentals

- Law of Large Numbers for Estimation
- Importance Sampling
- Rejection Sampling
- Ancestral Sampling

Markov Chain Monte Carlo (MCMC)

- Markov Chain Construction
- Stationary Distribution
- Detailed Balance
- Ergodicity and Mixing
- Burn-In Period
- Thinning
- Convergence Diagnostics (R-hat, Trace Plots)

Metropolis-Hastings

- Proposal Distribution
- Acceptance Ratio
- Random Walk Metropolis
- Independent Metropolis-Hastings
- Tuning the Proposal (Acceptance Rate ~0.234)

Gibbs Sampling

- Full Conditional Distributions
- Systematic vs Random Scan
- Block Gibbs Sampling

- Collapsed Gibbs Sampling
- Special Case of Metropolis-Hastings

Advanced MCMC Methods

- Hamiltonian Monte Carlo (HMC)
- No-U-Turn Sampler (NUTS)
- Slice Sampling
- Reversible Jump MCMC
- Parallel Tempering

Hidden Markov Models (HMMs)

Model Components

- Hidden States
- Observations
- Transition Probabilities (A Matrix)
- Emission Probabilities (B Matrix)
- Initial State Distribution (π)

Three Fundamental Problems

- Evaluation (Likelihood): Forward Algorithm — Forward Variables (α), Backward Variables (β), Forward-Backward Algorithm
- Decoding (Best State Sequence): Viterbi Algorithm — Dynamic Programming, Backtracking, MAP Sequence Estimation
- Learning (Parameter Estimation): Baum-Welch — EM for HMMs, E-Step: γ and ξ Variables, M-Step: Re-estimation Formulas

Applications

- Speech Recognition
- Part-of-Speech Tagging
- Gene Finding
- Financial Time Series

Kalman Filters

Linear Dynamical System

- State Space Model
- State Transition Matrix
- Observation Matrix
- Process Noise (Q)
- Measurement Noise (R)

Filtering (Predict-Update Cycle)

- Prediction Step (Prior)
- Update Step (Posterior)
- Kalman Gain
- Gaussian Closed-Form Updates

Smoothing

- Rauch-Tung-Striebel (RTS) Smoother
- Fixed-Interval Smoothing

Extensions

- Extended Kalman Filter (EKF)
- Unscented Kalman Filter (UKF)
- Particle Filters (Sequential Monte Carlo)
- Ensemble Kalman Filter

Applications

- Robot Localization
- Object Tracking
- Navigation Systems (GPS)
- Signal Processing

Probabilistic Programming

Concepts

- Generative Model Specification
- Automatic Inference
- Posterior Sampling
- Model Criticism and Checking

Frameworks

- PyMC (Python)
- Stan (C++/R/Python)
- Edward / TensorFlow Probability
- Pyro (PyTorch)
- NumPyro (JAX)

Workflow

- Model Specification
- Prior Predictive Checks
- Inference (MCMC, VI)
- Convergence Diagnostics
- Posterior Predictive Checks
- Model Comparison

Gaussian Processes

Definition and Properties

- Distribution Over Functions
- Mean Function
- Kernel (Covariance) Function
- Marginal and Conditional Gaussians

GP Regression

- Prior Over Functions
- Posterior Predictive Distribution
- Uncertainty Bands
- Noise Modeling

Kernel Functions

- Squared Exponential (RBF)

- Matérn Kernels
- Periodic Kernel
- Kernel Composition (Addition, Multiplication)

Hyperparameter Learning

- Marginal Likelihood Optimization
- Bayesian Optimization of Hyperparameters

Scalability

- Sparse GP Approximations (Inducing Points)
- Variational Sparse GP
- Random Fourier Features

Bayesian Neural Networks

Foundations

- Distributions Over Weights vs Point Estimates
- Prior Specification for Neural Networks
- Weight Uncertainty
- BNN vs Standard Neural Networks

Inference Methods for BNNs

- Bayes by Backprop (Blundell et al.)
- Variational Inference for BNNs
- Stochastic Gradient Langevin Dynamics (SGLD)
- Stochastic Gradient Hamiltonian Monte Carlo (SGHMC)
- Laplace Approximation for NNs

Approximate Bayesian Methods

- Monte Carlo Dropout
- Deep Ensembles
- Snapshot Ensembles
- Comparison: MC Dropout vs Ensembles vs True BNNs

Uncertainty Quantification

- Epistemic vs Aleatoric Uncertainty in BNNs
- Calibration of Uncertainty Estimates
- Out-of-Distribution Detection with BNNs
- Predictive Uncertainty Decomposition

Theoretical Connections

- Cold Posterior Effect
- PAC-Bayes Bounds for Neural Networks
- Lottery Ticket Hypothesis and Bayesian Pruning
- BNNs and Neural Network Ensembles Relationship
- Infinite-Width NNs and Gaussian Processes (Neural Tangent Kernel)

Scalability and Practical Considerations

- Scalability Challenges for Large Architectures
- Implementation with PyTorch (Blitz, TyXe)
- Implementation with TensorFlow Probability

- BNNs in Safety-Critical Domains (Medical, Autonomous)

Bayesian Optimization

Framework

- Surrogate Model (GP, Random Forest, TPE)
- Acquisition Function
- Sequential Optimization Loop
- Exploration-Exploitation Tradeoff in BO

Acquisition Functions

- Expected Improvement (EI)
- Upper Confidence Bound (UCB / GP-UCB)
- Probability of Improvement (PI)
- Knowledge Gradient
- Thompson Sampling for BO
- Entropy Search and Predictive Entropy Search

Surrogate Models

- Gaussian Process Surrogates
- Tree-structured Parzen Estimator (TPE)
- Random Forest Surrogates (SMAC)
- Bayesian Neural Network Surrogates

Scalability and Extensions

- High-Dimensional BO (Random Embeddings, Trust Regions)
- Multi-Fidelity BO (Fabolas, Freeze-Thaw)
- Multi-Objective BO (Pareto Front)
- BOHB (Bayesian Optimization + HyperBand)
- Categorical and Conditional Hyperparameters
- Noisy Observations Handling
- Transfer Learning and Warm-Starting in BO
- BO for Neural Architecture Search (NAS)

Tools and Frameworks

- Optuna
- Hyperopt
- BoTorch (PyTorch)
- Ax (Adaptive Experimentation Platform)
- SMAC
- Spearmint

Comparison with Other Methods

- BO vs Grid Search vs Random Search
- When to Use BO vs Population-Based Methods
- Limitations of Bayesian Optimization

Hierarchical Bayesian Models

Hierarchical Structure and Hyperpriors

Partial Pooling and Shrinkage

Multilevel Models

Hierarchical Linear Models

Empirical Bayes vs Full Bayesian Hierarchical

Non-Parametric Bayesian Methods

Dirichlet Process

Chinese Restaurant Process

Indian Buffet Process

Dirichlet Process Mixture Models

Gaussian Process as Non-Parametric Model

Beta Process

Pitman-Yor Process

CHAPTER 15

Generative Models

Prerequisites: Training Deep Networks, Probabilistic & Bayesian ML

Outcomes: VAEs · GANs · Diffusion models

259 concepts · Max depth: 4

Autoencoders

Standard Autoencoder

- Encoder Architecture
- Latent Space Representation
- Decoder Architecture
- Reconstruction Loss
- Undercomplete vs Overcomplete

Denoising Autoencoder

- Input Corruption Strategies
- Noise Types (Gaussian, Masking, Salt-and-Pepper)
- Robust Feature Learning
- Denoising Score Matching Connection

Variational Autoencoder (VAE)

- Probabilistic Encoder (Recognition Model)
- Latent Prior Distribution
- Reparameterization Trick
- Evidence Lower Bound (ELBO)
- KL Divergence Regularization
- Posterior Collapse
- Conditional VAE
- VAE Extensions (add to VAE section) — Hierarchical VAEs (NVAE, VDVAE), Importance-Weighted Autoencoders (IWAE), VAEs for Molecular Generation / Drug Discovery

β-VAE

- Disentangled Representations
- Beta Hyperparameter Tuning
- Trade-off: Reconstruction vs Disentanglement
- Evaluation Metrics for Disentanglement

VQ-VAE

- Vector Quantization Layer
- Codebook Learning
- Straight-Through Estimator
- Commitment Loss
- VQ-VAE-2 (Hierarchical)
- Applications in Audio and Image Generation

Autoencoder Variants (add to Autoencoders)

- Sparse Autoencoder (L1 Regularization)
- Contractive Autoencoder (Jacobian Penalty)
- Autoencoders for Anomaly Detection

Generative Adversarial Networks (GANs)

Vanilla GAN

- Generator Network
- Discriminator Network
- Minimax Game Formulation
- Nash Equilibrium
- Binary Cross-Entropy Loss
- Training Loop Dynamics

DCGAN

- Convolutional Generator
- Convolutional Discriminator
- Batch Normalization in GANs
- Architecture Guidelines
- Latent Space Arithmetic

WGAN

- Wasserstein Distance (Earth Mover's Distance)
- Critic vs Discriminator
- Weight Clipping
- WGAN-GP (Gradient Penalty)
- Lipschitz Constraint
- Spectral Normalization

Conditional GAN

- Class-Conditional Generation
- Label Embedding
- Auxiliary Classifier GAN (AC-GAN)
- Projection Discriminator

StyleGAN

- Mapping Network
- Style Injection via AdaIN
- Progressive Growing
- Stochastic Variation (Noise Inputs)
- Style Mixing
- StyleGAN2 Improvements
- StyleGAN3 (Alias-Free)

CycleGAN

- Unpaired Image-to-Image Translation
- Cycle Consistency Loss
- Identity Loss
- Domain Adaptation Applications

Pix2Pix

- Paired Image-to-Image Translation
- U-Net Generator
- PatchGAN Discriminator
- L1 Reconstruction Loss

GAN Training Stability

- Mode Collapse
- Training Divergence
- Vanishing Gradients
- Label Smoothing
- Two-Timescale Update Rule (TTUR)
- Progressive Training
- Evaluation Metrics (FID, IS, LPIPS)
- GAN Additional Topics (add to GANs) — Minibatch Discrimination, Unrolled GANs, LSGAN (Least Squares GAN), Hinge Loss GAN, GAN Inversion, GAN Convergence Theory

Diffusion Models

Denoising Diffusion Probabilistic Models (DDPM)

- Forward Diffusion Process
- Gaussian Noise Addition Schedule
- Reverse Denoising Process
- Noise Prediction Network (ε-prediction)
- Variational Lower Bound
- Simplified Training Objective

Score Matching

- Score Function (∇ log p(x))
- Denoising Score Matching
- Sliced Score Matching
- Noise Conditional Score Networks (NCSN)
- Langevin Dynamics Sampling
- Score-Based SDE Framework

Noise Schedules

- Linear Schedule
- Cosine Schedule
- Learned Schedules
- Signal-to-Noise Ratio
- Continuous vs Discrete Time

Sampling Methods

- DDPM Sampling (Stochastic)
- DDIM (Deterministic Sampling)
- Accelerated Sampling
- DPM-Solver
- Ancestral Sampling
- Progressive Distillation

Classifier-Free Guidance

- Conditional vs Unconditional Training
- Guidance Scale
- Classifier Guidance (Original)
- CFG Mechanism
- Trade-off: Quality vs Diversity

Latent Diffusion Models

- Perceptual Compression (Autoencoder Stage)
- Latent Space Diffusion
- Cross-Attention Conditioning
- U-Net Backbone Architecture
- Text Encoder Integration

Stable Diffusion

- Architecture Overview
- CLIP Text Encoder
- VAE Encoder/Decoder
- U-Net Denoiser
- Img2Img Pipeline
- Inpainting
- ControlNet
- LoRA Fine-Tuning
- SDXL and Subsequent Versions
- Diffusion Personalization & Applications (add to Stable Diffusion) — Textual Inversion, DreamBooth, LoRA for Diffusion Models, Negative Prompting via CFG, Flux Architecture

Normalizing Flows

Foundations

- Change of Variables Formula
- Invertible Transformations
- Jacobian Determinant Computation
- Exact Likelihood Computation

NICE

- Additive Coupling Layers
- Volume-Preserving Transformations
- Triangular Jacobians

RealNVP

- Affine Coupling Layers
- Multi-Scale Architecture
- Squeeze Operations
- Batch Normalization in Flows

Glow

- 1x1 Invertible Convolutions
- Actnorm
- Multi-Scale Flow Architecture

- Latent Space Manipulation

Advanced Flow Models

- Continuous Normalizing Flows (Neural ODE)
- Residual Flows
- Autoregressive Flows (MAF, IAF)
- Flow Matching

Energy-Based Models

Foundations

- Energy Function Definition
- Boltzmann Distribution
- Partition Function
- Unnormalized Density

Training Methods

- Contrastive Divergence
- Score Matching
- Noise Contrastive Estimation
- Stein Discrepancy

Sampling from EBMs

- MCMC Methods
- Langevin Dynamics
- Hamiltonian Monte Carlo
- Persistent Contrastive Divergence

Applications

- Restricted Boltzmann Machines
- Deep Boltzmann Machines
- EBMs for Compositional Generation
- Joint Energy Models (Classification + Generation)

CUTTING-EDGE Modern Generative Breakthroughs (2024-2025)

Flow Matching & Rectified Flows

- Probability Flow ODE
- Continuous Normalizing Flow
- Flow Matching
- Conditional Flow Matching
- Consistency Models
- Rectified Flow
- Reflow Procedure
- Visual Autoregressive Modeling (VAR)

3D Generation

- Neural Radiance Fields (NeRF)
- 3D Gaussian Splatting

Video Generation

- Temporal Attention
- Video Diffusion

Scaling & Architecture

- Guidance Scale
- Self-Consistency Property
- Consistency Distillation
- Consistency Training
- Diffusion Transformer (DiT)
- MMDiT
- Next-Scale Prediction
- Latent Diffusion

Additional Topics

- Diffusion Forward Process
- Diffusion Reverse Process
- Score Function
- Denoising Score Matching
- DDPM
- DDIM
- Noise Schedule
- Velocity Field
- Optimal Transport
- Linear Interpolation Path
- Continuity Equation
- Path Crossing
- AdaLN-Zero
- Spacetime Patches
- Score Distillation Sampling (SDS)
- Multi-View Diffusion
- Logit-Normal Timestep Sampling
- EMA Target Network

Audio & Music Generation

Diffusion-Based TTS (Tortoise TTS, XTTS)

Mel-Spectrogram Diffusion

AudioLDM (Latent Diffusion for Audio)

Autoregressive Audio (MusicLM, MusicGen)

Neural Audio Codec (EnCodec, SoundStream)

Generative Model Evaluation

FID (Fréchet Inception Distance)

Inception Score (IS) and Limitations

LPIPS (Learned Perceptual Image Patch Similarity)

CLIP Score (Text-Image Alignment)

FID-CLIP Tradeoff

Human Evaluation Protocols

AI-Generated Image Detection / Forensics

CHAPTERS 16–17

Advanced Topics

Research frontiers, learning theory, and pushing the boundaries of what’s possible.

CHAPTER 16

Advanced Deep Learning

Prerequisites: CNNs & Computer Vision, Large Language Models

Outcomes: Graph neural networks · Interpretability · Multimodal models

474 concepts · Max depth: 3

Graph Neural Networks

Foundations

- Graph Representation (Nodes, Edges, Adjacency)
- Graph Laplacian
- Spectral vs Spatial Methods
- Message Passing Framework
- Neighborhood Aggregation
- Permutation Invariance/Equivariance

Graph Convolutional Network (GCN)

- Spectral Convolution
- Chebyshev Polynomial Approximation
- First-Order Approximation (Kipf & Welling)
- Symmetric Normalization
- Over-Smoothing Problem

Graph Attention Network (GAT)

- Attention Mechanism on Graphs
- Multi-Head Graph Attention
- Learnable Edge Weights
- GATv2 (Dynamic Attention)

GraphSAGE

- Inductive Learning on Graphs
- Neighborhood Sampling
- Aggregation Functions (Mean, LSTM, Pooling)
- Mini-Batch Training on Large Graphs

Advanced GNN Architectures

- Graph Isomorphism Network (GIN)
- Graph Transformer
- Heterogeneous Graph Networks
- Temporal Graph Networks
- Equivariant GNNs

GNN Tasks

- Node Classification
- Graph Classification

- Link Prediction
- Graph Generation
- Graph Pooling (Hierarchical Readout)

Interpretability and Explainability

Model-Agnostic Methods

- LIME (Local Interpretable Model-Agnostic Explanations)
- SHAP (SHapley Additive exPlanations)
- Kernel SHAP
- Tree SHAP
- Permutation Feature Importance
- Partial Dependence Plots
- Individual Conditional Expectation (ICE)

Gradient-Based Methods

- Vanilla Gradients (Saliency Maps)
- Grad-CAM
- Grad-CAM++
- Integrated Gradients
- SmoothGrad
- Guided Backpropagation
- Layer-wise Relevance Propagation (LRP)

Attention-Based Interpretability

- Attention Visualization
- Attention Rollout
- Attention as Explanation Debate
- BERTology (Probing Transformers)

Concept-Based Explanations

- TCAV (Testing with Concept Activation Vectors)
- Network Dissection
- Concept Bottleneck Models
- Prototype Networks for Interpretability

Evaluation of Explanations

- Faithfulness Metrics
- Stability and Robustness
- Human Evaluation
- Sanity Checks for Saliency Maps

Multimodal Models

Vision-Language Models

- CLIP (Contrastive Language-Image Pre-training)
- ALIGN
- BLIP / BLIP-2
- Flamingo
- LLaVA
- Image Captioning

- Visual Question Answering (VQA)

Text-to-Image Models

- DALL-E / DALL-E 2 / DALL-E 3
- Imagen
- Midjourney Architecture Concepts
- Text-Image Alignment
- Compositional Generation

Audio-Visual Models

- Audio-Visual Speech Recognition
- Sound-Image Correspondence
- Video Understanding
- Audio-Visual Source Separation

Cross-Modal Learning

- Shared Embedding Spaces
- Cross-Modal Attention
- Modality Fusion Strategies (Early, Late, Hybrid)
- Missing Modality Handling
- Cross-Modal Retrieval
- Multimodal Tokenization

Emerging Multimodal Architectures

- Unified Multimodal Transformers
- Any-to-Any Generation
- Multimodal Chain-of-Thought
- Document Understanding (OCR + Language)

Self-Supervised and Contrastive Learning

Pretext Tasks

- Rotation Prediction
- Jigsaw Puzzles
- Colorization
- Masked Image Modeling (MAE, BEiT)
- Masked Language Modeling
- Next Sentence Prediction

Contrastive Learning Foundations

- Positive and Negative Pairs
- InfoNCE Loss
- Temperature Parameter
- Data Augmentation Strategies
- Hard Negative Mining
- Representation Collapse Prevention

SimCLR

- Augmentation Pipeline
- Projection Head
- Large Batch Training

- NT-Xent Loss

MoCo (Momentum Contrast)

- Momentum Encoder
- Queue of Negatives
- MoCo v2 / v3 Improvements
- Dictionary as Queue

BYOL (Bootstrap Your Own Latent)

- No Negative Samples Required
- Online and Target Networks
- Exponential Moving Average
- Predictor Network

Other Self-Supervised Methods

- DINO / DINOv2
- SwAV (Clustering-Based)
- Barlow Twins
- VICReg
- I-JEPA (Joint Embedding Predictive Architecture)

Meta-Learning and Few-Shot Learning

Meta-Learning Foundations

- Learning to Learn
- Task Distribution
- Support Set and Query Set
- N-way K-shot Formulation
- Episodic Training

Metric-Based Methods

- Siamese Networks
- Matching Networks
- Prototypical Networks
- Relation Networks
- Embedding Space Learning

Optimization-Based Methods

- MAML (Model-Agnostic Meta-Learning)
- First-Order MAML (Reptile)
- Inner Loop / Outer Loop
- Task-Specific Adaptation
- Meta-SGD

Memory-Based Methods

- Neural Turing Machines
- Memory-Augmented Neural Networks (MANN)
- External Memory Modules

Modern Few-Shot Approaches

- In-Context Learning (LLMs as Few-Shot Learners)

- Prompt-Based Few-Shot
- Foundation Models for Few-Shot
- Cross-Domain Few-Shot Transfer

CUTTING-EDGE AI for Scientific Discovery

Drug Discovery & Molecular ML

- Protein folding problem
- Diffusion models for molecular structure
- Protein language models
- CASP (Critical Assessment of protein Structure Prediction)
- AlphaFold 3 (Diffusion-Based Protein Prediction)

Physics & Climate

- Numerical Weather Prediction (NWP)
- Probabilistic weather forecasting
- Autoregressive weather rollout
- GraphCast (GNN Weather Forecasting)

Mathematics & Formal Reasoning

- Formal verification / proof assistants
- Reinforcement learning for theorem proving
- Autoformalization
- AlphaProof (Formal Math Theorem Proving)

Biology & Genomics

- Neuro-symbolic AI
- Evolutionary program search
- Zero-shot generalization

Additional Topics

- Multiple Sequence Alignment (MSA)
- Evoformer
- Invariant Point Attention (IPA)
- Graph Neural Networks (GNNs)
- Ensemble forecasting
- Icosahedral mesh
- Message passing
- 3D Earth-Specific Transformer
- Formation energy
- Active learning
- Retrosynthesis
- Vision-Language-Action (VLA) models
- Sim-to-real transfer
- Domain randomization
- Triangle multiplicative updates
- Foundation models for science

Emerging 2025 Architectures

Test-Time Compute Scaling

- Best-of-N Sampling with Reward Models
- Inference-Time Search (Beam Search over CoT)
- Process Reward Models for Verification
- Compute-Optimal Inference (Model Size vs Inference Compute)
- OpenAI o1/o3 Reasoning Paradigm

Transformer Alternatives (Beyond Mamba)

- RWKV (Receptance Weighted Key Value)
- RetNet (Retentive Networks)
- Hyena Hierarchy
- Based (Linear Attention Variants)
- Jamba (Mamba + Transformer Hybrid)

Efficient Architecture Innovations

- Mixture-of-Depths (Dynamic Token Computation)
- BitNet / 1-Bit LLMs
- Ring Attention (Distributed Long Context)
- Matryoshka Representation Learning (Nested Embeddings)
- Kolmogorov-Arnold Networks (KAN)
- Byte-Level Models (No Tokenizer)

Advanced Reasoning

- Self-Consistency (Majority Voting over CoT)
- Faithful vs Unfaithful Chain-of-Thought
- Reasoning Distillation (Training on CoT Traces)
- Verification and Critic Models
- Formal Verification of Generated Code

Continual and Lifelong Learning

Problem Setting

- Catastrophic Forgetting
- Stability-Plasticity Dilemma
- Task-Incremental vs Domain-Incremental vs Class-Incremental
- Forward and Backward Transfer Metrics

Regularization-Based Methods

- Elastic Weight Consolidation (EWC)
- Fisher Information Matrix for Weight Importance
- Synaptic Intelligence (SI)
- Memory Aware Synapses (MAS)

Replay-Based Methods

- Experience Replay and Episodic Memory
- Gradient Episodic Memory (GEM)
- Averaged GEM (A-GEM)
- Generative Replay

Architecture-Based Methods

- Progressive Neural Networks

- PackNet (Pruning for Continual Learning)
- Dynamic Expandable Networks

Knowledge Distillation for Continual Learning

- Learning without Forgetting (LwF)
- Distillation-Based Regularization

Benchmarks and Evaluation

- Avalanche Framework
- Split CIFAR, Split MNIST
- Metrics (Average Accuracy, Forgetting Measure, BWT, FWT)

Equivariant Neural Networks and Geometric Deep Learning

Foundations

- Invariance vs Equivariance Definitions
- Group Theory Basics for ML (Symmetry Groups)
- CNN Translation Equivariance
- Data Augmentation vs Built-In Equivariance

Architectures

- Group-Equivariant Convolutions
- E(n)-Equivariant Neural Networks (EGNN)
- SE(3)-Transformers
- Steerable CNNs
- Spherical CNNs
- Tensor Field Networks (Spherical Harmonics)

Geometric Deep Learning Blueprint

- Unification of CNNs, GNNs, Transformers
- 5G Framework (Grids, Groups, Graphs, Geodesics, Gauges)
- Gauge Equivariance on Manifolds

Applications

- Molecular Property Prediction
- Protein Structure Prediction
- Physics Simulations
- Particle Physics

Neural Architecture Search (NAS)

Search Strategies

- RL-Based NAS (Zoph & Le)
- DARTS (Differentiable Architecture Search)
- Evolutionary NAS
- Bayesian Optimization for NAS

Efficiency Methods

- One-Shot and Weight-Sharing NAS
- Supernet Training
- Performance Predictors

- Early Stopping Strategies

Hardware-Aware NAS

- MnasNet
- ProxylessNAS
- Latency and FLOPs Optimization
- EfficientNet and Compound Scaling

Evaluation and Benchmarks

- NAS-Bench-101, 201, 301
- Search Space Design and Bias
- Reproducibility Challenges

Related Concepts

- Lottery Ticket Hypothesis
- Relationship to Pruning
- Network Morphisms

Domain-Specific ML Applications

ML for Healthcare

- Medical Image Analysis (X-ray, CT, MRI Architectures)
- FDA Approval and SaMD Regulatory Framework
- Electronic Health Records (EHR) and Temporal Encoding
- Clinical NLP (Medical Note Extraction)
- Drug Discovery (Virtual Screening, De Novo Design)
- Drug-Drug Interaction Modeling with GNNs
- Survival Analysis (Cox Proportional Hazards, DeepSurv)
- Genomics (Variant Calling, Gene Expression)
- Distribution Shift Across Hospitals and Scanners
- Synthetic Medical Data Generation
- Medical LLMs (Med-PaLM, BioGPT)
- U-Net and Variants for Medical Segmentation
- Multi-Modal Patient Outcome Prediction
- Label Noise and Annotator Disagreement
- Clinical Datasets (MIMIC, CheXpert)
- Federated Learning for Multi-Hospital Collaboration
- Few-Shot Learning for Rare Diseases
- RL for Adaptive Clinical Trials

ML for Finance

- Algorithmic Trading (Features, Architectures)
- Lookahead Bias and Data Leakage Prevention
- Non-Stationarity in Financial Time Series
- Credit Scoring and Fairness Constraints
- Fraud Detection (Extreme Class Imbalance)
- Risk Modeling (VaR, CVaR, Parametric vs Non-Parametric)
- Anomaly Detection in Financial Transactions
- RL for Portfolio Optimization
- Alpha Signals vs Risk Factors

- Backtesting Pitfalls (Overfitting, Survivorship Bias)
- NLP for Finance (Sentiment, Earnings Calls, SEC Filings)
- Regime Change Detection
- GNNs for Financial Networks
- Regulatory Constraints (Basel III, Fair Lending)
- High-Frequency Trading and Latency
- Market Microstructure and Order Book Modeling
- Bayesian Uncertainty in Financial Risk

ML for Autonomous Vehicles

- Perception Pipeline (Camera, LiDAR, Radar)
- Sensor Fusion (Early, Late, Mid-Level)
- 3D Object Detection (PointPillars, VoxelNet)
- SLAM (Classical and Deep Learning)
- BEV Representation and BEVFormer
- Multi-Camera 3D Detection Without LiDAR
- Motion Planning (Rule, Search, Learning-Based)
- Imitation Learning (Behavior Cloning, DAgger)
- Safety-Critical ML and ISO 26262
- Long-Tail Safety-Critical Scenarios
- Driving Simulation (CARLA, nuPlan, Sim-to-Real)
- HD Map Creation
- Occupancy Grid Prediction
- End-to-End Autonomous Driving
- Trajectory Prediction for Road Agents
- Lane Detection
- Uncertainty Estimation for Driving Decisions
- Ethics (Trolley Problem, Detection Bias)

ML for Recommender Systems

- Collaborative Filtering (User-Based, Item-Based)
- Matrix Factorization and SVD
- Cold-Start Problem Solutions
- Content-Based Filtering
- Hybrid Approaches
- Implicit vs Explicit Feedback
- BPR (Bayesian Personalized Ranking)
- Neural Collaborative Filtering (NCF)
- GNN-Based Recommendations (PinSage, LightGCN)
- Two-Tower Models (Retrieval and Ranking)
- Session-Based Recommendation (RNN, Transformer)
- YouTube Recommendation Architecture
- Position Bias and Debiasing
- Filter Bubble and Diversity-Serendipity Balance
- Multi-Task Recommendation
- Exploration-Exploitation with Bandits
- RL for Long-Term Engagement
- Evaluation (NDCG, MAP, Recall@K, A/B Testing)

ML for Speech

- Traditional ASR Pipeline (Feature Extraction → Acoustic → LM → Decoder)
- MFCC and Mel-Spectrogram Features
- CTC (Connectionist Temporal Classification)
- RNN-Transducer (RNN-T)
- Attention-Based ASR (Listen, Attend and Spell)
- Whisper (Robust Multi-Language ASR)
- TTS (Tacotron, WaveNet, FastSpeech)
- Neural Vocoders (WaveRNN, HiFi-GAN)
- Speaker Verification and Identification (x-vectors, d-vectors)
- Speaker Diarization
- Zero-Shot TTS and Voice Cloning
- Self-Supervised Speech (wav2vec 2.0, HuBERT)
- Keyword Spotting and Wake-Word Detection
- Code-Switching in Multilingual ASR
- Speech Emotion Recognition
- Diffusion Models for Speech (DiffWave, Grad-TTS)

ML for Tabular Data

- Why Trees Outperform DL on Tabular Data
- TabNet (Attention-Based Feature Selection)
- FT-Transformer for Tabular
- SAINT (Self-Attention and Intersample Attention)
- NODE (Neural Oblivious Decision Ensembles)
- CatBoost Categorical Handling
- High-Cardinality Categorical Features
- Target Encoding vs One-Hot vs Learned Embeddings
- AutoML for Tabular (Auto-sklearn, AutoGluon, H2O)
- Grinsztajn et al. (2022) DL vs Trees Benchmark
- Missing Data Imputation Strategies

ML for Time Series

- Classical vs Deep Learning (ARIMA, Exponential Smoothing)
- Temporal Fusion Transformer (TFT)
- N-BEATS (Interpretable Decomposition)
- PatchTST (Patching for Transformers)
- Informer (ProbSparse Attention)
- DeepAR (Probabilistic Forecasting)
- DLinear (Linear Models Beat Transformers Finding)
- Non-Stationarity Handling (RevIN)
- Global vs Local Models
- Foundation Models (TimesFM, Chronos, Lag-Llama)
- State-Space Models for Time Series (S4, Mamba)
- Conformal Prediction for Calibrated Intervals
- Wavelet-Neural Hybrid Approaches
- Time Series Anomaly Detection
- Point vs Probabilistic Forecasting
- Evaluation (Rolling Window, Temporal Cross-Validation)

AI for Science (Expanded)

- RoseTTAFold (Improvements on AlphaFold)
- Weather Forecasting (Pangu-Weather, FourCastNet)
- Materials Science (Crystal Structure, Property Prediction)
- Particle Physics (Jet Classification, LHC Event Reconstruction)
- Climate Science (Downscaling, Subgrid Parameterization)
- Fourier Neural Operator for PDE Families
- DeePMD Framework for Molecular Dynamics
- Astronomy (Galaxy Classification, Exoplanet Detection, Gravitational Waves)
- Enformer for Gene Expression from DNA Sequence
- Diffusion Models for Protein Design
- Physics Constraints and Conservation Laws in ML

ML for Code (Expanded)

- Fill-in-the-Middle (FIM) Training
- Code Understanding (Summarization, Bug Detection, Type Inference)
- AST (Abstract Syntax Tree) Representations
- Retrieval-Augmented Code Generation
- Program Synthesis vs Code Generation
- Execution Feedback (AlphaCode, CodeRL)
- Security Vulnerabilities in Generated Code
- Benchmark Contamination Problem
- Multi-Turn Code Generation Agents
- Static Analysis for LLM Code Verification
- Code Embeddings for Search and Clone Detection
- Tree-Sitter and Structural Code Parsing

Cross-Domain ML Concepts

- ML vs Rule-Based System Decision Framework
- Domain-Specific Tech Stacks (DICOM, FHIR, FIX, ROS)
- Digital Twin Applications
- Model Risk Management in Regulated Industries
- AI Readiness Assessment
- Formal Verification for Safety-Critical ML
- Small Language Models and Edge AI
- Conformal Prediction for Uncertainty Quantification
- Cost of Errors Across Domains
- Build vs Buy Decision for ML
- ML ROI Measurement
- Domain-Specific Benchmarks
- Graceful Degradation for OOD Inputs

Neural ODEs and Continuous-Depth Models

Neural ODE Formulation ($dx/dt = f(x,t,\theta)$)

Adjoint Sensitivity Method for Gradients

Memory Advantages Over Discrete ResNets

ResNet as Euler Discretization of Neural ODE

Augmented Neural ODE (State Space Augmentation)

Continuous Normalizing Flows

Neural SDEs (Stochastic Differential Equations)

Flow Matching as Alternative to Neural ODE

Adaptive Step-Size ODE Solvers

Applications to Irregular Time Series

Applications to Generative Modeling

CHAPTER 17

Learning Theory & Advanced ML

Prerequisites: Supervised Learning, Probabilistic & Bayesian ML

Outcomes: Generalization theory · Online learning · RL fundamentals

391 concepts · Max depth: 5

Computational Learning Theory

PAC Learning

- Probably Approximately Correct Framework
- Epsilon (Accuracy Parameter)
- Delta (Confidence Parameter)
- Sample Complexity Bounds
- Hypothesis Class
- Realizable vs Agnostic Setting
- Efficient PAC Learnability
- PAC-Bayes Bounds

VC Dimension

- Shattering
- Growth Function
- VC Dimension Definition
- Examples (Linear Classifiers: d+1)
- Sauer's Lemma
- VC Generalization Bound
- Structural Risk Minimization
- Infinite VC Dimension Classes

Rademacher Complexity

- Empirical Rademacher Complexity
- Rademacher Variables
- Tighter Data-Dependent Bounds
- Relationship to VC Dimension
- Contraction Lemma (Lipschitz Composition)
- Generalization via Rademacher Bounds

Generalization Bounds

- Hoeffding's Inequality
- Union Bound (Finite Hypothesis Classes)
- McDiarmid's Inequality
- Covering Numbers
- Margin-Based Bounds
- Stability-Based Bounds (Algorithmic Stability)
- Compression Bounds

Bias-Complexity Tradeoff

- Approximation Error (Bias)
- Estimation Error (Variance)
- Model Complexity Control
- No Free Lunch Theorem
- Double Descent Phenomenon
- Interpolation and Benign Overfitting

Neural Tangent Kernel (NTK) Theory

- Infinite-Width Limit
- Lazy Training Regime
- NTK and Generalization
- Mean-Field Theory

Online Learning

Framework

- Sequential Prediction
- Adversarial Setting
- Regret (Static and Dynamic)
- No Statistical Assumptions on Data

Algorithms

- Perceptron (Mistake Bound)
- Winnow (Multiplicative Updates)
- Online Gradient Descent
- Follow the Regularized Leader (FTRL)
- Online Mirror Descent
- Exp3 (Adversarial Bandits)

Expert Advice

- Weighted Majority Algorithm
- Multiplicative Weights Update
- Hedge Algorithm
- Regret Bound O(sqrt(T log N))

Online Convex Optimization

- Convex Loss Functions
- Sublinear Regret Guarantees
- Strongly Convex vs General Convex
- Adaptive Learning Rates (AdaGrad)

Multi-Armed Bandits

Problem Setting

- Exploration vs Exploitation Tradeoff
- Regret Definition
- Stochastic Bandits
- Adversarial Bandits

Algorithms

- Epsilon-Greedy
- Upper Confidence Bound (UCB1)
- Thompson Sampling
- EXP3 (Adversarial Setting)
- Bayesian UCB

Extensions

- Contextual Bandits
- LinUCB
- Combinatorial Bandits
- Best Arm Identification
- Non-Stationary Bandits

Theoretical Results

- Lai-Robbins Lower Bound
- Logarithmic Regret
- Minimax Regret (sqrt(KT))

Applications

- A/B Testing and Adaptive Experiments
- Recommendation Systems
- Clinical Trials
- Ad Placement

Reinforcement Learning

Markov Decision Process (MDP)

- States, Actions, Transitions
- Reward Function
- Discount Factor (Gamma)
- Markov Property
- Episodic vs Continuing Tasks
- Partially Observable MDPs (POMDPs)

Policy

- Deterministic vs Stochastic Policy
- Optimal Policy
- Policy Evaluation
- Policy Improvement Theorem

Value Functions

- State-Value Function V(s)
- Action-Value Function Q(s,a)
- Bellman Expectation Equation
- Bellman Optimality Equation
- Advantage Function A(s,a)

Dynamic Programming Methods

- Policy Iteration
- Value Iteration

- Requires Full Model Knowledge
- Convergence Guarantees

Model-Free Methods

- Monte Carlo Methods — First-Visit vs Every-Visit MC, MC Policy Evaluation, MC Control (Epsilon-Greedy)
- Temporal Difference (TD) Learning — TD(0) Update Rule, Bootstrapping, TD vs MC Comparison, TD(λ) and Eligibility Traces
- Q-Learning — Off-Policy TD Control, Q-Value Update Rule, Exploration Strategies, Convergence (Tabular), Deep Q-Networks (DQN), Experience Replay, *(+2 more)*
- SARSA — On-Policy TD Control, Expected SARSA, Comparison with Q-Learning

Policy Gradient Methods

- Policy Parameterization
- Score Function (Log-Derivative Trick)
- REINFORCE Algorithm
- Variance Reduction (Baselines)
- Actor-Critic Methods
- Advantage Actor-Critic (A2C/A3C)
- Proximal Policy Optimization (PPO)
- Trust Region Policy Optimization (TRPO)
- Additional Policy Gradient Methods — Generalized Advantage Estimation (GAE), Entropy Bonus for Exploration, Soft Actor-Critic (SAC), Twin Delayed DDPG (TD3), Deep Deterministic Policy Gradient (DDPG), Deterministic Policy Gradient (DPG) Theorem, *(+1 more)*

Key Challenges

- Credit Assignment Problem
- Exploration vs Exploitation
- Sample Efficiency
- Function Approximation Instability
- Reward Shaping

Model-Based Reinforcement Learning

- Foundations — Model-Based vs Model-Free RL, Sample Efficiency Advantages, Planning vs Learning, Model Inaccuracy and Compounding Errors
- Algorithms — Dyna-Q (Combining Model-Based and Model-Free), Model-Based Policy Optimization (MBPO), Dreamer (Latent Space World Models), World Models (Ha & Schmidhuber), Model Predictive Control (MPC), Monte Carlo Tree Search (MCTS)
- Notable Systems — AlphaGo / AlphaZero (MCTS + Neural Networks), MuZero (Learned Model + Planning)
- Sim-to-Real Transfer — Domain Randomization, System Identification, Adaptive Methods, Robotics Applications

Offline / Batch Reinforcement Learning

- Conservative Q-Learning (CQL)
- Decision Transformer
- Dataset Composition and Quality
- Pessimism Principle

Inverse RL and Imitation Learning

- Behavioral Cloning

- DAgger (Dataset Aggregation)
- Reward Learning from Demonstrations
- GAIL (Generative Adversarial IL)

Multi-Agent Reinforcement Learning

- Cooperative and Competitive Settings
- Communication Protocols
- Nash Equilibrium in Games
- Emergent Behaviors

Meta-Learning

Learning-to-Learn Framework

MAML (Model-Agnostic Meta-Learning)

Prototypical Networks

Matching Networks

Reptile

Meta-Learning for Hyperparameter Optimization

Sparse Models and Compressed Sensing

Sparse Representations

- Sparsity in Signal Processing
- L0 Norm (NP-Hard)
- L1 Relaxation (Basis Pursuit)
- Sparse Coding

Compressed Sensing

- Sub-Nyquist Sampling
- Restricted Isometry Property (RIP)
- Measurement Matrices (Random Gaussian, Bernoulli)
- Recovery Guarantees
- Basis Pursuit (L1 Minimization)
- Matching Pursuit and OMP
- Iterative Hard Thresholding

Algorithms for Sparse Recovery

- LASSO
- ISTA (Iterative Shrinkage-Thresholding)
- FISTA (Fast ISTA)
- ADMM (Alternating Direction Method of Multipliers)
- Proximal Gradient Methods

Applications

- MRI Reconstruction
- Single-Pixel Camera
- Genomics (Sparse Gene Selection)
- Network Tomography

Matrix Factorization

Low-Rank Approximation

- SVD (Singular Value Decomposition)
- Truncated SVD
- Eckart-Young-Mirsky Theorem
- Nuclear Norm Minimization

Non-Negative Matrix Factorization (NMF)

- Non-Negativity Constraints
- Parts-Based Representation
- Multiplicative Update Rules
- Divergence Objectives (Frobenius, KL)
- Topic Modeling Interpretation

Matrix Completion

- Missing Data Problem
- Low-Rank Assumption
- Nuclear Norm Regularization
- Alternating Least Squares (ALS)
- Stochastic Gradient Descent
- Incoherence Conditions

Applications

- Collaborative Filtering (Recommender Systems)
- Latent Semantic Analysis (LSA)
- Image Compression
- Topic Discovery

Tensor Decomposition

Tensor Fundamentals

- Tensors as Multilinear Arrays
- Tensor Order (Modes)
- Fiber, Slice, and Unfolding
- Tensor Rank
- Outer Product and Rank-1 Tensors

CP Decomposition (CANDECOMP/PARAFAC)

- Sum of Rank-1 Tensors
- Alternating Least Squares
- Uniqueness Properties
- Rank Determination

Tucker Decomposition

- Core Tensor and Factor Matrices
- Higher-Order SVD (HOSVD)
- Higher-Order Orthogonal Iteration (HOOI)
- Multilinear Rank

Tensor Train (TT) Decomposition

- Chain of 3rd-Order Tensors
- TT-Rank
- TT-SVD Algorithm
- Scalability for High-Order Tensors

Applications in ML

- Multiway Data Analysis
- Latent Variable Model Learning
- Knowledge Graph Embeddings
- Neural Network Compression
- Higher-Order Statistics

Kernel and Information-Theoretic Methods

Kernel Methods (Theory)

- Reproducing Kernel Hilbert Spaces (RKHS)
- Representer Theorem
- Kernel Ridge Regression
- Multiple Kernel Learning

Information-Theoretic Learning

- Minimum Description Length (MDL)
- Kolmogorov Complexity
- Rate-Distortion Theory
- Information Bottleneck Method

Convergence Rates of Optimization Algorithms

Gradient Descent Convergence

- Smooth Convex Functions: $O(1/t)$ Rate
- Lipschitz Gradient Condition
- Step Size Selection (1/L Rule)
- Strongly Convex Functions: Linear Convergence
- Condition Number and Convergence Speed

Stochastic Gradient Descent Convergence

- SGD for Convex: $O(1/\sqrt{t})$ Rate
- SGD for Strongly Convex: $O(1/t)$ Rate
- Learning Rate Schedules (Diminishing, Cosine)
- Mini-Batch Size and Critical Batch Size
- Variance Reduction (SVRG, SAGA)

Accelerated Methods

- Nesterov's Accelerated Gradient: $O(1/t^2)$ Rate
- Optimality of Nesterov (Lower Bounds)
- Momentum and Condition Number
- Heavy Ball Method

Adaptive Methods Convergence

- Adam Convergence (and Failure Cases)

- AMSGrad Fix
- AdaGrad for Sparse Gradients
- Coordinate Descent Convergence

Non-Convex Optimization

- Polyak-Łojasiewicz (PL) Condition
- Convergence to Stationary Points
- Escaping Saddle Points
- Landscape Analysis for Neural Networks

Lower Bounds

- Nesterov's Lower Bound for First-Order Methods
- Information-Theoretic Lower Bounds
- Oracle Complexity

Information-Theoretic Limits in ML

Fano's Inequality and Classification Error Lower Bounds

Le Cam's Method for Minimax Lower Bounds

Assouad's Lemma

Minimax Rate of Estimation (Gaussian Mean)

Data Processing Inequality for Representations

Rate-Distortion Theory and Lossy Compression

Mutual Information and Generalization Bounds

Differential Entropy and Generative Model Quality

Information-Theoretic Sample Complexity Lower Bounds

Connection to Entropy in Decision Tree Splitting

Causal Inference in ML

Structural Causal Models (SCM)

- Causal DAGs
- Structural Equations
- Exogenous and Endogenous Variables
- Interventions and the do-Operator

Identification of Causal Effects

- Backdoor Criterion and Adjustment Formula
- Frontdoor Criterion
- do-Calculus (Three Rules)
- Instrumental Variables
- Randomized Controlled Trials (RCTs)

Treatment Effect Estimation

- Average Treatment Effect (ATE)
- Conditional Average Treatment Effect (CATE)
- Propensity Score Matching

- Double Machine Learning (DML)
- CausalForest for Heterogeneous Effects

Causal Discovery

- PC Algorithm
- GES (Greedy Equivalence Search)
- Granger Causality (and Its Limitations)
- NOTEARS (Continuous Optimization)

Counterfactuals

- Pearl's Three-Step Counterfactual Procedure
- Counterfactual Fairness
- Counterfactual Explanations

Causal Phenomena and Paradoxes

- Simpson's Paradox
- Confounding and Selection Bias
- Independent Causal Mechanisms (ICM) Principle
- Causal Reasoning for OOD Generalization

Optimal Transport in ML

Monge and Kantorovich Formulations

Wasserstein Distance (Earth Mover's Distance)

Wasserstein vs KL Divergence

Sinkhorn Algorithm and Entropic Regularization

Sinkhorn Divergence as Training Loss

Sliced Wasserstein Distance

Wasserstein Barycenters

Optimal Transport for Domain Adaptation (OTDA)

Word Mover's Distance (NLP)

Connection to Attention Mechanisms

Connection to Maximum Entropy Principles

WGAN and Optimal Transport

CHAPTER 18

Responsible AI

Ethics, fairness, privacy, and the societal impact of AI systems.

CHAPTER 18

Responsible AI and Ethics

Prerequisites: ML Systems and Production

Outcomes: Fairness & bias · Privacy & security · Sustainability

264 concepts · Max depth: 5

Societal Impact

Generative AI Ethics

- Deepfakes and Synthetic Media
- Preventing Misuse of Generative Models
- Personalization vs Manipulation
- AI-Generated Content and Misinformation

Domain-Specific Ethics

- AI in Healthcare Diagnosis
- AI in Criminal Justice (COMPAS Case Study)
- AI in Hiring (Amazon Case Study)
- Autonomous Decision-Making Systems

Economic and Social Impact

- AI Impact on Employment
- Responsibility Assignment When AI Causes Harm
- Lessons from Past AI Failures (Amazon Hiring, COMPAS, Tay)
- Human Autonomy and Dignity

AI Surveillance and Biometric Privacy

- Facial Recognition Ethics and Bans
- Biometric Data Regulations
- Mass Surveillance Risks

Labor Market and Automation Impact

- Job Displacement and Reskilling
- Economic Inequality
- Gig Economy and AI

Fairness

Bias in Data

- Historical Bias
- Representation Bias
- Measurement Bias
- Sampling Bias
- Label Bias
- Aggregation Bias

Bias in Models

- Algorithmic Bias
- Amplification of Existing Bias
- Proxy Variables
- Embedding Bias (Word2Vec, BERT)
- Bias in Generative Models

Fairness Metrics

- Demographic Parity
- Equalized Odds
- Equal Opportunity
- Predictive Parity
- Individual Fairness
- Counterfactual Fairness
- Impossibility Theorem (Incompatibility of Fairness Metrics)

Debiasing Techniques

- Pre-Processing (Resampling, Reweighting)
- In-Processing (Adversarial Debiasing, Constraint Optimization)
- Post-Processing (Threshold Adjustment)
- Fairness-Aware Embeddings
- Causal Approaches to Fairness
- Fairness Toolkits (AI Fairness 360, Fairlearn)

Legal Foundations

- Disparate Impact and Disparate Treatment
- EEOC 80% Rule
- Protected Classes
- EU AI Act Risk Categories

Intersectional and Subgroup Fairness

- Compounding Discrimination
- Gerrymandering in Fairness
- Multi-Attribute Fairness Metrics

Explainability and Interpretability

Foundations

- Explainability vs Interpretability Distinction
- Explainability-Accuracy Tradeoff
- Importance in High-Stakes Domains (Healthcare, Finance, Justice)
- Right to Explanation (GDPR Article 22)

Inherently Interpretable Models

- Linear and Logistic Regression
- Decision Trees and Rule Lists
- Generalized Additive Models (GAMs)
- Explainable Boosting Machines (EBMs)

Post-Hoc Explanation Methods

- Feature Attribution Methods — SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), Integrated Gradients, DeepLIFT, SHAP vs LIME Comparison, Feature Attribution vs Feature Importance
- Concept-Based Explanations — TCAV (Testing with Concept Activation Vectors), Concept Bottleneck Models
- Counterfactual Explanations — Counterfactual Generation Methods, DiCE (Diverse Counterfactual Explanations), Actionable Recourse
- Attention-Based Explanations — Attention Weights in Transformers, Limitations of Attention as Explanation, Attention Rollout and Flow

Evaluation and Faithfulness

- Faithfulness of Explanations
- Stability and Consistency of Explanations
- Risks of Misleading Explanations
- Human Evaluation of Explanations

Production Explainability

- Real-Time Explanations in Serving Systems
- Explaining Predictions to Non-Technical Users
- Audit and Compliance in Regulated Industries
- Explainability Toolkits (Captum, SHAP, Alibi)

Privacy

Differential Privacy

- Epsilon-Delta Definition
- Privacy Budget
- Noise Mechanisms (Laplace, Gaussian)
- DP-SGD (Differentially Private Stochastic Gradient Descent)
- Local vs Global Differential Privacy
- Composition Theorems

Federated Learning

- Federated Averaging (FedAvg)
- Cross-Device vs Cross-Silo
- Communication Efficiency
- Non-IID Data Challenges
- Secure Aggregation
- Federated Learning Frameworks (Flower, PySyft)

Data Anonymization

- K-Anonymity
- L-Diversity
- T-Closeness
- Data Masking and Pseudonymization
- Synthetic Data for Privacy

Privacy Regulations and Compliance

- GDPR (Right to Erasure, Data Minimization)
- CCPA

- Machine Unlearning
- Privacy Impact Assessments
- Data Processing Agreements

Privacy-Preserving ML Techniques

- Homomorphic Encryption
- Secure Multi-Party Computation
- Trusted Execution Environments
- Private Information Retrieval

Informed Consent and Data Rights

- Consent Frameworks
- Data Sovereignty
- Right to be Forgotten
- GDPR Data Subject Rights

Security

Adversarial Attacks

- Evasion Attacks (FGSM, PGD, C&W)
- White-Box vs Black-Box Attacks
- Adversarial Examples in Images
- Adversarial Attacks on NLP
- Physical-World Adversarial Examples
- Universal Adversarial Perturbations

Robustness

- Adversarial Training
- Certified Robustness (Randomized Smoothing)
- Input Preprocessing Defenses
- Robust Architecture Design
- Robustness Benchmarks

Model Stealing and Extraction

- Query-Based Model Extraction
- Model Watermarking
- Fingerprinting
- API Rate Limiting and Defenses

Data Poisoning

- Training Data Poisoning
- Backdoor Attacks (Trojan Models)
- Label Flipping Attacks
- Data Sanitization Defenses
- Poisoning Detection

Privacy Attacks on ML Models

- Membership Inference Attacks
- Model Inversion Attacks
- Attribute Inference Attacks
- Training Data Extraction from LLMs

- Defenses Against Privacy Attacks

LLM-Specific Security

- LLM-Specific Attacks — Prompt Injection Defense Mechanisms (Input/Output Filtering, Instruction Hierarchy), Data Poisoning in Fine-Tuning, Backdoor Attacks on Language Models, Training Data Extraction Attacks, Membership Inference on LLMs
- LLM Watermarking & IP — Text Watermarking Algorithms (Kirchenbauer et al.), AI-Generated Text Detection, Model Fingerprinting, Copyright in Training Data
- Machine Unlearning — Exact Unlearning, Approximate Unlearning, Unlearning for Right to be Forgotten, Evaluation of Unlearning Effectiveness

Sustainability

Carbon Footprint of AI

- Training Energy Consumption
- Inference Energy at Scale
- Carbon Intensity of Compute Regions
- Lifecycle Analysis of ML Models

Efficient Training

- Efficient Architectures
- Training Efficiency Techniques
- Early Stopping Strategies
- Transfer Learning to Reduce Training
- Green AI Movement

Sustainable ML Practices

- Carbon Tracking Tools (CodeCarbon, ML CO2 Impact)
- Model Reuse and Sharing
- Compute-Optimal Training (Chinchilla Scaling)
- Reporting Energy and Carbon Metrics

Human-in-the-Loop

Active Learning

- Uncertainty Sampling
- Query-by-Committee
- Expected Model Change
- Batch Active Learning

Human Oversight

- Human Review Workflows
- Confidence Thresholds for Escalation
- Feedback Integration
- Annotation and Correction Loops

Human-AI Collaboration

- Augmented Intelligence
- Decision Support Systems
- Calibrated Confidence Communication
- User Trust and Transparency

AI Governance

Regulatory Landscape

- EU AI Act — EU AI Act (expanded)
- NIST AI Risk Management Framework
- Industry-Specific Regulations
- International AI Governance

Organizational Governance

- AI Ethics Boards
- Model Risk Management
- Documentation Standards (Model Cards, Datasheets)
- Audit Trails and Accountability
- Responsible AI Maturity Models

Intellectual Property

- Copyright and AI-Generated Content
- Training Data Licensing
- Open Source Model Licenses
- Patent Considerations

AI Safety and Alignment

Foundations

- AI Safety vs AI Ethics Distinction
- The Alignment Problem
- Value Alignment for LLMs
- Goodhart's Law in ML Systems
- Reward Hacking and Specification Gaming
- Instrumental Convergence

Safeguards

- Guardrails and Content Filters in Production
- Red-Teaming for AI Systems
- Constitutional AI
- RLHF for Safety
- Human-in-the-Loop for Safety-Critical Systems

Responsible Development Practices

- Open-Source vs Safety Tradeoff
- Dual-Use Dilemma in AI Research
- Corporate Transparency About Capabilities and Limitations
- Building Responsible AI Culture in Organizations
- Asilomar AI Principles

Zero to Hero AI

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